Not a Typical Firm: Capital–Labor Substitution and Firms' Labor Shares*

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Abstract

The US labor share has declined, especially in manufacturing and retail. Yet, the labor share of a typical firm in these sectors has risen. We introduce a model where firms incur fixed costs to automate tasks. A decline in the price of capital goods used for automation reproduces the observed patterns: large firms automate tasks, reducing the aggregate labor share, while the median firm continues to operate a labor-intensive technology. When calibrating the automation fixed costs to match the observed adoption heterogeneity, the model generates the aggregate and firm-level facts quantitatively in response to lower capital prices, especially in manufacturing.

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A striking fact of the recent past is the fall of labor's share of GDP in the United States and other countries.[1] After being stable for much of the last century, the US labor share declined from 62% in the 1980s to 55% in 2012. As Figure 1 shows, this fall is due to a sizable decline dating back to the 1980s of the labor share in manufacturing of 20 percentage points and in retail of 12 percentage points, and a more recent decline since the year 2000 for wholesale, utilities, and transportation.

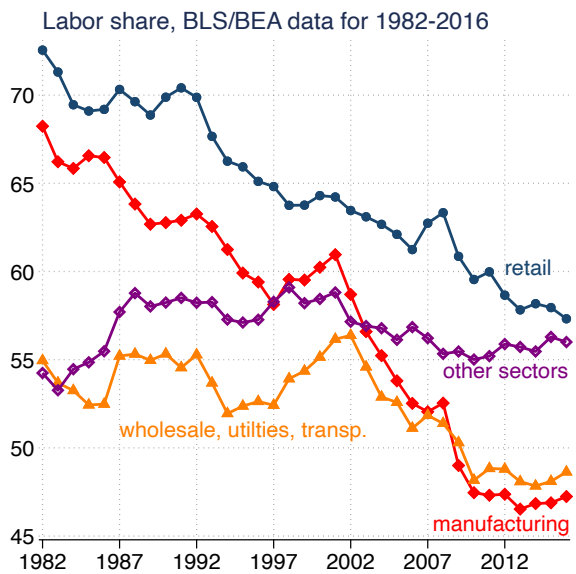

FIGURE 1: LABOR SHARE FOR MAJOR US ECONOMIC SECTORS. The figure plots the labor share in value added for retail, manufacturing, and other US economic sectors. Data from BEA-BLS integrated industry-level production accounts for US private non-farm sectors (see Eldridge et al., 2020).

One set of explanations of the labor share decline points to technological advances that favor the substitution of capital for labor. This substitution can take place along an aggregate production function with an elasticity of substitution greater than one (Karabarbounis and Neiman, 2013; Hubmer, 2023) or within tasks as more tasks are automated (Acemoglu and Restrepo, 2018). An extensive literature provides empirical support for these explanations, both across industries,[2] and also across firms within the same industry.[3] A common finding

[1] See Elsby, Hobijn and Şahin (2013); Karabarbounis and Neiman (2013); Dao, Das and Koczan (2019).

[2] Hubmer (2023) shows that the labor share decline is more pronounced in equipment and software intensive industries. Acemoglu and Restrepo (2019) and Acemoglu and Restrepo (2022) document that US industries with greater possibilities for automation saw larger labor share declines. Acemoglu and Restrepo (2020) and Graetz and Michaels (2018) report that manufacturing industries with greater adoption of robotics saw declining labor shares or rising value added per worker. Boustan, Choi and Clingingsmith (2022) also show that manufacturing industries with greater adoption of computer-numerically-controlled (CNC) machinery saw declining labor shares. Kogan et al. (2021) and Dechezleprêtre et al. (2023) find that US industries patenting labor-replacing technology see a reduction in labor shares.

[3] Acemoglu, Lelarge and Restrepo (2020), Koch, Manuylov and Smolka (2021), Humlum (2019), Bonfiglioli

in these studies is that industries or firms more exposed to automation experienced relatively larger labor share declines (see Restrepo (2023) for a recent review of the empirical evidence).

This evidence notwithstanding, recent studies show that the decline in labor shares is not uniform across firms. While the aggregate labor share has declined, the labor share of the *typical* US firm has risen or remained unchanged. In manufacturing, the median labor share across firms rose from 71% to 74% and the unweighted mean labor share decreased slightly (Autor et al., 2020; Kehrig and Vincent, 2021) since the 1980s.

The new firm-level facts cast doubt on explanations of the labor share decline based on capital–labor substitution, since a version of these theories in which firms operate the same technologies implies a uniform decline. The firm-level facts favor a second set of explanations that emphasize the reallocation of sales towards top firms. This reallocation, which could be the result of increased competition or winner-takes-all dynamics, reduces the aggregate labor share because top firms have higher markups and lower labor shares (Autor et al., 2020; Baqaee and Farhi, 2020b; De Loecker, Eeckhout and Unger, 2020).[4]

This paper shows that explanations of the labor share decline based on capital-labor substitution can generate the aggregate and firm-level patterns documented by Autor et al. (2020) and Kehrig and Vincent (2021). All that is needed is the adoption of capital-intensive technologies for automating additional tasks to entail fixed costs. We make this case in three steps. We first review evidence showing that the adoption of capital-intensive technologies used for automation (i.e., robots, specialized software, and dedicated equipment) concentrates on large firms, as expected from the presence of fixed costs. We then consider a task-based model where firms incur fixed costs to automate additional tasks in response to declining capital prices. We conclude by showing that a version of this model with fixed costs of automation calibrated to match observed heterogeneity in adoption can generate the aggregate and new firm-level facts quantitatively as the result of lower capital prices in manufacturing. Our model can generate these patterns even when capital and labor are complements. This case requires the decline in capital prices to be more pronounced for the

et al. (2020), and Cheng et al. (2021) document that firms adopting robots (or CNC machinery) in various countries reduced their labor shares. Finally, Dinlersoz and Wolf (2023) and Acemoglu et al. (2022) show that adopters of automation technologies have lower labor shares both in manufacturing and across all US economic sectors.

[4]These papers view concentration as a byproduct of increased competition (i.e., *good* concentration). Other studies, including Philippon (2019) and Barkai (2020), adopt the opposite view and have argued that firm-level markups have increased as a result of top firms achieving a more dominant position (i.e., *bad* concentration). We do not consider this alternative in our quantitative exercise because the evidence, both in Baqaee and Farhi (2020b) and De Loecker, Eeckhout and Unger (2020), as well as from our markup estimates in Section 3.3, suggests that the markup increase is not due to within-firm changes, as one expects from reduced competition, but due to reallocation across firms.

types of equipment and software used for automating additional tasks than for other types of capital (for example, existing equipment and structures already in use).

Evidence: Section 1 reviews evidence showing that the use of automation technologies concentrates on large firms. The most comprehensive evidence comes from the *Annual Business Survey* (ABS), and is reported in Acemoglu et al. (2022). Their estimates show that the largest firms in an industry are 1.7 times more likely to use automation technologies than the median firm. We complement this evidence with estimates of output elasticities and measures of investment for firms in Compustat. These estimates suggest that the largest firms in each industry have become more capital-intensive over time, especially in manufacturing.

Theory: Section 2 proposes a firm-dynamics model with monopolistic competition augmented with costly automation decisions across tasks. In these models, firms decide how to produce complementary tasks by using labor or capital (equipment or software) specifically designed for it (as in Zeira, 1998; Acemoglu and Restrepo, 2018). Our new assumption is that firms incur fixed costs to design and integrate the capital needed to automate tasks, which allows us to match the empirical evidence from the ABS.

We characterize the effects of a decline in the effective price of capital—or a q shock—, which captures technological advances increasing the productivity of capital or lowering the cost of producing it. We distinguish two advances. First, we have declines in the price of capital at the *intensive margin*. These refer to advances in (or reductions in the cost of producing) the equipment in use at tasks already produced with capital. Think for example of improvements in industrial furnaces or boilers, improvements in electric generators and transmission equipment, and cheaper vintages of an existing machine. These advances take place at the intensive margin since these tasks were already produced with capital. This form of technological progress is reminiscent of the investment-specific technical change literature, where one has cheaper vintages of capital replacing older ones (as in Greenwood, Hercowitz and Krusell, 1997). Second, we have declines in the price of capital at the *extensive margin*. These refer to the development of (or reductions in the cost of producing) new equipment and software for tasks that are currently assigned to labor, which lead to the automation of these tasks (as in Acemoglu and Restrepo, 2019). Think for example of advances in robotics, computer-assisted manufacturing, and software, which enabled firms to automate a widening range of tasks previously assigned to workers, such as welding and assembly, metalworking, and clerical jobs in offices.[5]

[5]The distinction between intensive and extensive margin advances is a natural one in task models. Ace-

Our main theoretical contribution is to show that a q shock at the extensive margin can reduce the aggregate labor share *at the same time* as the median firm labor share rises. The new mechanism driving this result is as follows: in response to this q shock, large and growing firms incur the fixed cost to automate additional tasks, becoming more capital-intensive. In these firms, capital substitutes for labor at more tasks, driving the decline of the aggregate labor share. Instead, due to the fixed cost of adoption, the median firm does not automate tasks and continues to operate a labor-intensive technology. For the median firm, capital and labor remain complements, explaining why their labor share rises.[6] This holds irrespective of whether the elasticity of substitution between capital and labor is below one (as in Oberfield and Raval, 2021) or above one (as in Karabarbounis and Neiman, 2013; Hubmer, 2023).

Quantitative exploration: Section 3 presents quantitative exercises designed to illustrate the implications of our model. One shortcoming we face is that we cannot measure capital advances at the extensive and intensive margin separately, precluding a traditional exercise where one measures these shocks, feeds them in the model, and computes their effects. We pursue a different approach. We feed in shocks of different magnitudes and ask: (i) Can q shocks generate labor share declines of the size we see in the data? (ii) Are the required q shocks of a reasonable magnitude and in line with available measures of capital prices? And most importantly, (iii) are the implied firm labor share dynamics in line with the new evidence in Autor et al. (2020) and Kehrig and Vincent (2021)?

Our first exercise focuses on manufacturing. It assumes a CES demand system with fixed markups, isolating the effect of q shocks operating through technology. In a second exercise, we extend this to a log-concave demand system with endogenous markups. This allows us to study how q shocks affect labor shares through markups. We treat the economy in 1982 as being in a steady state and calibrate relevant parameters to match the distribution of firm sales in manufacturing, average markups, and other moments. We also calibrate a small fixed cost of automation to match the adoption gradient by size in the ABS.

We show that one can generate the 20 pp labor share decline in manufacturing via a

moglu and Restrepo (2019) introduced the distinction but used a different terminology from us. Contemporary work by Jones and Liu (2022) and Drozd, Taschereau-Dumouchel and Mendes Tavares (2022) emphasizes this distinction and shows how advances at the extensive and intensive margin can combine to generate balanced growth. These contributions, however, do not explore the implications of both types of q shocks for the behavior and dynamics of firm-level labor shares, which is our main focus.

[6] A different mechanism introduced by Houthakker (1955) and explored by Jones (2005) emphasizes the possibility that a decline in capital prices reallocates economic activity to more capital-intensive firms, even if firms do not change their factor intensities. Kaymak and Schott (2018) argue that this mechanism generates a third of the manufacturing labor share decline in response to lower corporate taxes. On the other hand, Oberfield and Raval (2021) show this reallocation has a small effect on the US manufacturing labor share.

combination of q shocks. Our model achieves this by having a 60–67 log point decline in capital prices at the intensive margin and a more pronounced decline at the extensive margin. The implied decline in average capital prices matches available estimates of the total decline in capital prices for 1980–2012 in manufacturing of 108 log points.

More novel, in response to these q shocks, the model reproduces the firm labor share moments observed in the data, even though these are not targeted. While the manufacturing labor share declines, the model generates rising or constant median and unweighted mean labor shares. As in the Melitz-Polanec decomposition in Autor et al. (2020), the decline in the manufacturing labor share is driven by a more negative covariance between sales and labor shares. The more negative covariance reflects the fact that firms automate during periods of expanding sales, which aligns with the dynamics of changes in sales and labor shares for manufacturing firms in Kehrig and Vincent (2021). The CES model tracks these firm-level moments well, but the version with endogenous markups provides the best fit.

In the last exercise, we study the role of rising competition—the main alternative explanation for the labor share decline put forth by Autor et al. (2020)—jointly with q shocks. We model rising competition as the result of an increase in the mass of customers a firm can access—a λ shock. This shock can also generate the pattern seen in the data of rising concentration, declining aggregate labor share, and a rise in a typical firm labor share. With log-concave demand, a λ shock leads to an increase in sales concentration, a reduction in within-firm markups (and an increase in firms' labor shares), and a reallocation towards large firms with high markups (and low labor shares). When the distribution of firm productivity is more log-convex than Pareto, reallocation dominates and the aggregate labor share falls.

To gauge magnitudes and put both explanations on equal footing, we provide a decomposition exercise using our model where we recover the q and λ shocks required to match the trends in manufacturing and separate their contributions.

Our decomposition attributes most of the labor share decline in manufacturing and the divergence from the median firm labor share to q shocks. Our decomposition attributes half of the increase in concentration to λ shocks, but also shows that these had a limited impact on the manufacturing labor share, as the reallocation and within-firm changes in markups roughly cancel out. This conclusion aligns with markup estimates for Compustat firms, at the end of this section.

For this exercise we also look at retail and other sectors. We show that one can match sectoral trends in labor shares and sales concentration with different combinations of q and λ shocks across sectors. Our decomposition points to a more important role for λ shocks in driving the observed sectoral trends outside of manufacturing. In retail, for example, our

decomposition attributes 90% of the increase in sales concentration and 25% of the labor share decline to rising market access. Yet, most of the labor share decline continues to be attributed to q shocks and the resulting automation of tasks by large firms.

1 Motivating evidence on the adoption of automation technologies

The most comprehensive evidence on the uneven adoption of automation technologies by large firms comes from the 2019 module of the *Annual Business Survey*, which contains information on the adoption of robotics, dedicated equipment, specialized software, and artificial intelligence for 300,000 firms across all US economic sectors during the period from 2016 to 2018. Using the ABS, Acemoglu et al. (2022) show that, within detailed 6-digit industries, firms in the top percentile of the employment size distribution are 1.71 times more likely to use these technologies than firms between the 50th and 75th percentile. This strong size gradient is visible across sectors and when one conditions on firm age.

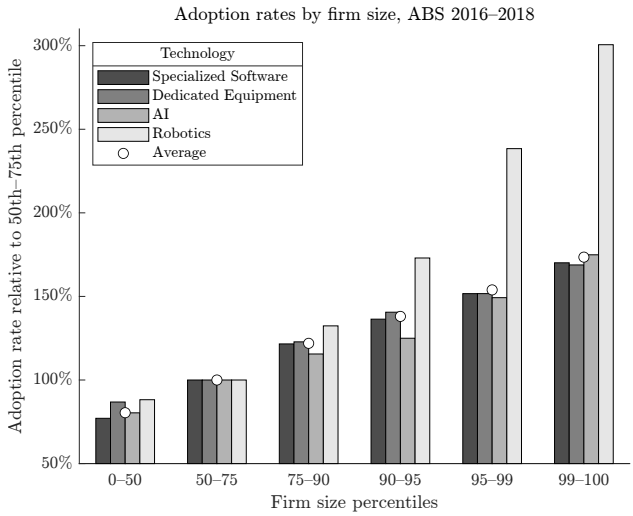

Figure 2: Adoption rates by firm size, ABS 2016–2018. The figure plots adoption rates for various automation technologies relative to adoption rates of firms in the 50th–75th percentiles of the employment size distribution. Data from Figure 4 in Acemoglu et al. (2022), reproduced with their permission.

Figure 2 reproduces their evidence and plots for each of the technologies in the ABS its adoption rate across firms in different size percentiles (relative to that of firms between the 50th and 75th employment percentiles) within detailed 6-digit industries. Adoption rates at the top are 1.68 (for dedicated equipment) to 3 times higher (for robotics) than in the middle of the firm-size distribution, with an average across technologies of 1.71.[7]

[7] The size gradient is visible for all technologies and stronger for technologies with limited adoption. For robotics, adoption rates rise from 1.7% (for the median bracket) to 5.1% (in the highest bracket). For AI, adoption rates rise from 3.1% to 5.5%. For dedicated equipment, adoption rates rise from 18.6% to 31.4%.

This pattern for robot adoption is also visible for manufacturing firms in Canada and in European countries (see Koch, Manuylov and Smolka, 2021; Humlum, 2019; Bonfiglioli et al., 2020; Acemoglu, Lelarge and Restrepo, 2020).

Evidence from other datasets shows that the intensity of use also rises with size. Dinlersoz and Wolf (2023) use the *Survey of Manufacturing Technologies* from 1993 to document that the adoption and intensity of use for various automation technologies among US manufacturing plants rises with size. Brynjolfsson et al. (2023) use the new Annual Survey of Manufactures (ASM) to document that the number of robots per worker rises with establishment and firm size.

We complement this evidence with estimates of the output-to-capital elasticity across firms in Compustat. This elasticity—the percent increase in output generated by a percent increase in the quantity of capital—summarizes the capital intensity of a firm and the extent to which it has automated its production processes. Our production function estimation approach is detailed in Appendix E. Relative to the existing literature, we make two modifications. First, we estimate a revenue production function (given the lack of price data in Compustat), but show that one can recover output elasticities by assuming constant returns to scale. Second, we allow output elasticities to vary across firm-size classes to account for differences in technology by size.

Figure 3 plots the estimated output-to-capital elasticities by firm size and time period. The left panel shows averaged elasticities for manufacturing industries, while the right panel does so for non-manufacturing. In the 1970s, firms had similar output-to-capital elasticities ranging from 0.08 to 0.12 in both sectors. In the following decades, we estimate a pronounced increase in output-to-capital elasticities among firms in the top quintiles of the size distribution. For the largest manufacturing firms, the elasticity increases by 0.2, going from 0.11 to 0.31. Outside of manufacturing, we estimate a less pronounced increase for large firms from 0.09 to 0.17. In line with these estimates, Appendix E documents that large firms in Compustat have higher investment rates and capital-output ratios, and that these differences have become more pronounced over time.

We view the evidence from the ABS, Compustat, and all sources cited above as complementary. The ABS represents the universe of US firms and contains explicit measures of technology adoption, but these are only available for 2016–2018 and miss the intensity of

For specialized software, adoption rates rise from 39.6% to 67.4%. These technologies are relevant because they are typically used by firms for automation. The statistics reported in Acemoglu et al. (2022) imply that 84% of manufacturing firms (in an employment weighted sense) using robotics, 53% of firms using dedicated equipment, 41% of the firms using specialized software systems, and 78% of firms using AI systems employ these technologies to automate tasks previously assigned to labor.

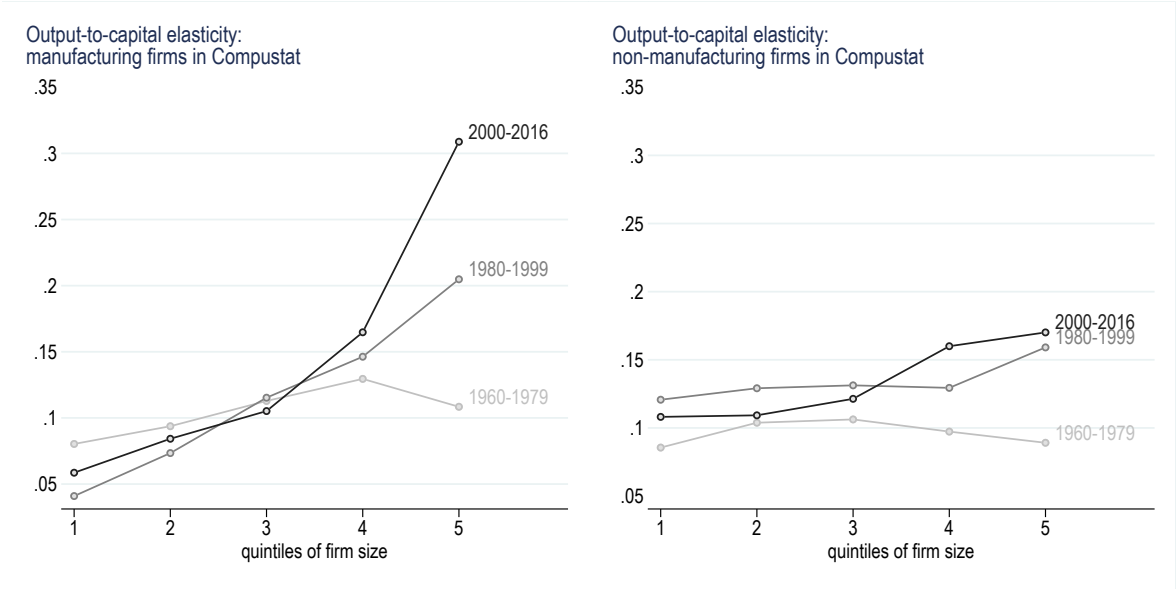

FIGURE 3: OUTPUT-TO-CAPITAL ELASTICITIES FOR COMPUSTAT FIRMS. Left panel presents estimates for Compustat manufacturing firms. Right panel presents estimates for Compustat non-manufacturing firms.

use. Compustat contains information on the intensity of capital use and allows us to trace changes in capital intensity over time, but represents a selected sample of firms. While each data source on its own has limitations, together they portray a consistent picture where large firms are more capital-intensive and have automated more processes than small competitors.

Our interpretation of these findings is that adopting and integrating automation technologies involves fixed costs. This assumption is plausible. Consider a car-manufacturing firm that wishes to automate welding. Besides purchasing the industrial robots required to complete this task, the firm must also hire a team of engineers to reorganize its plant, redesign some of their products, and program the robots so that they can be integrated. This is fixed cost that does not scale with the quantity of cars produced. The same software routine and plans can be used across factories and once more robots are added to the assembly line. If the firm wants to automate a different process, it must then pay another fixed cost to develop the software and integrate robots to that additional task. These upfront integration costs can be sizable. For example, in the case of industrial robots, integration costs far exceed the cost of a robot (see Acemoglu and Restrepo, 2020). Likewise, a firm that wants to deploy a new software to automate its logistics and inventory management must pay a fixed cost for developing the software and rearranging its operations to use it. Here too, the cost of developing the software is independent of firm scale (i.e., the same software system can handle a different number of package deliveries).[8]

[8] A complementary interpretation of these fixed costs is as capturing the possibility that small firms face

In our quantitative exercise, we use the ABS data in Figure 2 to calibrate the fixed cost of adopting automation technology. We treat Compustat estimates as *untargeted moments* and explore if our model matches the rotation in output-to-capital elasticities over time.

2 A task model of capital–labor substitution with adoption costs

This section introduces our task model of capital-labor substitution. We augment a firm-dynamics model (as in Hopenhayn, 1992) to include firms' decisions to automate tasks (as in Acemoglu and Restrepo, 2018). Our main innovation is to endogenize firms' automation decisions as determined by payment of a fixed cost per task.

2.1 Environment and equilibrium

Environment: Consider an economy in discrete time t with firms f in monopolistic competition. Firms produce differentiated varieties y_{tf} combined via a constant-returns to scale aggregator F into a final good y_t, whose price we normalize to 1:

$$y_t = F\left(\{y_{tf}\}\right).$$

The aggregator F is assumed symmetric so that all firms face the same demand curve.

Firms differ in their productivity z_{tf} and in the fraction of tasks or processes they have automated, $\alpha_{tf} \in [0,1]$. A firm produces output y_{tf} by combining a continuum of complementary tasks indexed by x with task substitution elasticity $\eta \in [0,1)$:

$$y_{tf} = z_{tf} \cdot \left(\int_0^1 y_{tf}(x)^{\frac{\eta-1}{\eta}} \cdot dx \right)^{\frac{\eta}{\eta-1}}.$$

Tasks in $[0, \alpha_{tf}]$ are automated and can be produced by capital. Non-automated tasks in $(\alpha_{tf}, 1]$ must be produced by labor. The quantity of task x is given by

$$y_{tf}(x) = \begin{cases} \psi_k(x) \cdot k_{tf}(x) & \text{if } x \leq \alpha_{tf} \\ \psi_\ell(x) \cdot \ell_{tf}(x) & \text{if } x > \alpha_{tf} \end{cases}$$

where $\ell_{tf}(x)$ and $k_{tf}(x)$ denote labor and capital employed to produce task x with task-specific productivity $\psi_\ell(x)$ and $\psi_k(x)$.[9]

tighter financial constraints and investments in automated systems are sizable and indivisible.

[9] Our notation assumes firms produce all tasks below α_{tf} with capital. This is the relevant scenario in our simulations. The appendix handles the general case where firms can allocate these tasks with labor.

Firms compete for ℓ workers hired at a wage w_t. The capital used for task x is produced from the final good, with an unit of transformed into $q_{I,t}(x)$ units of capital. We treat capital as an intermediate, which is produced instantly and fully depreciates after use. We refer to

$$q_t(x) = q_{I,t}(x) \cdot \psi_k(x)$$

as the productivity of capital and $1/q_t(x)$ as its (quality adjusted) price. We assume $\psi_\ell(x)/q_t(x)$ is increasing in x, so labor has comparative advantage at high-index tasks.

Incumbents begin a period with productivity z_{tf} and automation level α_{tf}, make optimal employment and capital utilization decisions, and collect profits. Firms then draw a fixed operating cost $c_o \cdot y_t$, where $c_o \sim G(c_o)$, and decide whether to exit or continue. Continuing incumbents can expand the set of automated tasks to include $(\alpha_{tf}, \alpha_{t+1,f}]$ at a cost $c_a \cdot y_t \cdot (\alpha_{t+1,f} - \alpha_{tf})$, which implies a fixed cost of automation per task of $c_a \cdot y_t$.[10] Continuing incumbents also draw next period's productivity level $z_{t+1,f}$, which follows an exogenous first-order Markov process with $z_{t+1,f}$ increasing in z_{tf} (in a stochastic sense).[11]

Every period a unit mass of potential entrants draw a productivity z and fixed operating cost c_o. They then decide whether to pay the fixed cost and enter. Entrants start with a common level of automation $\bar{\alpha}_t$, equal to the unweighted average of α_{tf} among incumbents. This common specification in models of diffusion (see Perla, Tonetti and Waugh, 2021) captures the standardization of automation technologies. Diffusion simplifies our analytical derivations, but is not required for our quantitative results.

Finally, when making entry and adoption decisions, incumbents and new entrants discount the future at a constant interest rate r, which we take as exogenous. We assume that $r > g_t$, where g_t is the growth rate of output between two consecutive periods.

Equilibrium: Along the equilibrium path, denote by $p_t(\alpha, z)$ the price charged by a firm with automation level α and productivity z, by $c_t(\alpha, z)$ its cost, by $\pi_t(\alpha, z)$ its profits, and by $y_t \cdot d_t(\alpha, z)$ its quantity produced ($d_t(\alpha, z)$ is the demand per unit of the final good, which depends on the distribution of all firms' actions via t).

Given a path for capital productivity $q_t(x)$ and an initial distribution of firms $\{\alpha_{0f}, z_{0f}\}$,

[10] Fixed costs are scaled by aggregate output to avoid a mechanical relationship between GDP and firms' incentives to produce or automate tasks. We denominate fixed costs in units of the final good to avoid a mechanical relationship between firm size and labor shares.

[11] We impose two restrictions, which are used in our proofs and are satisfied by commonly used processes. First, we require that for any increasing unbounded function M, $\mathbb{E}[M(z_{t+1,f})|z_{tf}]$ converges to infinity when $z_{tf} \to \infty$ and to $M(0)$ when $z_{tf} \to 0$. Second, we require that for any $0 < \underline{z} < z_{t_0 f} < \bar{z}$, firms reach $z_{tf} \geq \bar{z}$ and $z_{tf} \leq \underline{z}$ from $z_{t_0 f}$ with positive probability over time.

an equilibrium is given by a path for wages w_t and output y_t, and a path for the distribution of firms $\{\alpha_{tf}, z_{tf}\}$, such that for all $t \geq 0$:

E1. The ideal-price index condition holds $\int_f p_t(\alpha_{tf}, z_{tf}) \cdot d_t(\alpha_{tf}, z_{tf}) \cdot df = 1$.

E2. The labor market clears $y_t \cdot \int_f d_t(\alpha_{tf}, z_{tf}) \cdot \frac{\partial c_t(\alpha_{tf}, z_{tf})}{\partial w_t} \cdot df = \ell$.

E3. Automation and exit decisions maximize the value function of incumbents

$$\begin{aligned} V_t(\alpha, z) =& \pi_t(\alpha, z) \\ &+ \int \max\left\{0, -c_o \cdot y_t + \max_{\alpha' \epsilon [\alpha, 1]} \left\{-c_a \cdot y_t \cdot (\alpha' - \alpha) + \frac{1}{1+r} \mathbb{E}\left[V_{t+1}(\alpha', z')|z\right]\right\}\right\} dG(c_o). \end{aligned}$$

E4. Entry decisions maximize the value function of entrants

$$V_t^e(z) = \int \max\left\{0, -c_o \cdot y_t + \max_{\alpha' \epsilon [\bar{\alpha}_t, 1]} \left\{-c_a \cdot y_t \cdot (\alpha' - \bar{\alpha}_t) + \frac{1}{1+r} \mathbb{E}\left[V_{t+1}(\alpha', z')|z\right]\right\}\right\} dG(c_o),$$

where z denotes an entrant's productivity signal, and $\bar{\alpha}_t \equiv (\int_f \alpha_{tf} \cdot df)/(\int_f df)$.

E5. Starting from a distribution $\{\alpha_{0f}, z_{0f}\}$, the evolution of $\{\alpha_{tf}, z_{tf}\}$ is governed by the exogenous process for z, the endogenous process for α, and entry and exit.

For the analytical results below, we assume a CES demand with elasticity $\sigma > 1$. This isolates the role of technology on the labor share, since firms charge a fixed markup $\mu = 1 - 1/\sigma$. Later, we explore the implications of a log-concave demand system numerically.

2.2 Optimal automation decisions and factor shares

The main innovation in our model is to endogenize automation decisions. These decisions are important because they determine costs and factor shares. We can see this from the unit cost for a firm with productivity z and automation level α:

$$c_t(\alpha, z) = \frac{1}{z} \cdot \left(\Gamma_t^k(\alpha) + \Gamma^\ell(\alpha) \cdot w_t^{1-\eta}\right)^{\frac{1}{1-\eta}}. \tag{1}$$

This resembles a CES cost function, but with endogenous shares $\Gamma_t^k(\alpha) = \int_0^\alpha q_t(x)^{\eta-1} \cdot dx$ and $\Gamma^\ell(\alpha) = \int_\alpha^1 \psi_\ell(x)^{\eta-1} \cdot dx$ given by functions of the degree of automation α. Shephard's lemma implies that the share of capital in cost ε_{tf}^k—the output-to-capital elasticity—and the share of labor in cost ε_{tf}^ℓ—the output-to-labor elasticity—are

$$\varepsilon_{tf}^k = \frac{\Gamma_t^k(\alpha_{tf})}{\Gamma_t^k(\alpha_{tf}) + \Gamma^\ell(\alpha_{tf}) \cdot w_t^{1-\eta}}, \qquad \varepsilon_{tf}^\ell = \frac{\Gamma^\ell(\alpha_{tf}) \cdot w_t^{1-\eta}}{\Gamma_t^k(\alpha_{tf}) + \Gamma^\ell(\alpha_{tf}) \cdot w_t^{1-\eta}}.$$

These expressions show that firms can control their capital-intensity by investing in automating additional tasks. The more tasks a firm automates, the more capital intensive its production becomes and the lower its labor share in costs and value added.

Let α_t^* denote the automation level that minimizes $c_t(\alpha, z)$. This is the automation level firms would choose if automating tasks entailed no fixed cost. This is defined implicitly as the point at which the cost of producing a task with labor and capital are equalized:

$$\frac{w_t}{\psi_\ell(\alpha_t^*)} = \frac{1}{q_t(\alpha_t^*)}. \tag{2}$$

With fixed costs, not all firms automate all tasks up to α_t^*. Proposition 1 shows that, in line with the evidence in Section 1, firms automate more tasks as they expand, with the largest firms automating all tasks up to α_t^*.

Proposition 1 *For $\alpha < \alpha_{t+1}^*$, optimal automation decisions are given by $\alpha' = \tilde{\alpha}_t(\alpha, z)$, an increasing function of z that satisfies $\lim_{z\to 0} \tilde{\alpha}_t(\alpha, z) = \alpha$ and $\lim_{z\to\infty} \tilde{\alpha}_t(\alpha, z) = \alpha_{t+1}^*$.*

2.3 The effects of q shocks

This section characterizes the effects of q shocks—technological advances that increase the productivity of capital or reduce its cost. For constant capital prices, the economy admits a stationary equilibrium with wage w^* and common automation level α^* (an implication of diffusion). We focus on an economy in this equilibrium and derive the first-order effects of a permanent decrease in the price of capital $1/q_t(x)$.

We consider two types of q shocks described in the introduction: advances in capital at the intensive margin and at the extensive margin. The first involves a reduction in the effective price of capital for tasks below α^*. The second involves a reduction in the effective price of capital for tasks above α^*, causing the automation of tasks assigned to labor.

The effects of these shocks can be expressed using two elasticities: the elasticity of substitution across tasks η, and the *induced elasticity of substitution between capital and labor*:

$$\eta_t^* = \eta + \frac{\partial \ln[\Gamma_t^k(\alpha)/\Gamma^\ell(\alpha)]}{\partial \ln \alpha} \Bigg/ \frac{\partial \ln[\psi_\ell(\alpha)/q_t(\alpha)]}{\partial \ln \alpha}.$$

The induced elasticity corresponds to the elasticity of substitution that an econometrician exploiting permanent differences in wages across labor markets would estimate. It accounts for substitution across tasks (given by η) and the automation of additional tasks (given by

the change in α_t^*) in response to factor price changes. Because of this second margin, the induced elasticity exceeds η.

We denote the common cost shares for labor and capital across firms at the initial equilibrium by ε^ℓ and ε^k, and the induced elasticity of substitution by η^*.

Proposition 2 (Effects of q shock at intensive margin) *A permanent increase in $q(x)$ by $d\ln q(x) = d\ln q_{int} > 0$ for all $x \leq \alpha^*$ pushes the economy to a new steady state with wages rising by $d\ln w = (\varepsilon^k/\varepsilon^\ell) \cdot d\ln q_{int} > 0$ and automation rising by $d\ln \alpha^* > 0$. The aggregate labor share in cost (or value added) changes by*

$$d\ln \varepsilon^\ell = \varepsilon^k \cdot (1-\eta^*) \cdot d\ln w + \varepsilon^k \cdot (1-\eta) \cdot d\ln q_{int}.$$

Along the transition, the labor share of an incumbent firm that automates no additional tasks increases by $d\ln \varepsilon_{tf}^\ell = \varepsilon^k \cdot (1-\eta) \cdot (d\ln w_t + d\ln q_{int}) > 0$.

The proposition shows that the effects of the shock on the aggregate labor share can be decomposed in two parts: a general equilibrium effect coming from the change in wages and a part coming from the change in capital prices. The effect of a change in wages is well understood: it depends entirely on the elasticity of substitution η^* and on whether $\eta^* \lesseqgtr 1$.

The novel aspect has to do with the effect coming from capital price changes at the intensive margin. This form of technological progress reduces the price of capital-intensive tasks below α^*. Firms respond by substituting towards these tasks with an elasticity η. Because tasks are complements, this substitution is limited and capital-intensive tasks become a smaller share of costs, increasing the labor share.[12]

The proposition also shows that, along the transition, incumbent firms might see an increase in their labor shares. These firms see their capital-intensive tasks below α^* becoming cheaper and their labor-intensive tasks above α^* becoming expensive. These price changes increase the labor shares of small firms that do not automate tasks in response.

In sum, for $\eta^* \leq 1$ (as in our calibration), q shocks at the intensive margin raise the labor share of all firms along the transition and in the new steady state.

[12]This result provides a reasonable interpretation of recent papers documenting a positive effect of investment on firm-level labor shares. For example, Aghion et al. (2023a) show that, when suppliers of equipment become more productive, their existing downstream clients see an increase in the labor share of sales, and no systematic changes in their labor shares of value added. Their strategy of looking at existing clients isolates the effect of advances at the intensive margin.

Proposition 3 (Effects of q shock at extensive margin) *A permanent increase in $q(x)$ by $d\ln q(x) = d\ln q_{ext} > 0$ for all $x \geq \alpha^*$ pushes the economy to a new steady state with wages rising by $d\ln w > 0$ and automation rising by $d\ln \alpha^* > 0$. The aggregate share of labor in costs (or value added) changes by*

$$d\ln \varepsilon^\ell = \varepsilon^k \cdot (1-\eta^*) \cdot d\ln w - \varepsilon^k \cdot (\eta^* - \eta) \cdot d\ln q_{ext},$$

which is always negative for small $d\ln q_{ext}$. Along the transition, the labor share of an incumbent firm that automates no additional tasks increases by $d\ln \varepsilon^\ell_{tf} = \varepsilon^k \cdot (1-\eta) \cdot d\ln w_t > 0$.

Here too, the effects of the shock on the aggregate labor share can be decomposed in two parts: a general equilibrium effect coming from the change in wages (with the same logic as before) and a part coming from the change in capital prices at the extensive margin.

The novel aspect has to do with the effect coming from capital price changes at the extensive margin. These technological advances reduce the cost of producing tasks above α^* with capital. Firms respond by automating some of these tasks, which reduces their labor share by $\varepsilon^k \cdot (\eta^* - \eta) \cdot d\ln q_{ext}$. The difference $\eta^* - \eta > 0$ is informative of how much substitution firms can do at the extensive margin by automating tasks.[13]

Due to this substitution at the extensive margin, our model can generate a decline in the aggregate labor share in response to q shocks even in scenarios with $\eta^* \leq 1$.[14] The proof in the appendix shows that the increase in wages due to an extensive margin q shock is second order, which implies weak general equilibrium effects due to wage changes. As a result, small extensive margin q shocks always reduce the aggregate labor share, independently of η^*.

The proposition also shows that, along the transition, incumbent firms might see an increase in their labor shares. These firms see their labor-intensive tasks above α^* becoming expensive. These price changes increase the labor shares of small firms that do not automate

[13] This result aligns with the evidence in footnotes 2 and 3. These empirical studies focus on capital advances at the extensive margin that led to the automation of tasks previously assigned to labor. The studies show that the deployment of these technologies has been associated with declining labor shares, as in our proposition.

[14] This result runs against common intuitions and deserves explanation. In most analysis of the labor share, researchers assume a single capital aggregate "K" and study the decline in its price. In our model, this is equivalent to assuming a uniform decline in capital prices across tasks, or $d\ln q_{int} = d\ln q_{ext} = d\ln q$. The effect of uniform declines is to change wages by $d\ln w = (\varepsilon^k/\varepsilon^\ell) \cdot d\ln q_{int} > 0$ and the labor share by

$$d\ln \varepsilon^\ell = \varepsilon^k \cdot (1-\eta^*) \cdot d\ln w + \varepsilon^k \cdot (1-\eta^*) \cdot d\ln q = (\varepsilon^k/\varepsilon^\ell) \cdot (1-\eta^*) \cdot d\ln q.$$

This shows that the elasticity of substitution mediates the labor share effects of a change in wages (as emphasized above) and of a uniform capital price decline. If $\eta^* > 1$, a uniform capital price decline reduces the labor share. If $\eta^* < 1$, it increases it. This logic no longer holds when considering q shocks at the intensive and extensive margin of different magnitudes.

tasks in response. For these firms, labor and capital are effectively complements, causing the labor share to increase in response to the higher real wages.

Propositions 2 and 3 show that one can in principle explain the manufacturing data as resulting from a combination of q shocks. To illustrate, set $\eta^* = 1$—the baseline in our calibration. We can write the effect of a combined shock on the aggregate labor share as

$$d\ln\varepsilon^{\ell} = \varepsilon^{k}\cdot(1-\eta)\cdot(d\ln q_{int} - d\ln q_{ext}),$$

and on the labor share of incumbents that have not automated along the transition as

$$d\ln\varepsilon^{\ell}_{tf} = \varepsilon^{k}\cdot(1-\eta)\cdot(d\ln w_t + d\ln q_{int}).$$

These expressions show that q shocks that are more pronounced at the extensive margin can reduce the aggregate labor share and at the same time increase the labor share for small incumbents along the transition, explaining the decoupling of the aggregate and a typical firm's labor share in the data.

3 Quantitative exploration

The theory section showed that q shocks can generate a decline in the aggregate labor share accompanied by a rise in a typical firms' labor share. This section explores this mechanism quantitatively in a calibrated version of our model.

3.1 Exercise I: manufacturing with a CES demand

Our first exercise focuses on the US manufacturing sector and assumes a CES demand system with elasticity σ and fixed markups. This choice isolates the contribution of capital prices to the labor share via technology.

We take the economy in 1982 to be in steady state with all firms having the same level of automation α_0. We calibrate the parameters of our model to match key moments of the US manufacturing sector in this steady state and study a reduction in effective capital prices from 1982 to 2012. The timing is motivated by the fact that the decline of the manufacturing labor share starts in 1982, after being constant in the decades before.[15]

[15] Appendix G shows this also holds for other sectors. After having stable labor shares during 1947–1980, manufacturing and other sectors experienced a decline in their labor shares starting in the early 80s. Available measures of capital prices also point to a more rapid decline starting in the early 80s (see Hubmer, 2023). Finally, as shown in Section 1, firms of different sizes had similar capital intensities before 1980, which

Calibration: As emphasized in the theory, a q shock at the extensive margin can generate a decline in the manufacturing labor share even for an elasticity of substitution $\eta^* \leq 1$. To demonstrate this novel aspect of our model, we consider a parametrization with $\eta < 1$ and $\eta^* = 1$ as our baseline and report results for a calibration with $\eta^* > 1$ in the appendix.

We normalize initial capital prices to 1 and set capital and labor task productivities to

$$\psi_k(x) = x^{\frac{1-\gamma_k}{1-\eta}} \cdot (1-x)^{\frac{1+\gamma_k}{1-\eta}}, \qquad \psi_\ell(x) = A \cdot x^{\frac{1+\gamma_\ell}{1-\eta}} \cdot (1-x)^{\frac{1-\gamma_\ell}{1-\eta}},$$

where $\gamma_k, \gamma_\ell > 0$ control the comparative advantage of labor at higher-index tasks.

With this parametrization, the cost function of firm f in the initial steady state becomes

$$c_0(\alpha, z) = \frac{1}{z} \cdot \left(\frac{1}{\gamma_k} \cdot \left(\frac{\alpha}{1-\alpha} \right)^{\gamma_k} + \frac{1}{\gamma_\ell} \cdot \left(\frac{\alpha}{1-\alpha} \right)^{-\gamma_\ell} \cdot \left(\frac{w_0}{A} \right)^{1-\eta} \right)^{\frac{1}{1-\eta}}. \tag{3}$$

To show that $\eta^* = 1$, minimize (3) with respect to α. This yields the induced cost function

$$c_0(z, \alpha_0^*) = \frac{1}{z} \cdot \mathcal{M} \cdot w_0^{\frac{\gamma_k}{\gamma_k + \gamma_\ell}}, \qquad \text{with } \mathcal{M} \text{ a constant.}$$

This cost function features a constant elasticity with respect to wages, which shows that the induced elasticity of substitution η^* equals 1. Moreover, this derivation shows that all firms start with a common labor share in costs of $\gamma_k/(\gamma_k + \gamma_\ell)$ in the initial steady state.

We calibrate the production function parameters as follows. We pick $\gamma_k = 1$ and set $\gamma_\ell = 0.3$ to match an initial manufacturing labor share of 67%. We pick the labor productivity level A to normalize wages in 1982 to 1. We also set $\eta = 0.5$ to match the task-level elasticity of substitution estimate in Humlum (2019). These parameters are listed in Panel I of Table 1. Our parametrization of the production side is consistent with estimates of the elasticity of substitution between capital and labor below 1. For example, exploiting differences in wages across regions, Oberfield and Raval (2021) estimate elasticities of substitution in the 0.5–1 range. In our model, these estimates lie between η—for firms that have not adjusted their α's—and $\eta^* = 1$—for firms that set α optimally in response to wages in their region.

We calibrate the demand system, productivity process, and operating costs to match moments from the US manufacturing sector in 1982. Panel II of Table 1 lists the calibrated parameters and moments targeted.

We set the demand elasticity to $\sigma = 7.67$, which generates a constant markup of 1.15.

points to comparable automation levels in this initial steady state.

Table 1: Calibration of the CES demand model for manufacturing

	Parameter		Moment	Data	Model
	I. Parameters related to the production function				
η	Task substitution elasticity	0.5	From Humlum (2019)	0.5	0.5
γ_ℓ	Comparative advantage	0.30	Manufacturing labor share (BLS/BEA)	0.67	0.67
	II. Parameters governing firm dynamics and productivities				
σ	Demand elasticity	7.67	Aggregate markup	1.15	1.15
σ_z	Std. dev. of $\ln z$ innovations	0.105	Top 4 firms' sales share in 1982	40.0%	40.0%
$\underline{c}_o$	Scale operating cost	$4.0 \cdot 10^{-7}$	Entry (=exit) rate	0.062	0.063
ξ_o	Tail index operating cost	0.250	Relative exiter size	0.490	0.490
μ_e	Entrant productivity	0.905	Relative entrant size	0.600	0.600
ρ_z	Productivity persistence	0.95	Revenue TFP persistence among manufacturing firms		

Notes: The table reports the calibrated parameters and targets for our model with CES demand in manufacturing. The annual entry rate, as well as relative sizes of entrants and exiters, are from Lee and Mukoyama (2015) and based on the Annual Survey of Manufactures. The data on top firms' sales share comes from Autor et al. (2020).

We assume that firm productivity follows an AR1 process in logs:

$$\ln z_{t+1,f} = \rho_z \cdot \ln z_{t,f} + \varepsilon_{t+1,f},$$

where $\rho_z \in (0,1)$, $\varepsilon_{t+1,f} \sim N(\mu_z, \sigma_z)$, and $\mu_z = -\frac{(1-\rho_z)\cdot\sigma_z^2}{2\cdot(1-\rho_z^2)}$ so that the mean of z_{tf} is normalized to one. We pick the dispersion of firm productivity to match the share of sales among the top 4 firms within 4-digit manufacturing industries in Autor et al. (2020), which corresponds to the top 1.1% of firms in each industry (based on their reported number of firms per industry). We set the persistence of productivity ρ_z to 0.95, which we obtained from our estimates for the persistence of revenue TFP for manufacturing firms from the production function estimates in Section 1. This aligns with estimates using US Census data in Foster, Haltiwanger and Syverson (2008).

Following Clementi and Palazzo (2016), we impose a Pareto distribution for the operating and entry cost with scale parameter $\underline{c}_o$ and tail coefficient ξ_o to match the frequency and relative size of firms that exit. We also model entrants productivity distribution as a log-normal with mean μ_e. We calibrate these objects to match entry rates, exit rates, and the relative size of firms that exit and enter in Lee and Mukoyama (2015).

Parametrizing and inferring q shocks: The key driving force in this exercise is a decline in capital prices at the extensive and intensive margin. We model these as advances in the productivity of the investment sector, $q_{I,t}(x)$, varying across tasks.

For intensive-margin advances, we let the productivity of the investment sector for tasks below α_0^* vary over time according to

$$q_{I,t}(x) = q_{int,t} \text{ if } x \leq \alpha_0^*.$$

The term $q_{int,t}$ captures the extent of capital advances at the intensive margin. It is assumed to increase log-linearly from a level of 1 in 1982 to a new steady state level $\bar{q}_{int}$ in 2012, with $d \ln q_{int} = \ln \bar{q}_{int} > 0$ summarizing these technological improvements.

For extensive-margin advances, we let the productivity of the investment sector for tasks above α_0^* vary over time according to

$$q_{I,t}(x) = \min\{q_{int,t} \cdot \psi_\ell(x)/\psi_k(x), q_{ext,t}\} \text{ if } x > \alpha_0^*.$$

The term $q_{ext,t}$ captures the extent to which capital advances are more pronounced at the extensive margin. It is assumed to increase log-linearly from a level of 1 in 1982 to a new steady state level $\bar{q}_{ext}$ in 2012, with $d \ln q_{ext} = \ln \bar{q}_{ext} > 0$ summarizing these technological improvements. The formulation of this component mimics a sequence of small extensive-margin shocks as the one considered in Proposition 3. The term $q_{int,t}$ is also included inside the minimum operator to preserve the monotonicity of $\psi_\ell(x)/q_t(x)$.

The parametrization captures a wide range of possibilities. Setting $d \ln q_{int} > 0$ and $d \ln q_{ext} = 0$ yields a pure intensive-margin decline in the price of capital with no change for tasks above α_0^*. Setting $d \ln q_{ext} > 0$ and $d \ln q_{int} = 0$ yields a pure extensive-margin decline in the price of capital with no change for tasks below α_0^*. Setting $d \ln q_{int} = d \ln q_{ext} > 0$ yields a uniform decline in capital prices at all tasks.

As explained in the introduction, we do not observe capital prices or advances in capital technology separately for capital used at the intensive and extensive margin. National accounts report data on capital prices for aggregate asset classes and make no such distinction. From these data, we estimate a 108 log point decrease in total for quality-adjusted prices of capital used in manufacturing.[16] However, one cannot apportion this decrease across the

[16] Appendix F provides details for these calculations. We use capital prices from the BEA, adjusting for quality improvements using the methodology of Cummins and Violante (2002), extended to recent years by DiCecio (2009). Cummins and Violante (2002) estimate a statistical model explaining Gordon (1990)'s price indices as a function of those by the BEA/NIPA, their lags, and time trends. They extrapolate this model to produce quality-adjusted price indices for 1947–2000. We use the updated estimates from DiCecio (2009) to compute a quality adjustment for equipment and software of 2.6% per year for 1980–2011. We then compute the decline in investment-good prices for manufacturing using a user-cost weighted Törnqvist index of the prices for equipment and software (quality adjusted) and non-residential structures from the BEA Fixed Asset Tables. We also account for changes in capital taxes from Acemoglu, Manera and Restrepo (2020).

intensive and extensive margin in a straightforward manner.

To overcome this measurement limitation, we pursue a different approach. We assume that the manufacturing labor share decline is driven by q shocks and infer their magnitude to match the observed labor share decline. To put restrictions on this exercise, we also require the total decline in capital prices in our model to match the total decline in quality-adjusted capital prices observed for manufacturing during this period.

Our approach then recovers $d\ln q_{int}$ and $d\ln q_{ext}$ from: (i) the *average* decline in quality-adjusted capital prices in manufacturing of 108 log points; and (ii) the 20 pp decline in the manufacturing labor share from 1982 to 2012. These are *possibility results*: they show that one can have a coherent description of the manufacturing labor share decline driven by q shocks of reasonable size, and allow us to inspect the firm-level implications of these shocks.

A central objective of this exercise is showing that our model also generates the firm-level labor share dynamics in the data in response to q shocks. These depend on the fixed cost of automating tasks. We calibrate this fixed cost to match the uneven adoption of automation technologies in the ABS. As discussed in Section 1, firms in the top percentile of the employment distribution are 1.7 times more likely to adopt new automation technologies than firms between the 50th and 75th percentile. We calibrate c_a so that by 2018

$$\frac{\mathbb{E}[\Delta\alpha_{tf}|\text{firm } f \text{ in employment P99+}]}{\mathbb{E}[\Delta\alpha_{tf}|\text{firm } f \text{ in employment P50–75}]} = 1.7,$$

where the increase in α_{tf} relative to the 1982 steady state measures the adoption of new automation technologies from 1982 to 2018.[17]

Results: Panel I in Table 2 reports the inferred shocks and fixed cost for manufacturing.

We calibrate a fixed cost of automation of 0.19 to match the adoption rates in the ABS data. This is a small cost of reasonable magnitude. For example, in the model, aggregate spending on automation fixed costs as a share of manufacturing output reaches a peak of 1.9% by 2005. One can think of these fixed costs as an investment in R&D required to design and integrate automation equipment or software. The 1.9% estimate is a fifth of total R&D expenditure in manufacturing during this period.

For the q shocks, we recover values of $d\ln q_{int} = 0.67$ and $d\ln q_{ext} = 5.48$ to match (i) and

[17] In our model, automating a task requires adopting task-specific equipment and software designed for this purpose. Thus, $\Delta\alpha_{tf}$ measures that mass of new equipment adopted by firms and the ratio of $\Delta\alpha_{tf}$ for top to medium firms measures their relative adoption rates. We interpret the ABS data as saying that, for most types of task-specific equipment and software used for automation since the 1980s, large firms are 1.7 more likely to have adopted them than small firms, which justify the use of this target.

Table 2: Effects of lower capital prices, Manufacturing 1982–2012

		CES demand system				Log-concave demand
	Data	Both q shocks	Intensive margin	Extensive margin	No fixed costs	Both q shocks
	(1)	(2)	(3)	(4)	(5)	(6)
	I. Parameters and inferred aggregate shocks					
$d \ln q_{int}$		0.67	0.67	0 (below α_0^*)	0.67	0.66
$d \ln q_{ext}$		5.48	0	5.48	5.48	5.08
c_a		0.19	0.19	0.19	0	0.18
	II. Targeted moments, 1982–2012					
Δ aggregate labor share	-0.20	-0.20	0.05	-0.24	-0.20	-0.20
Δ log average capital price	-1.08	-1.08	-0.64	-0.55	-1.10	-1.08
Relative adoption (P99+ vs. P50-75 firms)	1.71	1.71	9.32	1.98	1.00	1.71
	III. Concentration 1982–2012 (from Autor et al., 2020; Decker et al., 2020)					
Δ log 4 firms' sales share	0.140	0.105	0.001	0.111	-0.022	0.084
Δ log 20 firms' sales share	0.072	0.104	0.001	0.109	-0.017	0.112
Δ log productivity dispersion (from 1980–2000)	0.050	0.061	0.000	0.043	-0.001	0.063
	IV. Typical firm labor share from Kehrig and Vincent (2021), 1982–2012					
Δ median labor share	0.030	-0.005	0.056	-0.051	-0.202	0.031
Δ unweighted mean	-0.017	-0.039	0.056	-0.080	-0.202	-0.012
	V. Melitz–Polanec decomposition from Autor et al. (2020)					
Δ aggregate labor share	-0.185	-0.198	0.049	-0.241	-0.202	-0.195
Δ unweighted incumbent mean	-0.002	-0.015	0.056	-0.060	-0.202	0.012
Exit	-0.055	-0.006	-0.001	-0.007	0.000	-0.015
Entry	0.059	0.006	0.001	0.008	0.000	0.012
Covariance term	-0.187	-0.183	-0.007	-0.182	0.000	-0.204
	VI. Covariance decomposition from Kehrig and Vincent (2021)					
Market share dynamics	0.047	0	0	0	0	0.071
Labor share by size dynamics	-0.043	-0.078	-0.002	-0.076	0	-0.023
Cross-cross dynamics	-0.232	-0.066	-0.005	-0.070	0	-0.200
	VII. Change in markups					
Δ log aggregate markup	-0.011	0	0	0	0	0.015
Within-firm change in markup	-0.076	0	0	0	0	-0.026
Reallocation to high-markup firms	0.065	0	0	0	0	0.041

Notes: Column (2) reports the findings from our CES model. Column (3) feeds in only advances in capital at the intensive margin. Column (4) feeds in only advances in capital at the extensive margin. Column (5) feeds in both q shocks but shuts down the fixed cost of automation. Column (6) reports the findings from our (re-calibrated) log-concave demand model. The data for markups comes from Compustat estimates and is described in Section 3.3.

(ii). These estimates translate into a decrease in capital prices at the intensive margin of 67 log points and a bigger decrease at the extensive margin of 164 log points.[18]

[18] We parametrize the extensive-margin shock as a function of $d \ln q_{ext}$, but this is not the decline in capital prices that one would measure. The functional form assumed and the fact that capital is only used when its price crosses some threshold imply that the measured decline is smaller. The 164 log point number is the price decline at tasks above α_0^* computed as a Törnqvist index from the point they switch to capital.

In line with theory, the inferred decline in capital prices has been limited at the intensive margin and more pronounced at the extensive margin. This uneven shock is needed to generate the large drop in the manufacturing labor share despite the fact that we calibrated the model to have an elasticity of substitution between capital and labor η^* of one.

The result of more pronounced advances in capital at the extensive margin in manufacturing aligns with two pieces of evidence. First, it aligns with evidence on the widespread use of automation technologies in manufacturing. The survey data in Acemoglu et al. (2022) shows that 50% of manufacturing firms (in an employment-weighted sense) used robotics, software systems, and dedicated equipment to automate tasks previously assigned to workers. Indeed, manufacturing has seen multiple developments in robotics, software systems, and computer numerically controlled (CNC) machinery that arguably took place at the extensive margin: these technologies allowed firms to substitute for blue-collar workers in a widening range of tasks, such as welding, painting, coating, assembly, and metalworking. In line with this interpretation, the evidence summarized in footnotes 2 and 3 shows that the deployment of these technologies was associated with declines in the labor share of adopting firms and industries.

Second, the fact that capital advances have been limited at the intensive margin aligns with data on capital prices for broad asset classes from the BEA Fixed Asset Tables. One way to see this is by ranking all non-residential assets in manufacturing by their net investment rates in the 70s (a measure of whether their stocks are growing or have reached their desired levels). Asset classes with below-average net investment rates saw a subsequent price decline of 48.6 log points from 1980 to 2012. Presumably, price declines in these "established" assets took place at the intensive margin, given their high rates of usage in the 70s. The price of "novel" types of capital—those with above-average net investment rates in the 70s—declined by 166 log points from 1980 to 2012. These novel types of capital were only gaining traction in the 70s, and so presumably these advances were more likely to take place at the extensive margin. These calculations reflect the fact that the decline in capital prices has been uneven across assets, and much more pronounced for software and computer equipment. To the extent that these are more likely to power automation technologies than other forms of capital, the divergent price trends point to more sizable advances at the extensive margin.[19]

Effects on concentration: Although not targeted, our model generates most of the increase in sales concentration in manufacturing since 1982 in response to q shocks.

[19]The details behind these and a related calculation are presented in Appendix F. Figure A13 in this appendix shows price trends by asset class from the BEA Fixed Asset Tables.

As shown in Panel III of Table 2, by 2012, the uneven adoption of automation technologies by the most productive firms leads to a 10.5 log points (4.4 pp) increase in the sales share of the top 1.1% firms and a 10.4 log points (7.6 pp) increase in the sales share of the top 5.5% firms in manufacturing. Autor et al. (2020) document increases of 14 log points for the sales share of the top 4 firms (the top 1.1% in our model) and 7.2 log points for the sales share of the top 20 firms (the top 5.5% in our model) in each manufacturing industry.

The model also generates an increase in productivity dispersion across firms of 6.1 log points over 1980–2000, which matches the 5 log points increase estimated by Decker et al. (2020) using US manufacturing data for this period.

Effects on firm labor shares: A key point of our exercise is showing that our model generates the firm-level labor share patterns in the data in response to q shocks. Although these are not targeted, our model matches several firm-level labor share facts.

Column 2 of Panel IV shows that our model generates the decline in the manufacturing labor share at the same time as the median labor share remains unchanged and the unweighted mean labor share decreases by 3.9 pp. This is comparable to the data, where it decreased by 1.7 pp.

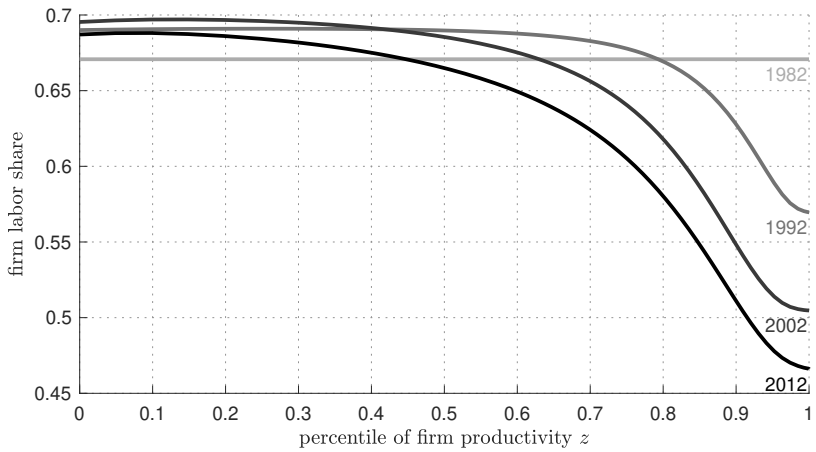

FIGURE 4: CROSS-SECTIONAL LABOR SHARES OVER THE TRANSITION. Firm labor shares in value added by firm productivity z_{tf} from the benchmark model with CES demand.

To illustrate this point, Figure 4 depicts the evolution of the labor share across the distribution of firm productivity z_{tf}. The lines trace the average labor share for firms at each percentile of the distribution over time. In 1982, all firms had the same steady-state labor share. In response to q shocks, we see a clockwise rotation of this curve, with the labor share rising at the middle and the bottom of the firm-productivity distribution, but decreasing at the top, in line with Proposition 3.

This clockwise rotation in labor shares matches the counter-clockwise increase in output-to-capital elasticities for Compustat firms in Figure 3. In fact, our model generates an increase in the output-to-capital elasticity for large firms in manufacturing of 20 pp, which matches the 20 pp increase in Compustat.

Our model also generates the firm-level labor share patterns documented by Autor et al. (2020) and Kehrig and Vincent (2021) along the transition in response to q shocks. Autor et al. decompose the decline in the manufacturing labor share a-la Melitz–Polanec:

$$
\begin{aligned}
\Delta s_t^\ell = & \Delta \bar{s}_t^\ell && \text{(Unweighted incumbents' mean)} \\
& + \omega_{tX} \cdot (s_{tS}^\ell - s_{tX}^\ell) && \text{(Contribution of exit)} \\
& + \omega_{t'E} \cdot (s_{t'E}^\ell - s_{t'S}^\ell) && \text{(Contribution of entry)} \\
& + \Delta \sum_f (\omega_{tf} - \bar{\omega}_t) \cdot (s_{tf}^\ell - \bar{s}_t^\ell) && \text{(Change in covariance).}
\end{aligned}
$$

Here, Δs_t^ℓ denotes the change in the manufacturing labor share between two periods, t and t', corresponding to five-year intervals. This can be decomposed into the change in the unweighted mean of labor shares among continuing firms, $\Delta \bar{s}_t^\ell$; two terms accounting for the contributions of exit and entry; and the change in the covariance among continuing firms between their share of value added, ω_{tf}, and their labor share, s_{tf}^ℓ.[20]

We conduct the same decomposition in our model for the 1982–2012 period. Panel V in Table 2 reproduces Autor et al.'s data and reports the decomposition from our model. In data and model, the covariance term fully accounts for the aggregate decline in the labor share, with exit and entry and the change in the unweighted mean of incumbents' labor shares playing minor roles.[21] Our model matches this decomposition because the adoption of automation technologies and the decline of the labor share concentrates on large firms, as seen in Figure 4, and this shows up in the covariance term.

[20] The contribution of exiting firms equals their value added share, ω_{tX}, times the difference in the mean labor share of continuing firms, s_{tS}^ℓ, and firms that exit, s_{tX}^ℓ. The contribution of entrants equals their value added share, $\omega_{t'E}$, times the difference between their mean labor share, $s_{t'E}^\ell$, and continuing firms', $s_{t'S}^\ell$.

[21] Exit and entry exhibit the same qualitative patterns as in the data, with entering and exiting firms having higher labor shares than incumbents. These differences are larger in the data, where many entering or exiting firms have labor shares above 1, reflecting features of their life cycle not in our model.

The covariance term can be further decomposed into three terms:

$$\begin{aligned}\Delta \sum_f (\omega_{tf} - \bar{\omega}_t) \cdot (s^\ell_{tf} - \bar{s}^\ell_t) = &\sum_f \Delta(\omega_{tf} - \bar{\omega}_t) \cdot (s^\ell_{tf} - \bar{s}^\ell_t) && \text{(market share dynamics)} \\ &+ \sum_f (\omega_{tf} - \bar{\omega}_t) \cdot \Delta(s^\ell_{tf} - \bar{s}^\ell_t) && \text{(labor share by size dynamics)} \\ &+ \sum_f \Delta(\omega_{tf} - \bar{\omega}_t) \cdot \Delta(s^\ell_{tf} - \bar{s}^\ell_t) && \text{(cross-cross dynamics).}\end{aligned}$$

These terms capture the joint dynamics of firms' labor shares and sales. A decrease in the covariance can be driven by a reallocation of value added towards firms with lower labor shares at base (the "market share dynamics"), a disproportionate reduction in large firms' labor shares (the "labor share by size dynamics"), or the possibility that firms reduce their labor shares and expand sales at the same time (the "cross-cross dynamics").

Using a balanced sample of firms for 1982–2012, Kehrig and Vincent (2021) document that the cross–cross dynamics contributed -23.2 pp to the decline in the manufacturing labor share, the labor share dynamics by size accounted for a 4.3 pp decline, and the market share dynamics increased the manufacturing labor share by 4.7 pp.[22]

Panel VI in Table 2 compares the contributions of these components in data and model. In our model, the labor share dynamics by size contributed 7.8 pp to the labor share decline and the cross-cross dynamics contributed 6.6 pp (by construction, the market share dynamics is zero since firms start with the same labor shares). Our model with CES demand and constant markups matches the covariance decomposition qualitatively, but does not generate the large cross-cross term seen in the data of -23 pp. Our exercise with endogenous markups solves this problem and generates cross-cross dynamics of this size.

The reason why our model generates a negative cross-cross term is that firms automate tasks precisely as they expand in response to high productivity draws. This is an implication of Proposition 1: automation occurs in spurts triggered by firm productivity crossing thresholds. During these spurts, firms gain market share and reduce their labor shares, which shows up as part of the cross–cross term in the decomposition in Kehrig and Vincent.

Decomposing the shocks: Columns 3 and 4 in Table 2 report the individual effects of the intensive and extensive margin shocks. Column 3 feeds in the inferred decline in capital

[22] Kehrig and Vincent compute these objects as 30-year changes in a balanced sample over 1982-2012. This is why these numbers do not add up to the covariance term in Autor et al.'s decomposition (computed over five-year periods and accumulated). A second minor difference is that Kehrig and Vincent's estimates of the labor share dynamics by size (their "big-player scenario") do not subtract the small change in the unweighted mean labor share of incumbents (as in the exact decomposition shown above).

prices below α_0^*. Column 4 feeds in the inferred decline in capital prices above α_0^*.

The intensive margin shock causes a similar increase in the aggregate and median labor shares of 5.0 pp and 5.6 pp. This shock generates no change in sales concentration, since firms do not pay fixed costs to use capital at the intensive margin.

The extensive margin shock drives the aggregate labor share decline, the rise in sales concentration, and the relevant firm-level labor share dynamics. This shock reduces the aggregate labor share by 23.9 pp and the labor share of the median firm by a modest 5.1 pp.

The role of fixed costs: Column 5 reports the effects of q shocks assuming firms face no fixed costs of automating tasks. q shocks now reduce the aggregate, the median, and the unweighted mean labor share by about 20 pp, which is at odds with the firm data. The small calibrated fixed cost of automating tasks is needed to make sense of the firm-level patterns.

Other checks: Appendix C reports complementary checks. First, we show results shutting down the diffusion of automation. In one exercise, we start all firms with the same level of automation and shut down diffusion along the transition, so that q shocks push the economy to a stationary equilibrium with firms having different levels of automation. In a second exercise, we also allow firms to have different levels of automation in the initial steady state too. Both exercises show that diffusion plays no role for our findings for 1982–2012.

We also report results from a parametrization of our model with $\eta^* = 1.45$, as estimated by Karabarbounis and Neiman (2013) and Hubmer (2023). The q shocks required to explain the data are now more uniform: 86 log points at the intensive margin and 137 log points at the extensive margin. We also present an exercise for $\eta^* = 1.45$ where we assume a uniform q shock, in which case one can match the observed labor share decline in manufacturing as the result of a 168 log point decline in capital prices (above the 108 log points in the data). In all these scenarios, our model continues to generate the relevant firm-level labor share patterns seen in the data, even though the exact magnitudes for the inferred q shocks varies.

3.2 Exercise II: manufacturing with a log-concave demand

Our second exercise extends our analysis to an economy with a log-concave demand system with endogenous markups. This allows us to study the interaction between lower capital prices and endogenous responses in markups. The model with a log-concave demand system does a better job at matching the firm-level dynamics in Kehrig and Vincent (2021) and generates a stronger increase in the labor share of a typical firm.

To model this, we assume that firms face a demand derived from a Kimball aggregator (Kimball, 1995). The quantity of the final good is defined implicitly by

$$\int_f \lambda \cdot H\left(\frac{y_{tf}}{\lambda \cdot y_t}\right) \cdot df = 1, \tag{4}$$

where H is a concave and increasing function and λ is a scaling parameter.

We assume that the aggregator H is such that the demand curve faced by firms satisfies Marshall's weak and strong second laws. This means that demand and marginal revenue are log concave.[23] The appendix proves three well-known properties of these demand systems:

(P1) Firms with lower costs (i.e., more productive) charge lower prices but higher markups.

(P2) More productive firms have lower passthroughs of marginal cost into prices.

(P3) Sales are a log-concave function of costs.

Calibration: We use the same parametrization of the production function and task productivities from the first exercise, summarized in Panel I of Table 3.

We make two modifications relative to the previous exercise. First, following Edmond, Midrigan and Xu (2022), we parametrize H using the specification from Klenow and Willis (2016). This specification yields a demand elasticity

$$\text{demand elasticity}_{tf} = \sigma \cdot q_{tf}^{-\frac{\nu}{\sigma}}, \tag{5}$$

where $q_{tf} = y_{tf}/(\lambda \cdot y_t)$ measures the relative quantity produced by a firm. This demand elasticity decreases as quantities produced rise, so that more productive firms face a more inelastic demand.[24] Here, σ controls the average demand elasticity faced by firms, and the super-elasticity ν/σ controls the extent to which markups rise for more productive firms. (If $\nu = 0$, the demand system simplifies to the CES aggregator used above.)

Second, we adopt a new process for firm-level productivities z_{tf}. We can no longer assume a log-normal productivity distribution, since, under Marshall's second laws, sales are a log-concave function of costs (P3 above). Because the sales distribution is approximately Pareto, we need the productivity distribution to be more log-convex than Pareto to fit the data.

[23] Throughout, we say that a function $y = f(x)$ is log-concave if $\ln y$ is concave in $\ln x$. These assumptions receive empirical support (see Baqaee and Farhi, 2020a) and capture the pricing dynamics of oligopolistic competition models, where large firms face more inelastic demand.

[24] The full specification for H and the derivation of equation (5) are provided in Appendix B. The Appendix also shows this specification satisfies Marshall's laws.

To achieve this, we assume that productivity is determined by a latent factor $\tilde{z}_{tf}$ that follows an AR(1) process as before and determines productivity as:

$$z_{tf} = \exp\left(F^{-1}_{Weibull(n,\zeta)}\left(\Phi\left(\tilde{z}_{tf}\right)\right)\right), \quad \text{where} \quad \tilde{z}_{t+1,f} = \rho_z \cdot \tilde{z}_{tf} + \varepsilon_{tf}$$

Here, Φ denotes the normal CDF, and $F^{-1}_{Weibull(n,\zeta)}$ the inverse CDF of a Weibull distribution with shape parameter $n > 0$ and scale $\zeta > 0$. The innovations are drawn from $\varepsilon_{tf} \sim N\left(\mu_z, \sigma_z\right)$, where μ_z and σ_z are normalized so that the long-run distribution of $\tilde{z}_{tf}$ is a standard normal. As before, entrants draw $\tilde{z}_{tf}$ from a normal with unit variance and mean $\ln \mu_e$.

The limit distribution of log productivity, $\ln z_{tf}$, is a Weibull with CDF

$$F_{Weibull(n,\zeta)}(x) = 1 - e^{-\left(\frac{x}{\zeta}\right)^n}.$$

The Weibull distribution generalizes the exponential by introducing the shape parameter n, which controls its log-convexity. When $n = 1$, the density of $\ln z_{tf}$ is log-linear, or equivalently, the limit distribution for z_{tf} is Pareto. For $n < 1$, the density of $\ln z_{tf}$ is log-convex, or equivalently, the distribution for z_{tf} is more log-convex than Pareto.

We calibrate the demand parameters $\{\sigma, \nu\}$ and the firm productivity distribution $\{\zeta, n\}$ to match the average manufacturing markup, the ratio of the (unweighted) mean firm labor share to the aggregate labor share, and the share of sales among the top 4 and top 20 firms in manufacturing in 1982. These parameters are jointly calibrated with the fixed cost of operation and its dispersion as well as with mean entrant productivity to also match the moments from Lee and Mukoyama (2015) used above, and setting $\rho_z = 0.95$. Panel II in Table 3 summarizes the parameters and moments informing their calibration.

We calibrate $\sigma = 6.0$ to match an aggregate markup of 1.15 and a super-elasticity ν/σ of 0.22 to match a 1.1 ratio between the unweighted average labor share of manufacturing firms and the aggregate manufacturing labor share before 1982 from Kehrig and Vincent (2021). Our calibration attributes all the differences in labor shares by firm size in the 1982 steady state to markups. The log-concave demand system implies that smaller firms have lower markups; therefore, the unweighted mean labor share across manufacturing firms exceeds the aggregate labor share by a factor that increases in the super-elasticity of demand. A super-elasticity of 0.22 matches the 1.1 ratio in the data.[25]

For the productivity distribution, we calibrate $\{\zeta, n\}$ to match sales shares of the top 4

[25] The calibrated demand super-elasticity is close to the value in Edmond, Midrigan and Xu (2022) of 0.16, who estimate it to match the change in markups with firm size in US Census data. Appendix C provides robustness checks imposing this lower super-elasticity.

Table 3: Calibration of the log-concave demand model for manufacturing

	Parameter		Moment	Data	Model
	I. Parameters related to production function				
η	Task substitution elasticity	0.5	From Humlum (2019)	0.5	0.5
γ_ℓ	Comparative advantage	0.30	Manufacturing labor share (BLS/BEA)	0.67	0.67
	II. Parameters governing firm dynamics and productivities in 1982 steady state				
ν/σ	Demand super-elasticity	0.22	Ratio of mean firm to aggregate labor share	1.10	1.09
σ	Demand elasticity	6.00	Aggregate markup	1.15	1.15
ζ	Weibull scale	0.077	Top 20 firms' sales share	69.7%	69.9%
n	Weibull shape	0.74	Top 4 firms' sales share	40.0%	40.0%
$\underline{c}_o$	Scale operating cost	$2 \cdot 10^{-7}$	Entry (=exit) rate	0.062	0.062
ξ_o	Tail index operating cost	0.24	Size of exiters	0.490	0.485
μ_e	Entrant productivity	0.882	Size of entrants	0.600	0.600
ρ_z	Productivity persistence	0.95	Revenue TFP persistence among manufacturing firms		

Notes: The table reports the calibrated parameters and targets for our model with log-concave demand in manufacturing. The ratio of the (unweighted) mean firm labor share to the aggregate manufacturing labor share is computed based on the replication data from Kehrig and Vincent (2021). The annual entry rate, as well as relative sizes of entrants and exiters, are from Lee and Mukoyama (2015) and based on the Annual Survey of Manufactures. The data on top firms' sales share comes from Autor et al. (2020).

and top 20 firms within manufacturing industries in 1982, which correspond to the top 1.1% and top 5.5% of firms in the sales distribution. A higher sales share among the top 5.5% firms indicates a thicker tail of the productivity distribution—a higher ζ. Conditional on the top 5.5% share, a higher sales share among the top 1.1% firms requires a lower value of n, which indicates a more than proportional increase in productivity as we move to the top of the sales distribution. We find that $n = 0.74$—a small deviation from Pareto with a moderate degree of log-convexity—fits the manufacturing sales concentration in 1982.

Parametrizing and inferring q shocks: We use the parametrization of q shocks introduced above. As before, we treat the economy in 1982 as being in steady state. We then recover the q shocks required to match (i) the observed decline in the manufacturing labor share (ii) the observed decline in average capital prices in this sector. In addition, and as we did in the previous section, we calibrate the fixed cost of automation per task to match (iii) the higher adoption rate of automation technologies among large firms in the ABS.

Column 6 in Table 2 reports the inferred shocks. To match (i)–(ii), our model requires a uniform decline in the effective price of capital of $d \ln q_{int} = 0.66$ and an extensive-margin change of $d \ln q_{ext} = 5.08$, which translates into a 168 log point decrease in capital prices at

the extensive margin. As before, we calibrate a small fixed cost of automating tasks $c_a = 0.18$ to match the ABS adoption data.

In response to these shocks, our model provides a good fit to the manufacturing data, matching the aggregate labor share decline by construction. More importantly, the model also matches all new firm-level labor share facts from Autor et al. (2020) and Kehrig and Vincent (2021), despite the fact that these are not targeted.

Effects on firm labor shares and the role of markups: The labor share of the typical manufacturing firm increases despite lower capital prices, exactly as in the data. Panel IV shows that the median firm labor share now rises by 3.1 pp and the unweighted mean decreases by 1.2 pp during 1982–2012, which match the 3 pp increase in the median and the 1.7 pp decline in the unweighted mean in the data.

The reason why the model with endogenous markups generates a more sizable increase in the typical firm labor share is that the median firm reduces its markups during the transition by 2.6%. This is because lower capital prices increase real wages. Firms respond to higher wages by raising prices with a less than one-to-one passthrough, which reduces markups.

Panel V of Table 2 describes the behavior of the labor share across firms using the Melitz–Polanec decomposition from Autor et al. (2020). As before, we find a crucial role for a decline in the covariance between firm sales and their labor share in explaining the decline in the labor share. The covariance term contributed 18.7 pp to the labor share decline in the data and 20.4 pp in our model.

The contributions of entry and exit are also two times more pronounced than in the CES model and closer to the data. This is because entrants and firms that exit are less productive on average. These firms have higher labor shares both because they automate fewer tasks and charge lower markups.

Panel VI shows that our model with endogenous markups quantitatively matches the covariance decomposition in Kehrig and Vincent (2021). The cross-cross dynamics (capturing the contribution of firms that expand at the same time as they reduce their labor shares) are now the dominant force behind the labor share decline in the model. The cross-cross dynamics term contributed 23.2 pp to the labor share decline in the data and 20 pp in our model. This term is more negative than in the CES model because firms that expand not only automate more tasks when they do so, but also raise their markups during periods of expansion—a feature of Marshall's second law.

As in the data, our model now produces a positive market share effect of 7.1 pp. This is

because firms that had low labor shares in 1982 were already large. Due to mean reversion in the productivity process, these firms experience a decline in their market shares over time, contributing to an increase in their labor shares.

The model also matches two extra features of firm labor share dynamics emphasized by Kehrig and Vincent (2021). The first is the transient behavior of firms' labor shares. Kehrig and Vincent document that firms experience cycles during which their labor shares decline and recover, and that these cycles have become more pronounced and asymmetric over time.

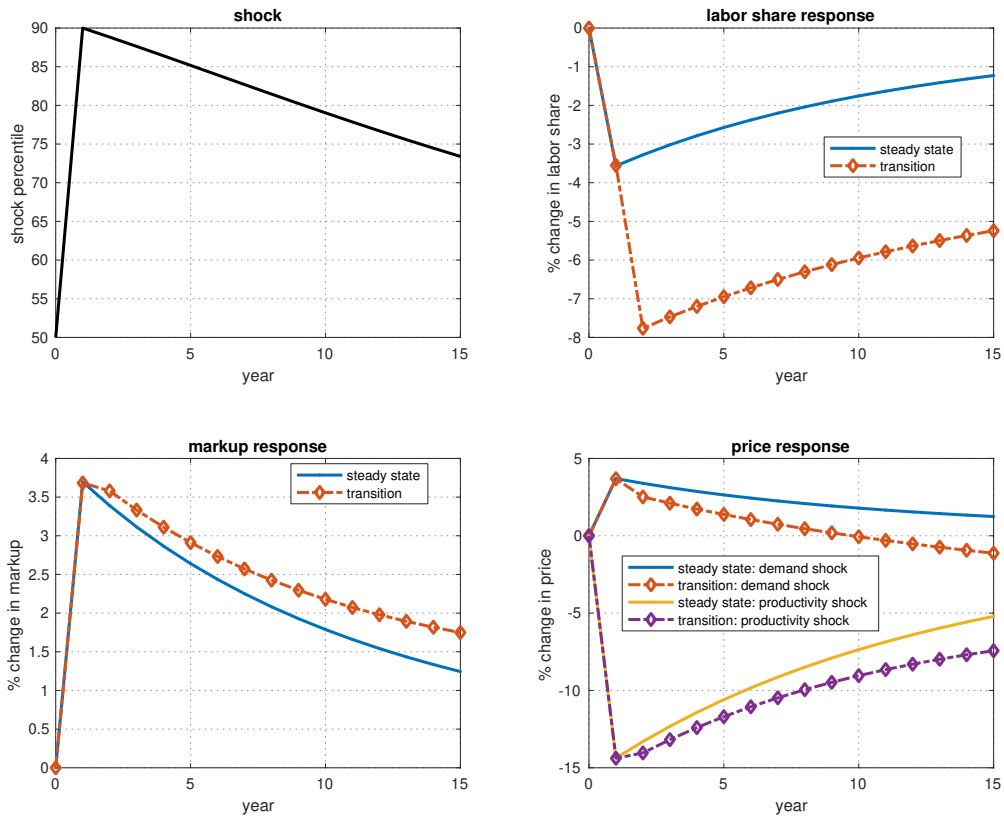

FIGURE 5: IMPULSE RESPONSES OF FIRM LABOR SHARE, MARKUP, AND PRICE TO PRODUCTIVITY AND DEMAND SHOCKS. Top left panel displays the shock: a firm's productivity z_{tf} or demand shock jumps from the 50th to the 90th percentile on impact, and then mean-reverts according to ρ_z. Top right panel displays firms' labor share response (relative to firm with median productivity), bottom left firms' markup response. As explained in the text, both productivity or demand shocks generate the same labor share and markup responses. The bottom right panel shows firms' price response, which differ across shocks.

In our model, these cycles are a natural implication of Marshall's second law. In the initial steady state, firms that receive a positive productivity shock raise markups and lower their labor shares. These firms then lower their markups and raise their labor shares as their productivity mean-reverts. Along the transition, these cycles become more pronounced and asymmetric, since firms also automate more tasks upon receiving a positive productivity shock. Figure 5 illustrates this result by plotting the impulse-response of firms' labor shares

and markups to a productivity shock in the initial steady state and along the transition.

The second is the possibility that these cycles are accompanied by price increases, though Kehrig and Vincent document this for a handful of industries with limited price data (chemicals, aluminum sheets, and others). One can capture this feature by allowing for firm-specific demand shocks. Appendix B shows that when firms face demand shocks, the resulting markup, automation decisions, and labor share responses are identical to those in response to productivity shocks. The key difference is that firms raise prices in response to demand shocks. Figure 5 illustrates this result by plotting the impulse-response of firms' labor shares, markups, and prices, to a demand shock in the initial steady state and along the transition.

Markups and the aggregate labor share: The exercises in this section and the comparison with our CES demand results show that endogenous markups are important for quantitatively matching the joint dynamics of labor and market shares documented in Kehrig and Vincent (2021). However, this does not imply that markups are an important driver of the secular labor share decline in manufacturing.[26]

To illustrate this point, Panel VII of Table 1 reports the contribution of markups to the labor share decline for the log-concave demand model. The labor share in an industry can be written as $s^\ell = \varepsilon^\ell/\mu$, where ε^ℓ is the share of labor in costs and μ is the aggregate industry markup, defined as the harmonic sales-weighted mean of markups across firms:

$$\frac{1}{\mu} = \sum_f \omega_f \cdot \frac{1}{\mu_f}.$$

Our model predicts a mild increase in the manufacturing markup of 1.5% (from 1.150 to 1.166) in response to q shocks, which implies that markups account for only 1 pp of the 20 pp decline in the manufacturing labor share.

This net effect masks two opposing forces. q shocks reallocate sales towards more productive firms with higher markups, which increased the manufacturing markup by 4.1%. This is mostly offset by a reduction in firm-level markups. Firms respond to higher wages by rising prices with a less than one-to-one passthrough, which reduces markups. These within-firm changes reduced the manufacturing markup by 2.6%.[27]

[26] The reason why this conclusion does not follow is that the Kehrig and Vincent moments are really about how firms change labor shares, prices, and sales as they get buffeted by idiosyncratic shocks. While informative of firms' pricing and timing of automation decisions, these moments are not directly informative of the underlying drivers of the labor share decline in manufacturing.

[27] These effects are computed as within-firm changes = $\sum_f \omega_f \cdot \Delta \ln \mu_f$ (where the sum is over all continuing firms) and reallocation component = $\Delta \ln \mu - \sum_f \omega_f \cdot \Delta \ln \mu_f$.

This exercise shows that while endogenous markups are important for matching some features of the data, the manufacturing labor share decline in our calibration continues to be driven by large firms using capital more intensively as they automate tasks.

3.3 Exercise III: capital prices vs. rising competition

Our previous exercises attributed the entire labor share decline in manufacturing to capital advances and inferred the magnitude of q shocks needed to match the data. We now consider the possibility that part of the labor share decline results from changes in market structure increasing competition among firms, as in Autor et al. (2020).

We model the increase in competition as an increase in λ in (4) over time—a λ shock. In this specification, λ can be interpreted as a measure of market access. Appendix B justifies this interpretation. It shows that the demand curve resulting from the Kimball aggregator in (4) is the same one would get in a model where identical consumers with Kimball preferences are matched to a mass λ of firms that compete over them. A higher λ generates increased competition over consumers. For example, one might think of advances in online marketing and platforms as allowing each firm to reach a larger set of customers (see, for example, Akerman, Leuven and Mogstad, 2022). Or advances in IT allowing firms to enter into additional markets at a lower overhead cost, raising competition for customers in those markets (see, for example, Aghion et al., 2023b; Hsieh and Rossi-Hansberg, 2023).

The appendix proves the following properties of the log-concave demand system under Marshall's weak and strong second laws:

(P4) The increase in λ causes competition for workers, raising real wages. The increase in marginal costs causes firms to increase their prices but lower their markups (from P1). It also causes a reallocation towards more productive firms with higher markups, as these firms raise prices the least in response to higher real wages (from P2).

(P5) The net effect of market access on the aggregate markup is ambiguous. Reallocation dominates if and only if the productivity distribution is log convex. These effects cancel when the distribution of firm productivity is log linear (i.e., Pareto).[28]

P4 and P5 show that λ shocks can also generate the pattern seen in the data of rising sales concentration, declining aggregate labor share, and a rise in a typical firm labor share.

To put both explanations on equal footing, we now assume that the trends in the manufacturing labor share and sales concentration are driven by a combination of q shocks and λ

[28] We refer to a distribution as log-convex (log-concave) if its PDF is log-convex (log-concave).

shocks. We use the calibrated model to infer the magnitudes of these shocks implied by the data and decompose their effects.

Results for manufacturing: We treat the economy in 1982 as being in steady state. We pick the q and λ shocks required to match (i) the observed decline in the sectoral labor share, (ii) the observed decline in average capital prices by sector, and (iii) the increase in sales concentration. This last moment disciplines the extent to which market access increased for manufacturing. In addition, we calibrate the fixed cost of automation per task to match (iv) the higher adoption rate of automation technologies among large firms documented in Acemoglu et al. (2022). Relative to our first exercise, we allow these trends to be explained by a combination of both q and λ shocks.

Column 1 in Table 4 summarizes the manufacturing data and Column 2 reports the inferred shocks and their effects. To match (i)-(iv), our model requires $d\ln q_{int} = 0.60$ and $d\ln q_{ext} = 5.17$ (these translate into a 60 log point intensive-margin and a 172 log point extensive-margin capital price decline), and a mild increase in market access of $d\ln\lambda = 0.06$.

The reason why we calibrate a small increase in λ is that the rise in sales concentration in manufacturing has been modest: manufacturing is the sector with the lowest increase in sales shares of top firms from 1982–2012 according to data in Table 1 of Autor et al. (2020).

Columns 3 and 4 decompose the effects of these two shocks. Column 3 reports the effect of q shocks holding λ constant. The decline in effective capital prices generates the observed decline in the manufacturing labor share, and 7.8 log points of the observed 14 log points increase in sales concentration among the top 4 firms in the sector. q shocks continue to generate the relevant firm-level labor share patterns in panels IV and V, including a rising labor share for the median firm.

Column 4 reports the effect of the λ shock holding q constant. Rising market access accounts for half of the increase in sales concentration, but has a negligible impact on the manufacturing labor share. This is because the inferred λ shock is small and creates offsetting effects via markups. As discussed above, market access heightens competition, which causes firms to lower their markups. This within-firm change in markups increases the manufacturing labor share by 1.4%. At the same time, rising market access reallocates sales towards more productive firms, which have higher markups. This reallocation reduces the manufacturing labor share by 1.5%. These effects cancel because the productivity distribution of manufacturing is close to Pareto ($n = 0.74$).

In sum, our exercise attributes most of the labor share decline and firm-level dynamics

Table 4: Effects of lower capital prices and rising competition, Manufacturing 1982–2012

		Results from log-concave demand model		
	Data	Both shocks	Only effects of $d\ln q$	Only effects of $d\ln\lambda$
	(1)	(2)	(3)	(4)
	I. Parameters and inferred aggregate shocks			
$d\ln q_{int}$		0.60	0.60	0
$d\ln q_{ext}$		5.17	5.17	0
$d\ln\lambda$		0.06	0	0.06
c_a		0.16	0.16	0.16
	II. Targeted moments, 1982–2012			
Δ aggregate labor share	-0.199	-0.199	-0.209	0.004
Δ log average capital price	-1.081	-1.074	-0.994	0
Δ log 4 firms' sales share	0.140	0.143	0.078	0.070
Relative adoption (P99+ vs. P50-75 firms)	1.71	1.71	1.62	9.74
	III. Typical firm labor share and other moments			
Δ median labor share	0.030	0.036	0.016	0.013
Δ unweighted mean	-0.017	-0.005	-0.025	0.012
Δ log 20 firms' sales share	0.072	0.142	0.105	0.046
Δ log productivity dispersion (from 1980-2000)	0.050	0.073	0.060	0.000
	IV. Melitz–Polanec decomposition from Autor et al. (2020)			
Δ aggregate labor share	-0.185	-0.195	-0.205	0.003
Δ unweighted incumbent mean	-0.002	0.017	-0.001	0.014
Exit	-0.055	-0.015	-0.016	-0.018
Entry	0.059	0.012	0.014	0.015
Covariance term	-0.187	-0.210	-0.203	-0.008
	V. Covariance decomposition from Kehrig and Vincent (2021)			
Market share dynamics	0.047	0.075	0.075	0.070
Labor share by size dynamics	-0.043	-0.027	-0.024	0.069
Cross-cross dynamics	-0.232	-0.206	-0.198	-0.147
	VI. Change in markups			
Δ log aggregate markup	-0.011	0.014	0.014	0.001
Within-firm change in markup	-0.076	-0.031	-0.025	-0.014
Reallocation to high-markup firms	0.065	0.045	0.039	0.015

Notes: Column (2) reports the findings from our benchmark model. Column (3) shows results when shutting down the market access shock, column (4) when shutting down instead the price of capital shocks. The data for markups comes from Compustat estimates and is described in Section 3.3.

in the manufacturing data to q shocks and the uneven and costly adoption of automation technologies at the extensive margin.

Other sectors: We conducted this decomposition for retail, wholesale, and utilities & transportation. This explores if increases in market access were more prominent outside manufacturing and can explain the firm and aggregate labor share trends in these sectors.

Appendix D describes our calibration and results. In all sectors, we follow the procedure for manufacturing. We first calibrate the log-concave demand model to match moments from the 1982 steady state. We then recover the q and λ shocks needed to match (i) the

observed decline in the sectoral labor share, (ii) the observed decline in average capital prices by sector, and (iii) the increase in sales concentration.[29]

We focus our discussion on retail, where we expected λ shocks to be more relevant. The labor share in retail decreased by 12.7 pp over 1982–2012, at the same time as the unweighted mean of payroll shares in the sector increased by 4.4 pp (see Autor et al., 2020). The sector also saw a vast increase in sales concentration, with the sales share of the top 4 firms in each industry rising by 14 pp and the sales share of the top 20 firms in each industry rising by 16 pp. This represents a large average increase in sales concentration of 54.6 log points. For comparison, sales concentration increased by 14 log points in manufacturing.

Our model generates the observed patterns for retail in response to a decline in the intensive-margin price of capital of $d\ln q_{int} = 0.48$, an estimated $d\ln q_{ext} = 3.25$, which translates to a extensive-margin decrease in capital prices of 144 log points, and an increase in market access of $d\ln\lambda = 0.30$. The λ shock is now an order of magnitude larger than in manufacturing, and is needed to generate the vast increase in concentration in retail.[30]

Over three quarters of the observed labor share decline continues to be attributed to q shocks and the automation of tasks by large firms. As in manufacturing, a small fixed cost of automating tasks calibrated to match the adoption data is both necessary and sufficient to generate the divergence between the aggregate labor share in the sector and the labor share of a typical firm. With the calibrated fixed cost, the unweighted mean labor share rises by 2.8 pp and the median by 4.8 pp, which matches the payroll data in Autor et al. (2020).

The main difference with manufacturing is that the λ shock and the resulting reallocation of sales to more productive and higher markup firms played a more prominent role in explaining the observed trends in retail.

The model now attributes 90% of the observed increase in sales concentration in retail to the λ shock and only 10% to q shocks. The model also generates a more sizable increase in the aggregate retail markup of 5.1% (from 1.15 to 1.21), which accounts for a quarter of the 12.7 pp decline in the labor share of retail. This is the result of a 1.5% decrease in the within-firm component (as firms respond to higher wages with an imperfect passthrough) and a 6.6% increase driven by reallocation to high-markup firms. Most of this reallocation is

[29]For non-manufacturing sectors, we set the persistence of productivity ρ_z to 0.86, which matches our estimates for revenue TFP obtained from our production function estimates in Section 1. For these sectors we lack data on exit and entry rates, as well as the relative size of entrants and exiting firms, and so we target the same moments from manufacturing. We also keep the demand super-elasticity of 0.22.

[30]The large increase in market access calibrated for retail aligns with other studies. For example, Hsieh and Rossi-Hansberg (2023) document that retail firms increased their number of establishments by 21.6% and locations by 18.6% during 1977–2013. Aghion et al. (2023b) provide a model where lower overhead costs increase the number of markets serviced by top firms in retail, wholesale, and services by 31.3% since 1990.

explained by the sizable λ shock. Reallocation now dominates because we calibrate a more log convex productivity distribution with $n = 0.47$ for retail.[31] The λ shock also plays a more important role in the decoupling of the typical firm labor share (which lowers its markups due to heightened competition) and the aggregate (which decreases due to reallocation).

Markups in the data: Our numerical exercises point to a role for markups in contributing to the labor share decline outside manufacturing and a limited role in manufacturing. We explore this prediction using estimates of markups from Compustat for 1980–2012.

As is the case with any paper that relies on Compustat to obtain markups, these estimates must be interpreted with caution since the sample of firms is selected and the data limited. We use the production function estimates from Section 1 and measure the markup for firm f at time t as $\mu_{tf} = \varepsilon^v_{tf}/s^v_{tf}$, where s^v_{tf} denotes the share of variable inputs in revenue and ε^v_{tf} denotes the variable input elasticity (one minus the output–capital elasticity in Section 1).

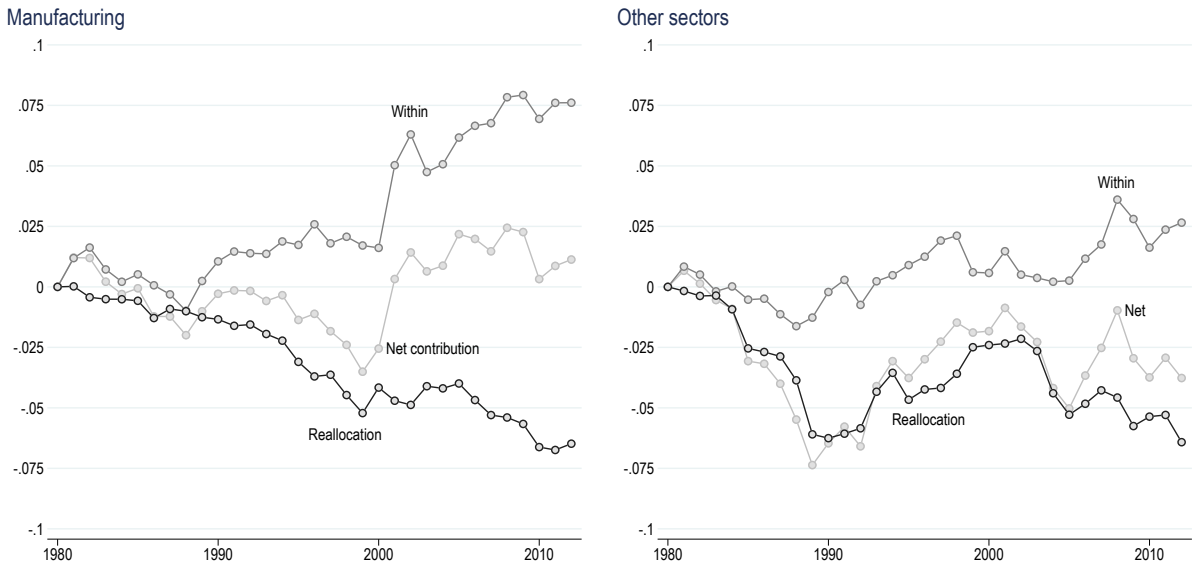

FIGURE 6: CONTRIBUTION OF WITHIN-FIRM CHANGES IN MARKUPS AND BETWEEN-FIRM REALLOCATION TO (PERCENT) CHANGES IN THE LABOR SHARE. The left panel provides the decomposition for manufacturing firms in Compustat. The right panel provides the decomposition for Compustat firms in other sectors.

Figure 6 plots the cumulative percent change in the inverse markup $1/\mu_t$ for manufacturing industries in the left and non-manufacturing industries in the right panel, decomposed into within-firm changes and the reallocation component. Positive values mean markups are increasing the labor share and negative values mean they are reducing it.

[31] Retail had high levels of concentration in 1982, which imply a more log-convex productivity distribution at baseline than in manufacturing.

The Compustat estimates support the conclusion that markups played no net role in driving the manufacturing labor share decline.[32] In this sector, the reallocation component accounts for a decline in the labor share of 6.5% between 1980 and 2012 (similar to the 4.5% decline in the model). However, the estimated within-firm changes in markups go in the opposite direction and generate no net effect.

The markup estimates from Compustat support the conclusion that markups played some role on net in driving the labor share decline outside manufacturing. In these sectors, the reallocation component reduced the labor share by 6.4% during 1980–2012. This is in line with the model, where we estimate a 6.6% reduction in the retail labor share due to the reallocation of sales towards high markup firms, most of it in response to the λ shock. Moreover, the rise in the within component has been weak in these sectors and only increased the labor share by 2.6% (similar to the 1.5% in our model).

4 Concluding remarks

This paper explored the forces driving the decline in the labor share and the rise in concentration in US manufacturing and other sectors. We considered a model where capital substitutes for labor at the task level and firms pay a fixed cost to automate tasks. Our exercises show that it is possible to have a coherent and quantitative description of the manufacturing labor share decline driven by lower capital prices that fits the relevant micro and macro facts:

1. Our model generates the large observed decline in the manufacturing and retail labor share in response to q shocks that are of a size consistent with data on capital prices.

2. Our model generates these declines even in scenarios where an econometrician would estimate an elasticity of substitution between capital and labor below 1. Our explanation is consistent with the limited substitution of capital for labor in response to higher wages estimated by Oberfield and Raval (2021) and others. Our model achieves this by having a more pronounced decline in capital prices at the extensive margin, triggering the substitution of capital for labor in a widening range of tasks among large firms.

[32] Relative to previous work by De Loecker, Eeckhout and Unger (2020), we find a smaller role for markups for two reasons: First, we compute markups as an harmonic sales weighted mean across firms, which is the relevant weight to quantify their contribution to the labor share decline. Second, our production function estimates allow for differences in the factor intensity of technology across firms in the same industry. Figure A9 in the Appendix show the implications of these two modifications. The conclusion of a limited role for markups in manufacturing also receives support from recent work by Foster, Haltiwanger and Tuttle (2022) using establishment-level data from the Annual Survey of Manufactures for 1972-2014.

3. The model reproduces the striking fact that while the manufacturing labor share declined by 20 pp, the labor share of a typical firm increased or remained unchanged.

4. In response to q shocks, the model generates the firm-level patterns in Autor et al. (2020) and Kehrig and Vincent (2021) for manufacturing, especially when allowing for a log-concave demand system. We conclude that the new firm-level facts documented in Autor et al. (2020) and Kehrig and Vincent (2021) cannot rule out explanations of the labor share decline driven by lower capital prices.

We also provided a decomposition exercise where we separate the contribution of capital advances from that of changes in market structure creating tighter competition between firms (such as rising market access). For manufacturing, our decomposition attributes most of the labor share decline and divergent trends in firm labor shares to lower capital prices and the uneven adoption of automation technologies at the extensive margin due to fixed costs. Outside of manufacturing, our decomposition shows that rising market access is needed to match the vast increase in sales concentration and that this shock also contributed to the decoupling of the aggregate and typical firms' labor share.

The paper leaves several areas for future research. First, our numerical exercises highlight the importance of distinguishing between capital advances at the intensive and extensive margin. An important research question is to develop direct measures of both margins. Second, our exercise focused on the substitution of capital for labor and its implications for aggregate and firm-level labor shares. One can imagine similar dynamics would apply for offshoring. If this also entails fixed costs, one would expect it to create similar effects.

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Online Appendix to "Capital–Labor Substitution and Firms' Labor Shares"

Joachim Hubmer and Pascual Restrepo

A Proofs and derivations for the model with CES demand

A.1 Policy functions and steady state

This sub-section proves Proposition 1 and characterizes the steady state of the economy. For these results, we work with a CES demand function

$$y_t = \left(\int_f y_{tf}^{\frac{\sigma-1}{\sigma}} \cdot df \right)^{\frac{\sigma}{\sigma-1}},$$

where $\sigma > 1$ denotes the elasticity of substitution across varieties.

Proof of Proposition 1. We first show that $\tilde{\alpha}_t(\alpha, z)$ weakly increases in z. We have

$$\tilde{\alpha}_t(\alpha, z) = \arg\max_{\alpha' \in [\alpha, 1]} -c_a \cdot y_t \cdot (\alpha' - \alpha) + \frac{1}{1+r} \mathbb{E}[V_{t+1}(\alpha', z')|z].$$

It is therefore sufficient to show that $\mathbb{E}[V_{t+1}(\alpha, z')|z]$ has increasing differences in (α, z). Let $\Omega_{t+1}(\alpha', z) = \partial_\alpha \mathbb{E}[V_{t+1}(\alpha', z')|z]$. The envelope theorem implies

$$\Omega_t(\alpha', z) = \mathbb{E}\left[\frac{\partial \pi_t(\alpha', z')}{\partial \alpha} | z \right] + \mathbb{E}\left[P_t(z') \cdot \min\left\{ c_a \cdot y_t, \frac{1}{1+r} \Omega_{t+1}(\alpha', z') \right\} \Big| z \right], \tag{A1}$$

where $P_t(z')$ denotes the probability of survival given z', and the minimum operator accounts for the fact that the restriction $\alpha' \geq \alpha$ binds in some states.

Let's define a sequence of functions $\Omega_t^{(n)}$ of (α', z) as:

$$\begin{aligned}
\Omega_t^{(1)}(\alpha', z) =& \mathbb{E}\left[\frac{\partial \pi_t(\alpha', z')}{\partial \alpha} | z \right] \\
\Omega_t^{(n+1)}(\alpha', z) =& \mathbb{E}\left[\frac{\partial \pi_t(\alpha', z')}{\partial \alpha} | z \right] + \mathbb{E}\left[P_t(z') \cdot \min\left\{ c_a \cdot y_t, \frac{1}{1+r} \Omega_{t+1}^{(n)}(\alpha', z') \right\} \Big| z \right].
\end{aligned}$$

We prove by induction in n that $\Omega_t^{(n)}(\alpha', z)$ weakly increases in z. The base case for $n = 1$ follows from the fact that $\mathbb{E}\left[\frac{\partial \pi_t(\alpha', z')}{\partial \alpha} | z \right]$ increases in z. Because the process is assumed to be increasing (in a stochastic sense), this is equivalent to showing that $\frac{\partial \pi_t(\alpha', z')}{\partial \alpha}$ increases in z'.

Let's write the marginal cost of firms as

$$c_t(\alpha, z) = \frac{1}{z} c_t(\alpha), \quad \text{with} \quad c_t(\alpha) = \min_{a \leq \alpha} \left(\Gamma_t^k(a) + \Gamma^\ell(a) \cdot w_t^{1-\eta} \right)^{\frac{1}{1-\eta}}.$$

This formulation is more general than the one from (1) in the main text. It allows for the possibility that firms might have paid the fixed cost to automate all tasks up to α, and yet, due to changing factor prices, choose to allocate some of these tasks to labor.

A second application of the envelope theorem (but now with respect to the optimal pricing decision of firms) implies

$$\frac{\partial \pi_t(\alpha', z')}{\partial \alpha} = -\frac{y_t(\alpha', z')}{z'} \cdot \frac{\partial c_t(\alpha')}{\partial \alpha},$$

where $y_t(\alpha', z')$ is the quantity sold by a firm with technology $\alpha_{tf} = \alpha'$ and $z_{tf} = z'$ at time t. Here, $\frac{\partial c_t(\alpha')}{\partial \alpha}$ is weakly negative (firms always get the option value of automating tasks if factor prices justify it). This means that $\frac{\partial \pi_t(\alpha', z')}{\partial \alpha}$ is increasing in z' if $\frac{y_t(\alpha', z')}{z'}$ increases in z', which holds in the CES demand systems when $\sigma > 1$.

For the inductive step, suppose that $\Omega_t^{(n)}(\alpha', z)$ is weakly increasing in z for all (t, α) with $n \leq N$. We have

$$\Omega_t^{(N+1)}(\alpha', z) = \mathbb{E}\left[\frac{\partial \pi_t(\alpha', z')}{\partial \alpha} | z \right] + \mathbb{E}\left[P_t(z') \cdot \min\left\{ c_a \cdot y_t, \frac{1}{1+r} \Omega_{t+1}^{(N)}(\alpha', z') \right\} \Big| z \right].$$

As before, $\mathbb{E}\left[\frac{\partial \pi_t(\alpha', z')}{\partial \alpha} | z \right]$ weakly increases in z. Moreover, $P_t(z') \cdot \min\{c_a \cdot y_t, (1/(1+r)) \cdot \Omega_{t+1}^{(N)}(\alpha', z')\}$ (weakly) increases in z' (due to the inductive hypothesis), and so the term $\mathbb{E}\left[P_t(z') \cdot \min\{c_a \cdot y_t, (1/(1+r)) \cdot \Omega_{t+1}^{(N)}(\alpha', z')\} | z \right]$ also (weakly) increases in z, which completes the inductive step.

Because the set of weakly increasing functions is closed, $\Omega_t(\alpha', z) = \lim_{n \to \infty} \Omega_t^{(n)}(\alpha', z)$ is also weakly increasing in z. It follows that $\mathbb{E}[V_{t+1}(\alpha', z') | z]$ has increasing differences in (α', z) as wanted.

We now turn to the limiting behavior of $\tilde{\alpha}_t(\alpha, z)$ as z grows to infinity. Automation decisions are guided by $\Omega_{t+1}(\alpha', z)$, which gives the marginal benefit to the firm of automating tasks up to α'. Suppose that $\alpha < \alpha_{t+1}^*$, and take any $\alpha' \in [\alpha, \alpha_{t+1}^*]$. With a CES demand system, $\frac{\partial \pi_t(\alpha', z')}{\partial \alpha}$ is an increasing and unbounded function of z', unless $\alpha' = \alpha_{t+1}^*$, in which case this is zero. As required in footnote 11, this implies that $\mathbb{E}\left[\frac{\partial \pi_t(\alpha', z')}{\partial \alpha} | z \right]$ and the right-hand side of equation (A1) converge to infinity as $z \to \infty$, unless $\alpha' = \alpha_{t+1}^*$. Optimal policy thus sets $\alpha' = \alpha_{t+1}^*$.

We conclude by exploring the limiting behavior of $\tilde{\alpha}_t(\alpha, z)$ as z goes to zero. The conditions in 11 imply that $\mathbb{E}\left[\frac{\partial \pi_t(\alpha', z')}{\partial \alpha}|z\right]$ and the right-hand side of equation (A1) converge to zero as $z \to 0$. This implies $\Omega_{t+1}(\alpha', z) = 0$ for all α' and optimal policy keeps $\alpha' = \alpha$. ■

Remark: the above proof shows that automation decisions and productivity levels are complementary if $\frac{y_t(\alpha,z)}{z}$ increases in z. A demand system satisfies this property if the product of the demand elasticity and the passthrough of marginal costs into prices exceeds 1 for all firms. This holds with a CES demand (demand elasticity σ and passthrough 1), but does not hold generically with a log-concave demand system. In any case, the proof above also shows that this complementarity will only break down for extremely large firms for which σ approaches 1. These firms reduce their use of labor and capital as their productivity increases and have no incentives to automate further.

The next Proposition shows that the model admits a stationary equilibrium where all firms allocate all tasks below a common α^* to capital and the behavior of aggregates is identical to that from a model where firms face no fixed costs of automation.

Consider a version of our model where firms face no fixed costs of automation (and so they set $\alpha_{tf} = \alpha_t^*$) and capital prices are fixed, so that $q_t(x) = q(x)$. This version of our model is equivalent to a standard Hopenhayn model where firms marginal costs only depend on their productivity z and are given by

$$c_t(z) = \frac{1}{z} \cdot \min_{\alpha} \left(\Gamma^k(\alpha) + \Gamma^\ell(\alpha) \cdot w_t^{1-\eta}\right)^{\frac{1}{1-\eta}}. \tag{A2}$$

As shown in Hopenhayn (1992), this model has a unique stationary equilibrium. Let w^* denote the wage in this stationary equilibrium an α^* the common level of automation that minimizes marginal costs for this wage level.

Proposition A1 *Suppose $q_t(x) = q(x)$. The economy admits a unique stationary equilibrium with wage w^*. In this stationary equilibrium, $\alpha_{tf} \geq \alpha^*$ almost surely (i.e. for all firms except a set of measure zero) and produce tasks below α^* with capital and tasks above α^* with labor.*

Proof. Suppose the wage converges to w and let α denote the level of automation that minimizes marginal costs for this wage level.

Consider the path for $\bar{\alpha}_t$. Define

$$\alpha_{inf} = \lim_{T\to\infty} \inf_{t\geq T}\{\bar{\alpha}_t\}.$$

We first show that we cannot have $\alpha_{inf} < \alpha$ by way of contradiction.

Suppose $\alpha_{inf} < \alpha$. For large t, firms alive at t started with a level of automation of at least α_{inf} almost surely. This follows from the restrictions in footnote 11, which imply that all firms exit with positive probability, and so the probability that firms alive at time t where born at T or later converges to 1 as $t \to \infty$.

For these firms, $\alpha_{tf} \geq \alpha_{inf}$. Moreover, a positive mass m of these firms received positive productivity shocks and increased α_{tf} above $\alpha_{inf} + \delta$, for a positive but small δ such that $\alpha_{inf} + \delta < \alpha$. This follows from Proposition 1 and the restrictions in footnote 11, which imply that firms reach the requisite size to justify investing in increasing α_{tf} up to $\alpha_{inf} + \delta$ with positive probability. However, This would imply $\bar{\alpha}_t \geq \alpha_{inf} + m \cdot \delta$, contradicting the definition of α_{inf}.

This means we must have $\alpha_{min} \geq \alpha$. For large t, firms alive at t started with a level of automation of at least α almost surely. This means that $\alpha_{tf} \geq \alpha$ almost surely and the economy converges to a standard firm-dynamics model where firms costs are given by (A2), as wanted.

As shown in Hopenhayn (1992), this model features a unique stationary equilibrium, with wage w^*. In this stationary equilibrium, $\alpha_{tf} \geq \alpha^*$ and firms produce all tasks below α^* with capital and all tasks above α^* with labor almost surely. Note that there might be a measure-zero set of firms with $\alpha_{tf} < \alpha^*$ that do not exit, and that is why the proposition claims $\alpha_{tf} \geq \alpha^*$ almost surely. ■

A.2 Effects of q shocks

This subsection proves Propositions 2 and 3. We first provide a technical lemma that helps characterize the impact of q shocks on real wage levels. This lemma is not central to this paper, but is proven here for completeness.

Lemma A1 (Effects of q shocks on the stationary equilibrium) *Let c denote the (real) marginal cost for a firm with unit productivity in the stationary equilibrium. The stationary distribution of firm productivities and c remain unchanged following a q shock.*

Proof. The proof is by construction and involves showing that this outcome satisfies the equilibrium conditions E1–E6.

Let $f(z)$ denote the mass of firms with productivity z in the initial stationary distribution and c the (real) marginal cost for a firm with unit productivity in this equilibrium. By construction, this equilibrium satisfies E1–E5.

Consider an arbitrary q shock. We guess and verify that the wage adjusts in the new equilibrium so as to keep $f(z)$ and c unchanged. In turn, output adjusts as to ensure labor-market clearing.

We verify this conjecture in steps:

- First, note that the ideal-price index depends on c and the distribution of z, both of which we conjectured remain unchanged. This shows that our conjecture satisfies E1.
- Second, note that firm entry and exit decisions conditional on z remain unchanged. This is because the demand faced by firm and their operating costs scale with output y, while their marginal cost remains constant and equal to $(1/z) \cdot c$. This means that firms value functions will scale with y. Notice that in the initial and final equilibrium, firms set $\alpha_{t+1,f} = \alpha_{tf}$ (a consequence of Proposition A1). This means that there are no costs incurred for automating additional tasks in a stationary equilibrium. This shows that our conjecture satisfies E3 and E4.
- Third, because entry and exit decisions conditional on z remain unchanged, the stationary distribution of productivity $f(z)$ also remains unchanged. This shows that our conjecture satisfies E5.
- Finally, the change in output is pinned down by labor market clearing, which shows that our conjecture satisfies E2.

The same proof applies to any homothetic demand system. ■

Proof of Proposition 2. Write $q(x) = q \cdot q_0(x)$ for tasks below α^* and consider a permanent increase in q by $d \ln q_{int}$ for these tasks. The equilibrium impact of this shock is to change wages by $d \ln w$ and automation decisions by a common amount $d \ln \alpha^*$ (a consequence of Proposition A1).

We first characterize the effect of this q shock on wages. Lemma A1 shows that wages adjust so as to keep unit costs unchanged. An application of Shephard's lemma implies that

$$d \ln c = \varepsilon^{\ell} \cdot d \ln w - \varepsilon^{k} \cdot d \ln q_{int},$$

where, in addition, the envelope theorem ensures that the effect of changes in α^* on c are second order and can be ignored. Because $d \ln c = 0$, we can solve for the change in wages as

$$d \ln w = \frac{\varepsilon^{k}}{\varepsilon^{\ell}} \cdot d \ln q_{int}.$$

We now turn to the behavior of cost shares (or equivalently, output elasticities). In steady state, all firms have the same labor cost share (a consequence of Proposition A1), given by

$$\varepsilon^{\ell} = \frac{\Gamma^{\ell}(\alpha^*) \cdot w^{1-\eta}}{\Gamma_0^k(\alpha^*) \cdot q^{\eta-1} + \Gamma^{\ell}(\alpha^*) \cdot w^{1-\eta}}.$$

This common cost share for labor varies with wages and α^*. Equation (2) implies that the change in the optimal threshold α^* satisfies

$$d \ln \alpha^* = \frac{1}{\partial \ln \psi_{\ell}(\alpha^*)/q(\alpha^*)/\partial \ln \alpha} \cdot d \ln w.$$

Using this expression for $d \ln \alpha^*$ and the definition of η^*, we obtain

$$\begin{aligned} d \ln \varepsilon^{\ell} &= \varepsilon^k \cdot d \ln \frac{\varepsilon^{\ell}}{\varepsilon^k} \\ &= \varepsilon^k \cdot (1-\eta) \cdot (d \ln q_{int} + d \ln w) + \varepsilon^k \cdot \frac{\partial \ln \Gamma^{\ell}(\alpha^*)/\Gamma_0^k(\alpha^*)}{\partial \ln \alpha} \cdot d \ln \alpha^* \\ &= \varepsilon^k \cdot (1-\eta) \cdot (d \ln q_{int} + d \ln w) + \varepsilon^k \cdot (\eta - \eta^*) \cdot d \ln w \\ &= \varepsilon^k \cdot (1-\eta^*) \cdot d \ln w + \varepsilon^k \cdot (1-\eta) \cdot d \ln q_{int}. \end{aligned}$$

Along the transition, firms will differ in the extent to which they will automate their tasks. Let $d \ln \alpha_{tf}$ denote the additional tasks automated by firm f at time t. We have that

$$d \ln \varepsilon_{tf}^{\ell} = \varepsilon^k \cdot (1-\eta) \cdot (d \ln q_{int} + d \ln w_t) + \varepsilon^k \cdot \frac{\partial \ln \Gamma^{\ell}(\alpha^*)/\Gamma_0^k(\alpha^*)}{\partial \ln \alpha} \cdot d \ln \alpha_{tf}.$$

The expression for incumbents that do not automate follows from taking $d \ln \alpha_{tf} = 0$. ■

Proof of Proposition 3. Write $q(x) = q \cdot q_0(x)$ for tasks above α^* and consider a permanent increase in q by $d \ln q_{ext}$ for these tasks. The equilibrium impact of this shock is to change wages by $d \ln w$ and automation decisions by a common amount $d \ln \alpha^*$ (a consequence of Proposition A1).

We first characterize the effect of this q shock on wages. Lemma A1 shows that wages adjust so as to keep unit costs unchanged. The envelope theorem ensures that the effect of changes in α^* on c are negative but second order, while the effects of changes in α^* on c are positive and first order. Because $d \ln c = 0$, wages must increase by a positive but second order amount, $d \ln w = \mathcal{O}(d \ln q_{ext}^2) > 0$.

We now turn to the behavior of cost shares (or equivalently, output elasticities). In steady

state, all firms have the same labor cost share (a consequence of Proposition A1), given by

$$\varepsilon^{\ell} = \frac{\Gamma^{\ell}(\alpha^*) \cdot w^{1-\eta}}{\Gamma_0^k(\alpha^*) + \Gamma^{\ell}(\alpha^*) \cdot w^{1-\eta}}.$$

This common cost share for labor will vary with prices and α^*. Equation (2) implies that the change in the optimal threshold α^* satisfies

$$d\ln\alpha^* = \frac{1}{\partial \ln \psi_{\ell}(\alpha^*)/q(\alpha^*)/\partial \ln \alpha}(d\ln w + d\ln q_{ext}).$$

Using this expression for $d\ln\alpha^*$ and the definition of η^*, we obtain

$$\begin{aligned}
d\ln\varepsilon^{\ell} &= \varepsilon^k \cdot d\ln\frac{\varepsilon^{\ell}}{\varepsilon^k} \\
&= \varepsilon^k \cdot (1-\eta) \cdot d\ln w + \varepsilon^k \cdot \frac{\partial \ln \Gamma^{\ell}(\alpha^*)/\Gamma_0^k(\alpha^*)}{\partial \ln \alpha} \cdot d\ln\alpha^* \\
&= \varepsilon^k \cdot (1-\eta) \cdot d\ln w - \varepsilon^k \cdot (\eta^* - \eta) \cdot (d\ln w + d\ln q_{ext}) \\
&= \varepsilon^k \cdot (1-\eta^*) \cdot d\ln w - \varepsilon^k \cdot (\eta^* - \eta) \cdot d\ln q_{ext}.
\end{aligned}$$

Along the transition, firms will differ in the extent to which they automate their tasks. Let $d\ln\alpha_{tf}$ denote the additional tasks automated by firm f at time t. We have that

$$d\ln\varepsilon_{tf}^{\ell} = \varepsilon^k \cdot (1-\eta) \cdot d\ln w_t + \varepsilon^k \cdot \frac{\partial \ln \Gamma^{\ell}(\alpha^*)/\Gamma_0^k(\alpha^*)}{\partial \ln \alpha} \cdot d\ln\alpha_{tf}.$$

The expression for incumbents that do not automate follows from taking $d\ln\alpha_{tf} = 0$. ■

A.3 The induced elasticity of substitution η^*

The text explains that η^* is the elasticity of substitution that one would estimate from permanent variation in wages. This subsection formalizes this connection.

The elasticity of substitution is defined as

$$\text{elasticity of substitution} = \frac{d\ln K/L}{d\ln w} = 1 - \frac{d\ln(\varepsilon^{\ell}/\varepsilon^k)}{d\ln w}.$$

Using the expression for cost shares in the text, the definition of η_t^* and the fact that

$$d\ln\alpha^* = \frac{1}{\partial \ln \psi_{\ell}(\alpha^*)/q(\alpha^*)/\partial \ln \alpha} \cdot d\ln w,$$

we get

$$\begin{aligned}\text{elasticity of substitution} =& 1-(1-\eta)-\frac{\partial \ln \Gamma^{\ell}(\alpha^*)/\Gamma_t^k(\alpha^*)}{\partial \ln \alpha}\cdot\frac{d\ln \alpha^*}{d\ln w}\\ =& 1-(1-\eta)-(\eta-\eta_t^*)\\ =& \eta_t^*,\end{aligned}$$

as wanted.

B Proofs and derivations for the model with log-concave demand

This section provides the details of the model with a log-concave demand system.

B.1 Micro-foundation for λ

We show that the demand system implied by (4) is isomorphic to one where firms compete against a growing mass of firms for a given consumer.

Let's first consider the demand system in the main text. Firm demand is derived from the following cost minimization problem:

$$\min_{y_{tf}} \int_f p_{tf} \cdot y_{tf} \cdot df \quad \text{s.t:} \quad \int_f \lambda \cdot H\left(\frac{y_{tf}}{\lambda_t \cdot y_t}\right) \cdot df = 1.$$

Let $\rho_t \cdot y_t$ denote the Lagrange multiplier on the constraint. The first-order condition for the choice of y_{tf} is then

$$y_{tf} = y_t \cdot \lambda \cdot D\left(\frac{p_{tf}}{\rho_t}\right), \tag{A3}$$

where D is decreasing and given by the inverse function of $H'(x)$. Plugging the demand for each variety in the constraint, we obtain

$$\int_f \lambda \cdot H\left(D\left(\frac{p_{tf}}{\rho_t}\right)\right) \cdot df = 1. \tag{A4}$$

Moreover, because the price of the final good is normalized to 1, we must have

$$1 = \int_f \lambda \cdot p_{tf} \cdot D\left(\frac{p_{tf}}{\rho_t}\right) \cdot df. \tag{A5}$$

In sum, the equilibrium for the Kimball demand system is summarized by equations (A3), (A4), and (A5).

Let's now show this system is equivalent to one with multiple consumers where firms can access and compete over a fixed mass of them.

There is a mass 1 of customers with equal incomes indexed by j whose flow utility u_{tj} from consuming a set of varieties $\mathcal{F}_{tj}$ is defined implicitly by

$$\int_{f \in \mathcal{F}_{tj}} H\left(\frac{y_{tfj}}{u_{tj}}\right) \cdot df = 1.$$

Consumers maximize their utility u_{tj} subject to their budget constraint. As above, consumer j demand from firm $f \in \mathcal{F}_{tj}$ is

$$y_{tfj} = y_t \cdot D\left(\frac{p_{tf}}{\rho_{tj}}\right),$$

where y_t is income per consumer and ρ_{tj} satisfies

$$\int_{f \in \mathcal{F}_{tj}} H\left(D\left(\frac{p_{tf}}{\rho_{tj}}\right)\right) \cdot df = 1.$$

Firms and customers are randomly matched to each other, with each customer matched to a mass λ of firms. Random matching implies that all consumers face the same distribution of prices, and so they share a common $\rho_{tj} = \rho_t$.

As a result, total demand for firm f is

$$y_{tf} = y_t \cdot \lambda \cdot D\left(\frac{p_{tf}}{\rho_t}\right),$$

which coincides with equation (A3). The equation for ρ_{tj} can then be written as

$$\int_f \lambda \cdot H\left(D\left(\frac{p_{tf}}{\rho_t}\right)\right) \cdot df = 1,$$

which coincides with (A4). Finally, adding revenue across firms we get $y_t = \int_f y_{tf} \cdot p_{tf} \cdot df$, which implies

$$1 = \int_f \lambda \cdot p_{tf} \cdot D\left(\frac{p_{tf}}{\rho_t}\right) \cdot df. \tag{A6}$$

This equation coincides with (A5), establishing the aggregation result.

B.2 Implications for prices, sales, and markups

This subsection shows that Marshall's second laws imply properties P1–P3 in the text.

The formal assumptions behind Marshall's second laws are:

- **Weak second law:** demand elasticity $-x \cdot D'(x)/D(x)$ exceeds 1 and increases in x.
- **Strong second law:** marginal revenue $x + D(x)/D'(x)$ is positive and log-concave.

Proposition A2 *Consider a firm with a constant marginal cost c and denote its optimal price by $p^*(c)$, markups by $\mu^*(c)$, and firm sales by $\omega^*(c)$. Under Marshall's weak second law, $p^*(c)$ is increasing and $\mu^*(c)$ is decreasing. Moreover, under Marshall's strong second law, markups and prices, $\mu^*(c)$ and $p^*(c)$, are log-convex functions of costs, which implies lower passthroughs for more productive firms. Finally, sales $\omega^*(c)$ are log-concave and decreasing functions of costs.*

Proof. Prices are given by

$$p^*(c) = \arg\max_p y \cdot \lambda \cdot D\left(\frac{p}{\rho}\right) \cdot (p - c).$$

This problem has increasing differences in p and c, which implies that $p^*(c)$ increases in c.

Moreover, the first order condition for this problem is

$$-\frac{1}{\rho} D'\left(\frac{p}{\rho}\right) \cdot (p - c) = D\left(\frac{p}{\rho}\right) \quad \Rightarrow \quad \frac{\mu^*(c)}{\mu^*(c) - 1} = -\frac{p^*(c)}{\rho} \frac{D'\left(\frac{p^*(c)}{\rho}\right)}{D\left(\frac{p^*(c)}{\rho}\right)}.$$

Marshall's weak second law combined with the fact that $p^*(c)$ increases in c implies that the right-hand side of the above equation increases in c. The left-hand side is a decreasing function of $\mu^*(c)$, which therefore implies that $\mu^*(c)$ is decreasing in c as wanted.

We can rewrite the first-order condition for prices as

$$\frac{p^*(c)}{\rho} + \frac{D(p^*(c)/\rho)}{D'(p^*(c)/\rho)} = \frac{c}{\rho}.$$

Differentiating this expression yields

$$\frac{\partial \ln p^*(c)}{\partial \ln c} = \frac{1}{d\left(\frac{p^*(c)}{\rho}\right)},$$

where

$$d(x) = \frac{\partial \ln\left(x + D(x)/D'(x)\right)}{\partial \ln x}$$

is a decreasing function according to Marshall's strong second law. It follows that $\ln p^*(c)$ is a convex function in $\ln c$. Moreover, $\ln \mu^*(c) = \ln p^*(c) - \ln c$ inherits this convexity.

Turning to sales shares, we have that $\omega^*(c)$ can be written as

$$\omega^*(c) = h(p^*(c))/y,$$

where $h(x) = xD(x)$ is a log-concave and decreasing function of x (from Marshall's weak second law). Thus, $\omega^*(c)$ is the composition of a log-concave and decreasing function ($h(x)$) with a log-convex and increasing function $p(c)$, which results in a log-concave and decreasing function. ■

B.3 Effects of rising market access

We now derive the effects of an increase in market access—Properties P4–P5 in the text.

We assume the economy starts from a stationary equilibrium where all firms have the same degree of automation. We then characterize the immediate effect of a λ shock. By design, this exercise does not account for any effect of λ shocks through subsequent entry, exit, or automation decisions. These effects are harder to study analytically, but are explored in our numerical exercises.

Let μ_z and p_z denote the markup and price charged by a firm of productivity z, and ω_z its sales share.

Proposition A3 *An increase in λ has the following immediate effects:*

- *μ_z decreases for all z;*
- *for $z > z'$, $\mu_z/\mu_{z'}$ decreases;*
- *for $z > z'$, $\omega_z/\omega_{z'}$ increases.*

Proof. Firms' marginal cost is $(1/z) \cdot c(w)$ for some common $c(w)$, which is a function of the equilibrium wage.

Let $c_{norm} = c/\rho$, where ρ is the value of ρ_t at the initial equilibrium. c_{norm} is an endogenous object pinned down by market access and the distribution of productivities, as we show later. It also summarizes the degree of competition in the economy.

We can rewrite firms' pricing problem as

$$\max_{p_{norm}} D(p_{norm}) \cdot \left(p_{norm} - \frac{1}{z} \cdot c_{norm}\right),$$

where $p_{norm} = p/\rho$ is a normalized firm price. Optimal firm prices are $p_z = \rho \cdot p^*(c_{norm}/z)$, markups are $\mu_z = \mu^*(c_{norm}/z)$, and sale shares are $\omega_z = \omega^*(c_{norm}/z)$.

We now show that λ increases c_{norm}. The implicit definition of ρ can be rewritten as

$$\int_z \lambda \cdot H\left(D\left(p^*\left(c_{norm}/z\right)\right)\right) \cdot m_z \cdot dz = 1.$$

From this equation we see that λ increases the equilibrium value of c_{norm} (keep in mind that, here, m_z is maintained constant, as we are characterizing only the immediate impact of a λ shock).

We now characterize the effects of the increase in c_{norm}.

First, for a given z, $\mu_z = \mu^*(c_{norm}/z)$ decreases in c_{norm}, as wanted.

Second, because $\mu^*(c)$ is log-convex, we have that, for $z > z'$,

$$\ln \mu_z - \ln \mu_{z'} = \ln \mu^*\left(c_{norm}/z\right) - \ln \mu^*\left(c_{norm}/z'\right)$$

is decreasing in c_{norm}, as wanted.

Third, because the function $\omega^*(c)$ is log-concave,

$$\ln \omega_z - \ln \omega_{z'} = \ln \omega^*\left(c_{norm}/z\right) - \ln \omega^*\left(c_{norm}/z'\right)$$

is increasing in c_{norm} for $z > z'$, as wanted. ■

Proposition A4 *Let $f(z)$ denote the mass of firms of productivity z. The aggregate labor share is $s^\ell = \varepsilon^\ell/\mu$, where the aggregate markup μ is a sales weighted harmonic mean of firm-level markups:*

$$\frac{1}{\mu} = \int_z \frac{1}{\mu_z} \cdot \omega_z \cdot f(z) \cdot dz.$$

The immediate effect of λ is to increase the aggregate markup if the distribution of productivity is log-convex (i.e., more convex than Pareto), lower it if the distribution of productivity is log-concave (i.e., less convex than Pareto), and leave it unchanged if the distribution of productivity is log-linear (i.e., Pareto).

Proof of Proposition A4. As before, we investigate the implications of an increase in c_{norm} holding the distribution of productivities constant at $f(z)$ and without accounting for subsequent automation decisions.

We can write the aggregate markup as

$$\frac{1}{\mu} = \int_z \frac{1}{\mu^* \left(c_{norm}/z\right)} \cdot \omega^* \left(c_{norm}/z\right) \cdot f(z) \cdot dz.$$

With the change of variable $x = c_{norm}/z$, we can rewrite this as

$$\frac{1}{\mu} = \int_x \frac{1}{\mu^* \left(x\right)} \cdot g(x, c_{norm}) \cdot dx,$$

where $g(x, c_{norm})$ is a density function given by

$$g(x, c_{norm}) = \omega^* \left(x\right) \cdot f(c_{norm}/x) \cdot \frac{c_{norm}}{x^2} \cdot dx.$$

First, suppose that $f(z)$ is log-concave. This implies that

$$\ln g(x, c_{norm}) = \ln \omega^* \left(x\right) + \ln f(c_{norm}/x) + \ln c_{norm} - 2 \ln x$$

has increasing differences in x and $\bar{c}$. This is equivalent to the following monotone likelihood ratio property (MLRP):

$$\frac{g(x, c_{norm})}{g(x', c_{norm})} \text{ increasing in } c_{norm} \text{ for } x > x'.$$

The MLRP property implies that an increase in c_{norm} generates a shift up (in the first-order stochastic dominance sense) in $g(x, c_{norm})$. Because the function $\frac{1}{\mu^*(x)}$ is increasing in x, the aggregate markup μ decreases in c_{norm} as wanted.

Second, suppose that $f(z)$ is log-convex. This implies that

$$\ln g(x, c_{norm}) = \ln \omega^* \left(x\right) + \ln f(c_{norm}/x) + \ln c_{norm} - 2 \ln x$$

has decreasing differences in x and c_{norm}. This is equivalent to the following monotone likelihood ratio property (MLRP):

$$\frac{g(x, c_{norm})}{g(x', c_{norm})} \text{ decreasing in } c_{norm} \text{ for } x > x'.$$

The MLRP property implies that an increase in c_{norm} generates a shift down (in the first-order stochastic dominance sense) in $g(x, c_{norm})$. Because the function $\frac{1}{\mu^*(x)}$ is increasing in x, the aggregate markup μ increases in c_{norm} as wanted.

Finally, suppose that $f(z)$ is log-linear. This implies that

$$\ln g(x, c_{norm}) = \ln \omega^*(x) + \ln f(c_{norm}/x) + \ln \bar{c} - 2 \ln x$$

is a linear function in $\ln c_{norm}$. Equivalently,

$$\frac{g(x, c_{norm})}{g(x', c_{norm})} \text{ is independent of } c_{norm}.$$

Thus, the integral defining μ is independent of c_{norm}. ■

B.4 Properties of the Klenow–Willis aggregator

As a functional form for the Kimball (1995) aggregator H we use the specification from Klenow and Willis (2016), defined as

$$H(q) \equiv 1 + (\sigma - 1) \cdot \exp\left(\frac{1}{\nu}\right) \cdot \nu^{\frac{\sigma}{\nu} - 1} \cdot \left[\Gamma\left(\frac{\sigma}{\nu}, \frac{1}{\nu}\right) - \Gamma\left(\frac{\sigma}{\nu}, \frac{q^{\frac{\nu}{\sigma}}}{\nu}\right)\right],$$

where $\Gamma(\cdot, \cdot)$ is the upper incomplete Gamma function,

$$\Gamma(s, x) \equiv \int_x^\infty t^{s-1} \cdot \exp(-t) dt.$$

This gives rise to the following (relative) demand function $D^{-1} = H'$:

$$D(x) = \left(1 - \nu \cdot \ln\left(x \cdot \frac{\sigma}{\sigma - 1}\right)\right)^{\frac{\sigma}{\nu}}.$$

The price elasticity of demand is

$$-\frac{x \cdot D'(x)}{D(x)} = \frac{\sigma}{1 - \nu \cdot \ln\left(x \cdot \frac{\sigma}{\sigma - 1}\right)} = \sigma \cdot D(x)^{-\frac{\nu}{\sigma}}, \tag{A7}$$

which reduces to the constant σ if $\nu = 0$ (the benchmark case of a CES aggregator). In general, equation (A7) shows that under this parametrization, the super-elasticity of demand is equal to the constant $-\frac{\nu}{\sigma}$, and that larger firms will face more inelastic demand curves.

To conclude, we show that the Klenow-Willis aggregator satisfies Marshall's second laws. Equation (5) shows that the demand elasticity is increasing in the relative price and greater than 1 (Marshall's weak second law), imposing the restriction that $\sigma > 1$ and $\nu > 0$. To see

that the strong law holds as well, write the logarithm of marginal revenue as

$$\begin{aligned}\ln\left(x+\frac{D(x)}{D'(x)}\right) &= \ln x + \ln\left(1+\frac{D(x)}{x\cdot D'(x)}\right)\\ &= \ln x + \ln\left(\frac{\sigma+\nu\cdot\ln x+\nu\cdot\ln\left(\frac{\sigma}{\sigma-1}\right)-1}{\sigma}\right),\end{aligned}$$

which is a concave function of $\ln x$ as desired.

B.5 Incorporating demand shocks

In our baseline model, firm dynamics are driven by productivity shocks z_{tf}. This subsection discusses the implications of allowing for firm-specific demand shocks z^d_{tf}. It shows that demand shocks produce identical responses to a productivity shock on revenue, markups, automation decisions, labor shares, and profits. The two shocks only differ in their implications for firm prices.

To introduce demand shocks, we modify the Kimball aggregator in (4) to

$$\int_f \lambda\cdot H\left(\frac{z^d_{tf}\cdot y_{tf}}{\lambda\cdot y_t}\right)\cdot df = 1,$$

where z^d_{tf} is a taste shifter for firm f's variety.

The demand curve faced by firms is now given by

$$y_{tf} = \lambda\cdot y_t\cdot\frac{1}{z^d_{tf}}\cdot D\left(\frac{p_{tf}}{\rho_t\cdot z^d_{tf}}\right),$$

and its profit maximization problem is modified to

$$\pi_t(\alpha_{tf}, z_{tf}, z^d_{tf}) = \max_{p_{tf}} \lambda\cdot y_t\cdot\frac{1}{z^d_{tf}}\cdot D\left(\frac{p_{tf}}{\rho_t\cdot z^d_{tf}}\right)\cdot\left(p_{tf}-\frac{1}{z_{tf}}c_t(\alpha_{tf})\right),$$

where $c_t(\alpha_{tf})$ is the unit cost of a firm with unitary productivity and $\pi_t(\alpha_{tf}, z_{tf}, z^d_{tf})$ the profit function.

Let $p^d_{tf} = p_{tf}/z^d_{tf}$ be a taste-adjusted price. We can rewrite profit maximization as

$$\pi_t(\alpha_{tf}, z_{tf}, z^d_{tf}) = \max_{p^d_{tf}} \lambda\cdot y_t\cdot D\left(\frac{p^d_{tf}}{\rho_t}\right)\cdot\left(p^d_{tf}-\frac{1}{z_{tf}\cdot z^d_{tf}}c_t(\alpha_{tf})\right).$$

This shows that firms' profits and optimal choices are functions of the composite $z_{tf} \cdot z^d_{tf}$. This implies that a productivity and a demand shock with the same persistence will generate the exact same responses in terms of firm profits, exit, and automation decisions, all of which depend entirely on the profit function.

Moreover, using the notation introduced in the proof of Proposition A2, we can write optimal taste-adjusted prices as

$$p^d_{tf} = p^* \left(\frac{1}{z_{tf} \cdot z^d_{tf}} c_t(\alpha_{tf}) \right),$$

prices as

$$p_{tf} = z^d_{tf} \cdot p^* \left(\frac{1}{z_{tf} \cdot z^d_{tf}} c_t(\alpha_{tf}) \right),$$

and markups as

$$\mu_{tf} = \frac{p_{tf}}{\frac{1}{z_{tf}} c_t(\alpha_{tf})} = \frac{p^d_{tf}}{\frac{1}{z_{tf} \cdot z^d_{tf}} c_t(\alpha_{tf})} = \mu^* \left(\frac{1}{z_{tf} \cdot z^d_{tf}} c_t(\alpha_{tf}) \right).$$

This shows that demand and productivity shocks have identical implications for markups but different implications for prices. Markups increase with both demand and productivity shocks. Instead, prices increase with demand shocks (since $p^*(c)$ is increasing in c but with a passthrough below one) and decrease with productivity shocks.

C Additional numerical exercises and details for manufacturing

C.1 Details of the q-shock:

A more general expression for the q shocks used in our quantitative exploration is

$$q_{I,t}(x) = \begin{cases} q_{int,t} & \text{if } x \leq \alpha_0^* \\ \min\{q_{int,t} \cdot \frac{\psi_\ell(x)/\psi_\ell(\alpha_0^*)}{\psi_k(x)/\psi_k(\alpha_0^*)}, q_{ext,t}\} & \text{if } x > \alpha_0^* \end{cases}$$

for some increasing $\{q_{int,t}, q_{ext,t}\}$ converging to $\{q_{int}, q_{ext}\}$.

The main text normalized baseline wages to 1 and assume that $q_0^I(x) = 1$. This implies $\psi_\ell(\alpha_0^*)/\psi_k(\alpha_0^*) = 1$, which simplifies the formulation of the shock.

Figure A1 represents the q shock graphically. Tasks are arranged in [0,1] in the horizontal axis. At time 0, we have $q_{I,t}(x) = 1$ for all tasks. Over time, the productivity with which the economy can produce the capital needed for task x rises. In the figure, we depict a case where capital advances are more pronounced at the extensive margin, as in our calibration.

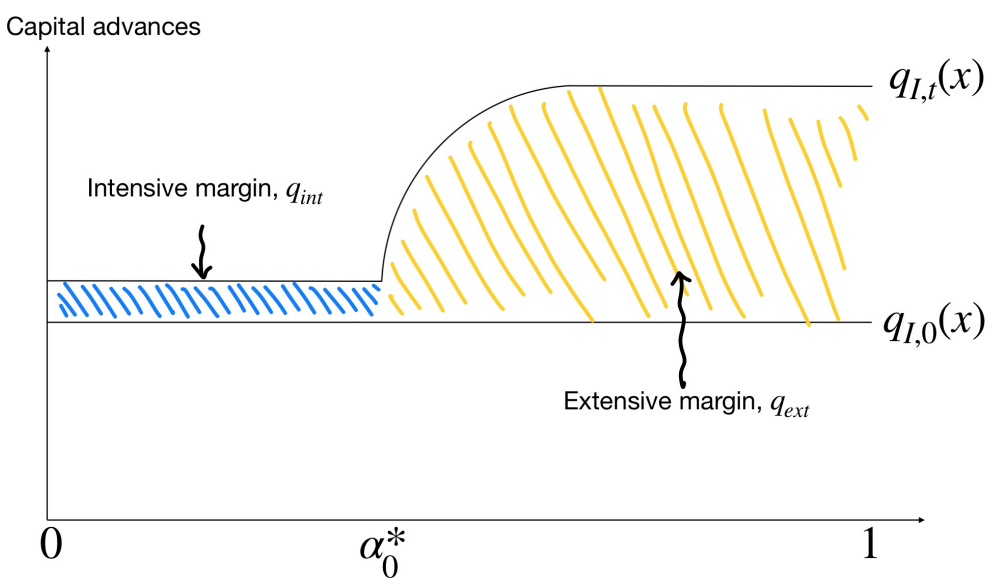

Figure A1: Representation of q shocks. Tasks are arranged in [0,1] in the horizontal axis. At time 0, we have $q_{I,0}(x) = 1$ for all tasks. Over time, the productivity with which the economy can produce the capital needed for task x rises to $q_{I,t}(x)$.

C.2 Shutting down diffusion

Columns (3) and (4) in Table A1 report two counterfactual exercises that assess the significance of the diffusion of automation assumption. In our baseline experiment (reproduced

in column 2), technology diffuses via entry as entrants' initial automation level equals the unweighted average automation level in use in the economy, $\alpha_{entrant,t} = \bar{\alpha}_t$. In column (3) we maintain the diffusion assumption for the initial steady state, such that firms have identical labor shares in 1982, but then remove it over the transition. I.e., entering firms in any year $t > 1982$ are assigned the automation level of the initial steady state $\alpha^\star$, which is lower than $\bar{\alpha}_t$. We do not re-calibrate the inferred aggregate shocks and parameters. The results are very similar. In fact, the main difference is that in the counterfactual economy, without diffusion to entrants, the relative adoption gradient naturally increases.

Column (4) explores the implications of shutting down diffusion for the initial steady state as well. For this experiment, we assume that entrants start with a fraction of the optimal automation level $\alpha_{tf} = m \cdot \alpha^\star$, with $m < 1$. We calibrate $m = 0.5$ to match the ratio of the unweighted mean firm to the aggregate labor share of 1.11 in 1982. This modified version of the model generates an initial steady state with the same amount of labor share dispersion by size as in the data. The resulting dynamics in response to q shocks remain very similar to our baseline findings.

C.3 Calibration with $\eta^\star = 1.45$

Columns (5) and (6) in Table A1 report results from an alternative parametrization with an induced elasticity of $\eta^\star = 1.45$, as estimated by Karabarbounis and Neiman (2013) and Hubmer (2023).

For this parametrization, we normalize initial capital prices to 1 and set

$$\psi_k(x) = 1, \qquad \psi_\ell(x) = A \cdot \left(x^{\frac{1-\eta^*}{\eta^*-\eta}} - 1 \right)^{\frac{1}{1-\eta^*}}.$$

Note that this requires $\eta^\star > 1$ so that $\psi_\ell(x)/\psi_k(x)$ is increasing in x.

With this specification, the cost function of firms in the initial steady state becomes

$$c_0(\alpha, z) = \frac{1}{z} \cdot \left(\alpha + \left(1 - \alpha^{\frac{\eta^*-1}{\eta^*-\eta}} \right)^{\frac{\eta^*-\eta}{\eta^*-1}} \cdot \left(\frac{w_0}{A} \right)^{1-\eta} \right)^{\frac{1}{1-\eta}}.$$

Minimizing with respect to α, we obtain

$$\alpha_0^* = \left(\frac{(w_0/A)^{\eta^*-1}}{1 + (w_0/A)^{\eta^*-1}} \right)^{\frac{\eta^*-\eta}{\eta^*-1}}$$

Table A1: Robustness checks manufacturing: no diffusion and induced elasticity $\eta^* = 1.45$

		Induced elasticity $\eta^* = 1$ (baseline)			$\eta^* = 1.45$	
	Data	Baseline	No diffusion over transition	No diffusion in initial st st	Both q shocks	Uniform q shock
	(1)	(2)	(3)	(4)	(5)	(6)
	I. Parameters and inferred aggregate shocks					
$d \ln q_{int}$		0.67	0.67	0.67	0.86	1.68
$d \ln q_{ext}$		5.48	5.48	5.48	1.63	1.68
c_a		0.19	0.19	0.19	0.32	0.35
	II. Targeted moments, 1982–2012					
Δ aggregate labor share	-0.20	-0.20	-0.20	-0.21	-0.20	-0.20
Δ log average capital price	-1.08	-1.08	-1.08	-1.09	-1.09	-1.68
Relative adoption (P99+ vs. P50-75 firms)	1.71	1.71	4.27	1.67	1.70	1.71
	III. Concentration 1982–2012 (from Autor et al., 2020; Decker et al., 2020)					
Δ log 4 firms' sales share	0.140	0.105	0.125	0.068	0.055	0.071
Δ log 20 firms' sales share	0.072	0.104	0.124	0.069	0.054	0.070
Δ log productivity dispersion	0.050	0.061	0.073	0.061	0.041	0.059
	IV. Typical firm labor share from Kehrig and Vincent (2021), 1982–2012					
Δ median labor share	0.030	-0.005	0.084	-0.005	-0.042	-0.002
Δ unweighted mean	-0.017	-0.039	0.004	-0.077	-0.055	-0.023
	V. Melitz–Polanec decomposition from Autor et al. (2020), 1982–2012					
Δ aggregate labor share	-0.185	-0.198	-0.199	-0.197	-0.202	-0.202
Δ unweighted incumbent mean	-0.002	-0.015	-0.023	-0.016	-0.029	0.006
Exit	-0.055	-0.006	-0.006	-0.005	-0.004	-0.004
Entry	0.059	0.006	0.007	0.006	0.006	0.006
Covariance term	-0.187	-0.183	-0.177	-0.183	-0.174	-0.210
	VI. Covariance decomposition from Kehrig and Vincent (2021), 1982–2012					
Market share dynamics	0.047	0	0	0	0	0
Labor share by size dynamics	-0.043	-0.078	-0.077	-0.078	-0.057	-0.069
Cross-cross dynamics	-0.232	-0.066	-0.066	-0.064	-0.083	-0.099

Notes: Column (2) reproduces the findings from our baseline CES model. Column (3) features the same initial steady state but then removes diffusion of technology via entry over the transition. Column (4) in addition removes the diffusion assumption in the initial steady state. Columns (5-6) feature a higher induced elasticity of $\eta^* = 1.45$. (5) infers the combination of intensive and extensive margin capital price declines as in the baseline. (6) instead imposes that capital prices decline uniformly.

and

$$c_0(\alpha_0^*, z) = \frac{1}{z} \cdot \left(1 + \left(\frac{w_0}{A}\right)^{1-\eta^*}\right)^{\frac{1}{1-\eta^*}} .$$

This shows that the induced elasticity of substitution is η^*, as wanted.

As in our baseline, we set $\eta = 0.5$, which means that tasks are complements. We also calibrate A so that the labor share in the initial steady state is 67%.

In column (5) we follow the same inference procedure as in the baseline model. To match the manufacturing labor share decline and average capital price decline, we back out a combination of q shocks that loads relatively less on the extensive margin. This is expected since with $\eta^* > 1$, even a uniform capital price decline generates an aggregate labor share decline.

Column (6) demonstrates this point by fitting a uniform q shock to the manufacturing labor share decline. This column shows that one could also generate the observed labor share decline as a result of a uniform decline in capital prices of 168 log points. The fact that this exceeds the decline in capital prices seen in the data highlights the importance of allowing for differences in capital advances at the intensive and extensive margin, since a uniform shock would require more technological progress than inferred from capital price data.

C.4 Changing the super-elasticity of demand in the log-concave demand model

In the main text, we calibrated a demand super-elasticity of $\frac{\nu}{\sigma} = 0.22$ by matching the ratio of the (unweighted) mean firm labor share to the aggregate sectoral labor share. This appendix reports results for manufacturing using a lower super-elasticity of 0.16 as estimated by Edmond, Midrigan and Xu (2022).

For this robustness check, we focus on the exercise in section 3.3. This shows how a different super-elasticity changes our inference and the effects of q and λ shocks.

First, we re-calibrate the parameters in the initial steady state. The main difference is that a lower value of the super-elasticity requires less convexity in the productivity distribution, since the mapping from productivity to firm sales is less log-concave. For manufacturing, we infer $n = 0.91$ (instead of $n = 0.74$ as in Table 3). Thus, the inferred z-distributions are closer to the log-linear case (Pareto).

Table A2 reports the main results over the transition (1982–2012) for both sectors. Relative to the results in the main text, the inferred rising competition shock is somewhat larger, with $d\ln\lambda = 0.09$ (instead of $d\ln\lambda = 0.06$ as in Table 4). However, the lower log-convexity of the z-distribution implies that the λ shock generates a smaller increase in the aggregate markup, and correspondingly a smaller decrease in the aggregate labor share. The contribution of falling capital prices to the labor share decline is similar across parametrizations. As expected, the labor share dynamics across firms are in between those obtained in the main text with a higher super-elasticity $\frac{\nu}{\sigma} = 0.22$ and the CES demand case.

Table A2: Robustness checks manufacturing: lower super-elasticity of $\frac{\nu}{\sigma} = 0.16$

		Results from log-concave demand model		
	Data	Both shocks	Only effects of $d \ln q$	Only effects of $d \ln \lambda$
	(1)	(2)	(3)	(4)
	I. Parameters and inferred aggregate shocks			
$d \ln q_{int}$		0.63	0.63	0
$d \ln q_{ext}$		5.29	5.29	0
$d \ln \lambda$		0.09	0	0.09
c_a		0.17	0.17	0.17
	II. Targeted moments, 1982–2012			
Δ aggregate labor share	-0.199	-0.198	-0.206	0.005
Δ log average capital price	-1.081	-1.084	-1.029	0
Δ log 4 firms' sales share	0.140	0.143	0.093	0.055
Relative adoption (P99+ vs. P50-75 firms)	1.71	1.71	1.66	14.23
	III. Typical firm labor share and other moments			
Δ median labor share	0.030	0.038	0.020	0.010
Δ unweighted mean	-0.017	-0.003	-0.019	0.010
Δ log 20 firms' sales share	0.072	0.139	0.111	0.035
Δ log productivity dispersion	0.050	0.074	0.065	0.000
	IV. Melitz–Polanec decomposition from Autor et al. (2020)			
Δ aggregate labor share	-0.185	-0.195	-0.204	0.003
Δ unweighted incumbent mean	-0.002	0.023	0.008	0.011
Exit	-0.055	-0.012	-0.012	-0.013
Entry	0.059	0.010	0.011	0.012
Covariance term	-0.187	-0.216	-0.210	-0.007
	V. Covariance decomposition from Kehrig and Vincent (2021)			
Market share dynamics	0.047	0.055	0.056	0.053
Labor share by size dynamics	-0.043	-0.046	-0.044	0.052
Cross-cross dynamics	-0.232	-0.174	-0.174	-0.113
	VI. Change in markups, 1982–2012			
Δ log aggregate markup	-0.011	0.010	0.011	0.000
Within-firm change in markup	-0.076	-0.025	-0.021	-0.010
Reallocation to high-markup firms	0.065	0.035	0.032	0.010

Notes: The table reports the equivalent of Table 4 in the main text but imposes a lower super-elasticity of $\frac{\nu}{\sigma} = 0.16$ (instead of $\frac{\nu}{\sigma} = 0.22$ as in the main text). The parameters of the respective economies are re-calibrated, both in the steady state to match all other targeted moments, as well as in regards to the inferred shocks $d \ln q_{int}$, $d \ln q_{ext}$, $d \ln \lambda$ and the automation fixed cost c_a over the transition.

D Calibration and results for other sectors

This appendix summarizes the results of our decomposition for non-manufacturing sectors.

D.1 Retail

We follow the same calibration approach as in manufacturing, with the calibrated parameters and targets listed in Table A3. As explained in the text, we lack detailed data for some of the moments used in retail and so we keep some of the parameters or moments from manufacturing.

Table A3: Calibration of the log-concave demand model for retail

	Parameter		Moment	Data	Model
	I. Parameters related to production function				
η	Task substitution elasticity	0.5	From Humlum (2019)	0.5	0.5
γ_ℓ	Comparative advantage	0.22	Retail labor share (BLS/BEA)	0.72	0.72
	II. Parameters governing firm dynamics and productivities in 1982 steady state				
ν/σ	Demand super-elasticity	0.22	Imputed from manufacturing		
σ	Demand elasticity	8.95	Aggregate markup	1.15	1.15
ζ	Weibull scale	0.0128	Top 20 firms' sales share	29.9%	29.9%
n	Weibull shape	0.47	Top 4 firms' sales share	15.1%	15.1%
$\underline{c}_o$	Scale operating cost	$4.6 \cdot 10^{-6}$	Entry (=exit) rate	0.062	0.062
ξ_o	Tail index operating cost	0.320	Size of exiters	0.490	0.494
μ_e	Entrant productivity	0.855	Size of entrants	0.600	0.600
ρ_z	Productivity persistence	0.86	Revenue TFP persistence among retail firms		

Notes: The table reports the calibrated parameters and targets for our model with log-concave demand in retail. The data on top firms' sales share is from Autor et al. (2020)'s estimates for the US retail sector. The annual entry rate, as well as relative sizes of entrants and exiters, are from Lee and Mukoyama (2015) and imputed from manufacturing.

Table A4 summarizes the results for retail. Column 1 reports the available data. Column 2 reports the effects of the q and λ shocks backed out to match trends in retail's labor share and concentration. Columns 3 and 4 report the effects of the increase in q and λ shocks separately. The decline in capital prices at the extensive margin continues to be the dominant force in driving the labor share decline. However, the model attributes only a small fraction of the increase in sales concentration to q shocks.

Table A4: Effects of lower capital prices and rising competition: Retail 1982–2012

		Results from log-concave demand model		
	Data	Both shocks	Only effects of $d\ln q$	Only effects of $d\ln\lambda$
	(1)	(2)	(3)	(4)
	I. Parameters and inferred aggregate shocks			
$d\ln q_{int}$		0.48	0.48	0
$d\ln q_{ext}$		3.25	3.25	0
$d\ln\lambda$		0.30	0	0.30
c_a		0.06	0.06	0.06
	II. Targeted moments, 1982–2012			
Δ aggregate labor share	-0.127	-0.127	-0.122	-0.022
Δ log average capital price	-0.865	-0.864	-0.690	0.000
Δ log sales concentration	0.546	0.546	0.065	0.480
Relative adoption (P99+ vs. P50-75 firms)	1.71	1.71	1.12	2.57
	III. Typical firm labor share and other moments			
Δ median labor share		0.048	-0.036	0.037
Δ unweighted mean		0.028	-0.046	0.035
Δ log productivity dispersion		0.033	0.016	0.001
	IV. Change in markups			
Δ log aggregate markup	0.038	0.051	0.013	0.038
Within-firm change in markup	-0.026	-0.015	-0.010	-0.016
Reallocation to high-markup firms	0.064	0.066	0.023	0.054

Notes: Column (2) reports the findings from our benchmark model. Column (3) shows results when shutting down the market access shock, and column (4) when shutting down instead the price of capital shock. The data for markups comes from Compustat estimates and is described in Section 3.3.

D.2 Wholesale and Utilities & transportation

Table A5 summarizes the steady state calibration of the model with log-concave demand for wholesale as well as the utilities & transportation sector. The calibration strategy is identical to manufacturing and retail. Here too, we lack detailed data for some of the moments used and so we keep some of the parameters or moments from manufacturing.

For these two sectors, the log-convexity of the z-distribution is rather mild in these two sectors (n slightly below 1), more in line with manufacturing than retail.

Table A6 shows the decomposition exercise. In wholesale and in utilities & transportation, the labor share decline is mild, while the observed increase in sales concentration is moderate. Consequently, the inferred capital price declines at the extensive margin are smaller than in manufacturing, while the inferred increases in competition ($d\ln\lambda$) are weaker than in retail but stronger than in manufacturing. The inferred automation fixed costs (c_a) are comparable.

Figure A2 summarizes our findings across sectors.

Table A5: Calibration of the log-concave demand model for Wholesale and Utilities & Transportation

	Parameter		Moment	Data	Model
	I. Wholesale: steady state parameters and moments (1982)				
γ_ℓ	Comparative advantage	0.63	Wholesale labor share	0.53	0.53
ν/σ	Demand super-elasticity	0.22	Imputed from manufacturing		
σ	Demand elasticity	9.4	Aggregate markup	1.15	1.15
ζ	Weibull scale	0.071	Top 20 firms' sales share	42.9%	42.9%
n	Weibull shape	0.75	Top 4 firms' sales share	22.3%	22.3%
$\underline{c}_o$	Scale operating cost	$3.2 \cdot 10^{-7}$	Entry (=exit) rate	0.062	0.062
ξ_o	Tail index operating cost	0.235	Size of exiters	0.490	0.493
μ_e	Entrant productivity	0.889	Size of entrants	0.600	0.601
ρ_z	Productivity persistence	0.86	Revenue TFP persistence among wholesale firms		
	II. Utilities & Transportation: steady state parameters and moments (1992)				
γ_ℓ	Comparative advantage	0.72	Util.&transp. labor share	0.51	0.51
ν/σ	Demand super-elasticity	0.22	Imputed from manufacturing		
σ	Demand elasticity	10.7	Aggregate markup	1.15	1.15
ζ	Weibull scale	0.066	Top 20 firms' sales share	59.1%	58.0%
n	Weibull shape	0.74	Top 4 firms' sales share	30.4%	31.3%
$\underline{c}_o$	Scale operating cost	$9.0 \cdot 10^{-8}$	Entry (=exit) rate	0.062	0.063
ξ_o	Tail index operating cost	0.212	Size of exiters	0.490	0.489
μ_e	Entrant productivity	0.891	Size of entrants	0.600	0.600
ρ_z	Productivity persistence	0.86	Revenue TFP persistence among ut. & transp. firms		

Notes: The table reports the calibrated parameters and targets for our model with log-concave demand in wholesale (panel I) and utilities and transportation (panel II). The data on top firms' sales share comes from Autor et al. (2020)'s estimates for each sector. The annual entry rate, as well as relative sizes of entrants and exiters, are from Lee and Mukoyama (2015) and imputed from manufacturing.

Table A6: Effects of lower capital prices and rising competition: Wholesale, Utilities & Transportation 1982–2012

		Results from log-concave demand model		
	(1) Data	(2) Benchmark	(3) Only $d\ln q$	(4) Only $d\ln\lambda$
A. Wholesale (1982–2012)	*I. Parameters and inferred aggregate shocks*			
$d\ln q_{int}$		1.30	1.30	0
$d\ln q_{ext}$		2.36	2.36	0
$d\ln\lambda$		0.19	0	0.19
c_a		0.18	0.18	0.18
	II. Targeted moments, 1982–2012			
Δ aggregate labor share	-0.045	-0.045	-0.048	0.005
Δ log average capital price	-1.596	-1.593	-1.563	0
Δ log sales concentration	0.202	0.209	0.048	0.163
Relative adoption (P99+ vs. P50-75 firms)	1.71	1.71	1.42	3.52
	III. Typical firm labor share and other moments			
Δ median labor share		0.187	0.156	0.033
Δ unweighted mean		0.150	0.122	0.031
Δ log productivity dispersion		0.182	0.165	0.102
	IV. Change in markups			
Δ log aggregate markup	0.038	0.009	0.006	0.003
Within-firm change in markup	-0.026	-0.043	-0.032	-0.015
Reallocation to high-markup firms	0.064	0.052	0.039	0.018
B. Utilities & Transportation (1992–2012)	*I. Parameters and inferred aggregate shocks*			
$d\ln q_{int}$		0.62	0.62	0
$d\ln q_{ext}$		1.06	1.06	0
$d\ln\lambda$		0.12	0	0.12
c_a		0.10	0.10	0.10
	II. Targeted moments, 1982–2012			
Δ aggregate labor share	-0.028	-0.028	-0.029	0.002
Δ log average capital price	-0.684	-0.683	-0.676	0
Δ log sales concentration	0.108	0.105	0.025	0.080
Relative adoption (P99+ vs. P50-75 firms)	1.71	1.71	1.47	4.53
	III. Typical firm labor share and other moments			
Δ median labor share		0.102	0.088	0.017
Δ unweighted mean		0.083	0.070	0.016
Δ log productivity dispersion		0.117	0.107	0.089
	IV. Change in markups			
Δ log aggregate markup	0.038	0.004	0.002	0.001
Within-firm change in markup	-0.026	-0.015	-0.010	-0.008
Reallocation to high-markup firms	0.064	0.018	0.012	0.010

Notes: Column (2) contains the benchmark model. Due to data availability, the transition is over 1982–2012 for wholesale, resp. 1992–2012 for utilities & transportation. Column (3) shuts down the market access shock, and column (4) shuts down the price of capital shock.

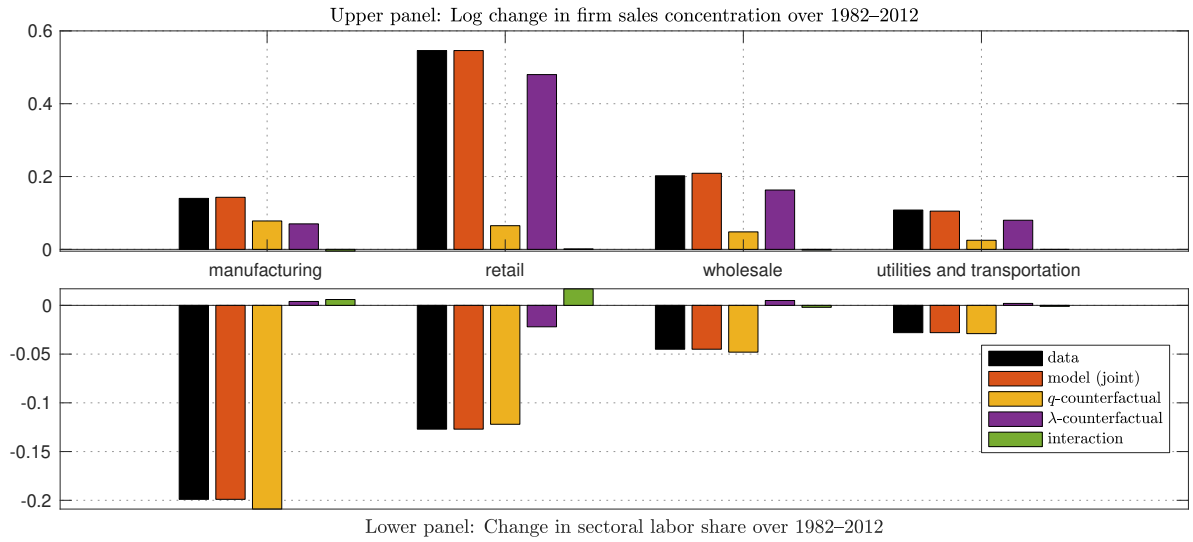

FIGURE A2: MODEL DECOMPOSITION OF LABOR SHARE AND SALES CONCENTRATION CHANGES.

For each sector, the upper panel displays the log change in firm sales concentration (i) in the data Autor et al. (from 2020), (ii) in the benchmark model with q and λ shocks jointly calibrated, (iii) in a model counterfactual that keeps only the q shock active, (iv) in a model counterfactual that keeps only the estimated λ shock active; (v) displays the interaction term, defined as (ii - iii - iv). The lower panel shows sectoral labor share changes in data (BEA-BLS) and model.

E Estimating Output Elasticities

E.1 Data description, sample, and definitions

We use data from Compustat for 1960–2016. We use the following variable definitions and conventions:

- Revenue y^R_{tf}: we measure revenue using firm sales—*SALES* in Compustat.
- Expenditures in variable inputs v_{tf}: we measure these expenditures using the cost of goods sold— *COGS* in Compustat.
- Stock of capital k_{tf}: we follow De Loecker, Eeckhout and Unger (2020) and measure the capital stock using the gross value of property, plants, and equipment—*PPEGT* in Compustat. We obtained similar results using an alternative measure of capital computed using the perpetual inventory method. For this measure, we use the gross value of property, plants, and equipment as our initial stock. We then measure net investment as the difference in the net capital stock—*PPENT* in Compustat—over consecutive periods and deflate this over time using the investment price deflator to compute the capital stock over time.
- Investment rate x_{tf}: we measure the investment rate as the percent change in capital; that is, $\ln x_{tf} = \ln k_{t+1,f} - \ln k_{tf}$
- Industry and firm groupings $c(f)$: we conduct our estimation separately for 23 NAICS industries, roughly defined at the 2-digit level. When grouping firms into size quintiles, we do so for each year and within each 3-digit NAICS industry. We also experimented with the classification of industries based on SIC codes used in Baqaee and Farhi (2020b) and obtained very similar results.
- Sample definition and trimming: following De Loecker, Eeckhout and Unger (2020), we trim the sample by removing firms in the bottom 5th and top 5th percentiles of the *COGS*-to-*SALES* distribution. In addition, following Baqaee and Farhi (2020b), we exclude firms in farm and agriculture, construction, real estate, finance, and utilities from our markup and labor share calculations in Figures 3 to 6.
- Winsorizing: we winsorize the obtained revenue elasticities at zero, and take 5-year moving averages to smooth them. Moreover, following Baqaee and Farhi (2020b), we winsorize our markup estimates at the 5th and 95th percentile of their distribution.

E.2 Estimation approach and details

Consider a firm that produces output by combining capital, k, and variable inputs, v, such as labor and materials. This section describes our approach for estimating the output-to-capital elasticity ε^k_{tf} and the output-to-variable-input elasticity ε^v_{tf} from firm-level data on revenue (y), expenditures in variable inputs (v), and capital (k). Following Olley and Pakes (1996) and Ackerberg, Caves and Frazer (2015), we make the following assumptions:[33]

A1 Differences across firms in the price of variable inputs reflect quality, which implies that we can treat expenditures in variable inputs as a measure of their quality-adjusted quantity.

A2 Revenue y^R_{tf} is given by a revenue production function of the form

$$\ln y^R_{tf} = z^R_{tf} + \varepsilon^{Rv}_{tc(f)} \cdot \ln v_{tf} + \varepsilon^{Rk}_{tc(f)} \cdot \ln k_{tf} + \epsilon_{tf},$$

where $c(f)$ denotes groups of firms with the same degree of automation and facing a common process for their revenue productivity, which only differ in their revenue productivity, z^R_{tf}, and an ex-post shock ϵ_{tf} that is orthogonal to k_{tf} and v_{tf}.

A3 Unobserved productivity z^R_{tf} evolves according to a Markov process of the form

$$z^R_{tf} = g(z^R_{ft-1}) + \zeta_{tf},$$

where ζ_{tf} is orthogonal to k_{tf} and v_{ft-1}, and the function g is common to all firms in the same group $c(f)$.

A4 True revenue, $\ln y^{R*}_{tf} = \ln y_{tf} - \epsilon_{tf}$ can be expressed as

$$\ln y^{R*}_{tf} = h(\ln x_{ft}, \ln k_{tf}, \ln v_{tf}),$$

where $\ln x_{tf} = \ln k_{t+1,f} - \ln k_{tf}$ denotes the investment rate of a firm and the function h is common to all firms in the same group $c(f)$.

[33] An alternative approach to estimating markups assumes constant returns to scale (as we do) and directly measures the user cost of capital as $R = r + \delta - \pi_k$, where r is a required rate of return inclusive of an industry-specific risk premium, δ is the depreciation rate, and π_k is the expected change over time in capital prices. One can then compute markups as revenue divided by total cost ($= V + RK$). The user-cost formula, which goes back to Hall and Jorgenson (1967) requires common and frictionless capital markets and assumes no adjustment costs for capital. This strikes us as restrictive when thinking about firms undergoing a costly automation process. Instead, the approach described below makes no assumptions about the marginal product of capital across firms, or the importance of adjustment costs.

A5 The gross output production function exhibits constant returns to scale in capital and variable inputs, which implies that output elasticities are given by

$$\text{(A8)} \qquad \varepsilon^{v}_{tf} = \varepsilon^{Rv}_{tc(f)} \Big/ \left(\varepsilon^{Rv}_{tc(f)} + \varepsilon^{Rk}_{tc(f)}\right) \qquad\qquad \varepsilon^{k}_{tf} = \varepsilon^{Rk}_{tc(f)} \Big/ \left(\varepsilon^{Rk}_{tc(f)} + \varepsilon^{Rk}_{tc(f)}\right).$$

Assumptions A1–A4 are standard in the literature. Assumption A4 justifies the use of the investment rate as a proxy variable. Economically, this assumption requires that all firms in a given group share the same investment policy function $k_{t+1,f} = \pi(k_{tf}, z^R_{tf})$, and that this common policy function is invertible. Under these assumptions, and given a grouping of firms $c(f)$, we can estimate revenue elasticities following the usual approach from Ackerberg, Caves and Frazer (2015), which uses the investment rate as a proxy variable to obtain true revenue and then estimates revenue elasticities by exploiting the orthogonality of ζ_{tf} to k_{tf} and $v_{t-1,f}$.

Assumption A5 is added to deal with the fact that we do not observe prices, such that the usual estimation procedure yields revenue elasticities, not the quantity elasticities that are relevant for computing markups (see Bond et al., 2021). Under Assumption A5 we can recover output elasticities from revenue elasticities using (A8). Suppose that revenue is given by $y^R = p(q) \cdot q$, where $p(q)$ is the inverse demand curve. Quantity elasticities and revenue elasticities are then linked according to $\varepsilon^{Rv} = (p'(q) \cdot q/p(q) + 1) \cdot \varepsilon^v$ and $\varepsilon^{Rk} = (p'(q) \cdot q/p(q) + 1) \cdot \varepsilon^k$, where $1/\mu = (p'(q) \cdot q/p(q) + 1)$. Assuming constant returns to scale implies that $\varepsilon^v = \varepsilon^{Rv}/(\varepsilon^{Rv} + \varepsilon^{Rk})$, as wanted.

Given a grouping of firms $c(f)$, we can estimate revenue elasticities following the usual approach from Ackerberg, Caves and Frazer (2015), which uses investment as a proxy variable for unobserved productivity. This requires a first-stage regression where we first compute "true" output as

$$\ln y^{R*}_{tf} = \mathbb{E}[\ln y^R_{tf} | \ln x_{tf}, \ln k_{tf}, \ln v_{tf}, t, c(f)] = h(\ln x_{tf}, \ln k_{tf}, \ln v_{tf}; \theta^h_{tc(f)}).$$

Here $\theta^h_{tc(f)}$ is a parametrization for a flexible function h that might vary over time and between groups of firms. For any pair of revenue elasticities $\varepsilon^{Rv}_{tc(f)}$ and $\varepsilon^{Rk}_{tc(f)}$, one can then compute revenue productivity as

$$z^R_{tf} = \ln y^{R*}_{tf} - \varepsilon^{Rv}_{tc(f)} \cdot \ln v_{tf} - \varepsilon^{Rk}_{tc(f)} \cdot \ln k_{tf},$$

estimate the flexible model

$$z_{tf}^R = g(z_{t-1,f}^R; \theta_{tc(f)}^g) + \zeta_{tf},$$

where $\theta_{tc(f)}^g$ is a parametrization for a flexible function g, and form the following moment conditions that identify the revenue elasticities:

$$\mathbb{E}\left[\zeta_{tf} \otimes (\ln k_{tf}, \ln v_{t-1,f})\right] = 0.$$

In our baseline approach, we parametrize the functions h and g using quadratic polynomials and conduct our estimation over 10-year rolling windows. More importantly, and in line with the emphasis in our model that large firms operate different technologies and face a different demand curve, we group firms by quintiles of sales in each industry. Thus, our estimation provides output elasticities that vary over time, by industry, and by quintiles of firm size in each industry. This represents a significant deviation from previous papers which assume that all firms in a given industry share the same output elasticities.

A byproduct of this estimation procedure are series for revenue TFP, z_{tf}^R. The estimated persistence of revenue TFP is 0.95 for manufacturing and 0.86 for retail, wholesale, utilities and transportation. These justifies the values of ρ_z used in our calibration approach.

Besides our main estimation approach, we also explored the following variations:

Estimates parametrizing g and h using cubic polynomials We estimate elasticities under the same assumptions outlined in the main text, but parametrize g and h using cubic polynomials. Figure A3 plots the behavior of the resulting output elasticities over time by firm size quintile. Figure A4 reports the contribution of within-firm changes in markups and between-firm reallocation to (percent) changes in the labor share.

Estimates assuming there are no ex-post shocks ϵ In the absence of ex-post shocks, we can treat observed revenue as true revenue and there is no need to use a proxy variable to recover productivity. Instead, we can compute revenue productivity directly as

$$z_{tf}^R = \ln y_{tf}^R - \varepsilon_{tc(f)}^{Rv} \cdot \ln v_{tf} - \varepsilon_{tc(f)}^{Rk} \cdot \ln k_{tf},$$

and proceed with the rest of the estimation in the same way as before.

Figure A5 plots the behavior of the resulting output elasticities over time by firm size quintile. Figure A6 reports the contribution of within-firm changes in markups and between-

firm reallocation to (percent) changes in the labor share.

Estimates assuming a linear Markov process for productivity Suppose that productivity follows a linear Markov process

$$z_{tf}^{R} = \beta z_{t-1,f}^{R} + \zeta_{tf}.$$

Define $\upsilon_{tf} = z_{tf}^{R} + \epsilon_{tf}$. Because ex-post shocks are i.i.d, we have that υ_{tf} also follows a linear Markov process

$$\upsilon_{tf} = \beta \upsilon_{t-1,f} + \underbrace{\zeta_{tf} + \epsilon_{tf} - \beta \epsilon_{t-1,f}}_{=\iota_{tf}}.$$

Estimation proceeds as follows. First, we can compute υ_{tf} directly as

$$\upsilon_{tf} = \ln y_{tf}^{R} - \varepsilon_{tc(f)}^{Rv} \cdot \ln v_{tf} - \varepsilon_{tc(f)}^{Rk} \cdot \ln k_{tf}.$$

Then we estimate the linear model

$$\upsilon_{tf} = \beta \upsilon_{t-1,f} + \iota_{tf},$$

and base estimation on the moment conditions

$$\mathbb{E}\left[\iota_{tf} \otimes (\ln k_{tf}, \ln v_{t-1,v})\right] = 0.$$

Figure A7 plots the behavior of the resulting output elasticities over time by firm size quintile. Figure A8 reports the contribution of within-firm changes in markups and between-firm reallocation to (percent) changes in the labor share.

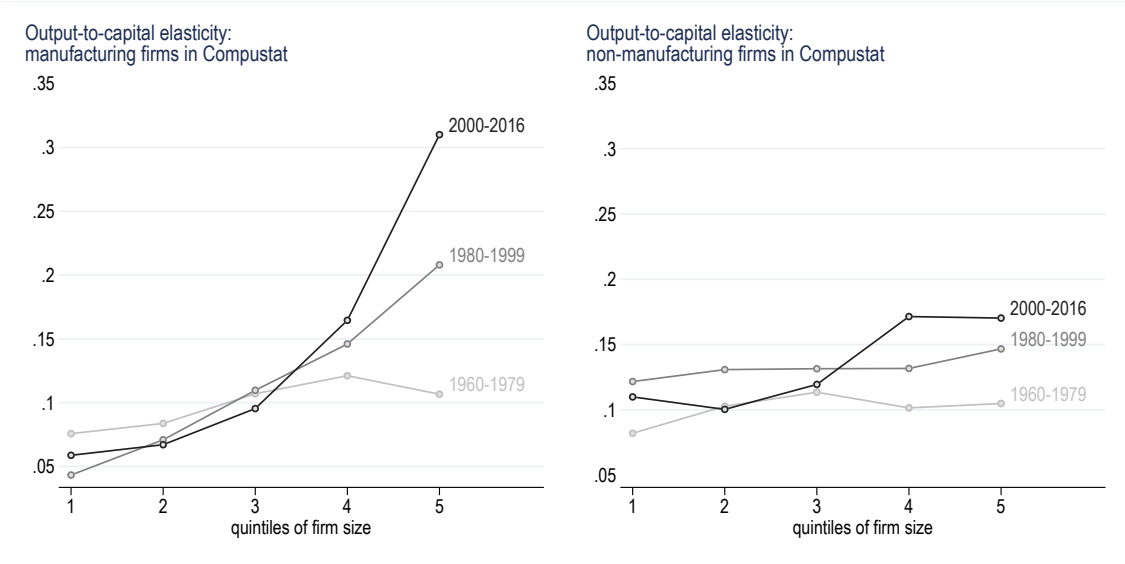

FIGURE A3: OUTPUT-TO-CAPITAL ELASTICITIES FOR COMPUSTAT FIRMS ESTIMATED USING A CUBIC PARAMETRIZATION OF g AND h. The left panel presents estimates for Compustat manufacturing firms. The right panel presents estimates for Compustat non-manufacturing firms. Firm-level elasticities are estimated using a cubic parametrization for g and h, as explained in Appendix E.

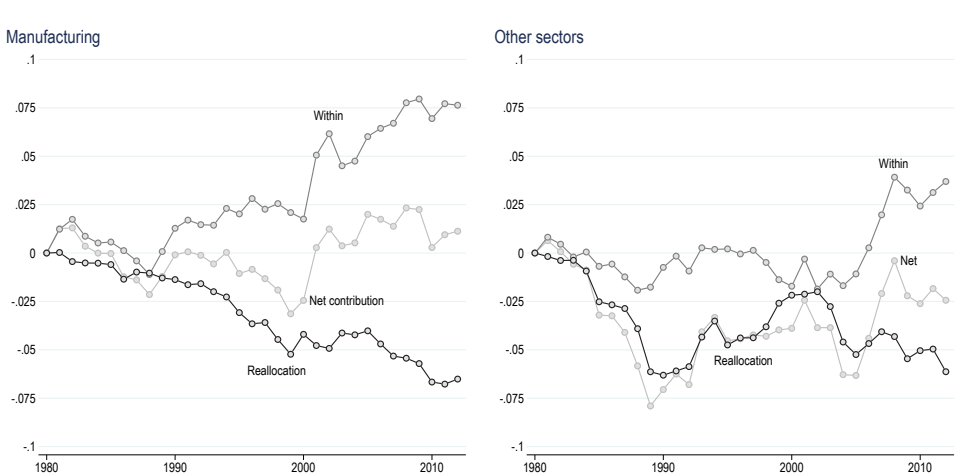

FIGURE A4: DECOMPOSITION OF THE CONTRIBUTION OF WITHIN-FIRM CHANGES IN MARKUPS AND BETWEEN-FIRM REALLOCATION TO (PERCENT) CHANGES IN THE LABOR SHARE. See the main text for details on this decomposition. Firm-level markups are estimated using a cubic parametrization for g and h, as explained in Appendix E. The left panel provides the decomposition for manufacturing firms in Compustat. The right panel provides the decomposition for Compustat firms in other economic sectors.

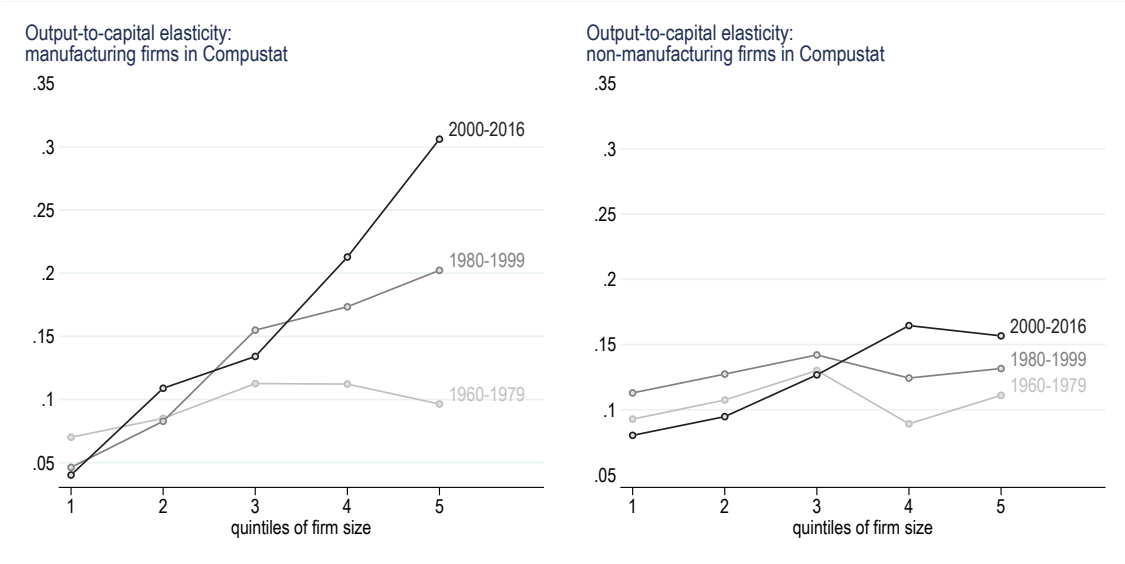

FIGURE A5: OUTPUT-TO-CAPITAL ELASTICITIES FOR COMPUSTAT FIRMS ESTIMATED UNDER THE ASSUMPTION THAT THERE ARE NO EX-POST SHOCKS. The left panel presents estimates for Compustat manufacturing firms. The right panel presents estimates for Compustat non-manufacturing firms. Firm-level elasticities are estimated under the assumption of no ex-post shocks, as explained in Appendix E.

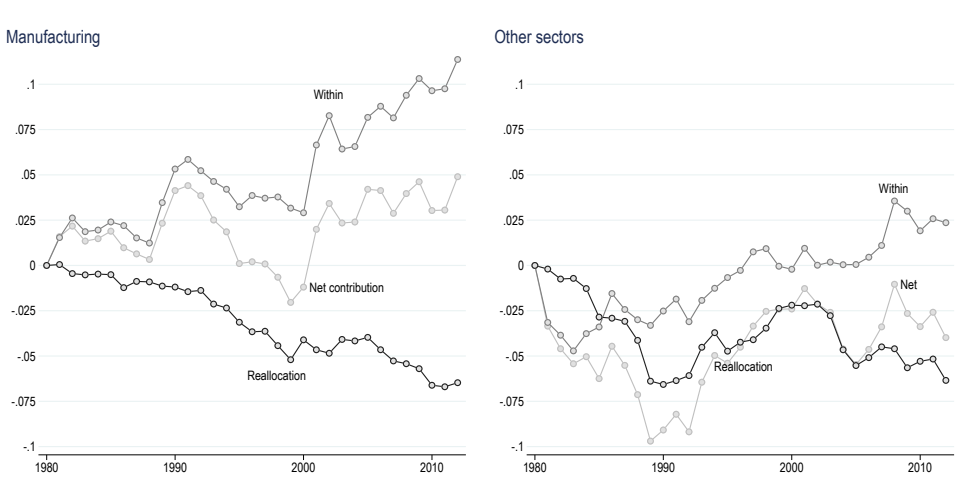

FIGURE A6: DECOMPOSITION OF THE CONTRIBUTION OF WITHIN-FIRM CHANGES IN MARKUPS AND BETWEEN-FIRM REALLOCATION TO (PERCENT) CHANGES IN THE LABOR SHARE. See the main text for details on this decomposition. Firm-level markups are estimated under the assumption of no ex-post shocks, as explained in Appendix E. The left panel provides the decomposition for manufacturing firms in Compustat. The right panel provides the decomposition for Compustat firms in other economic sectors.

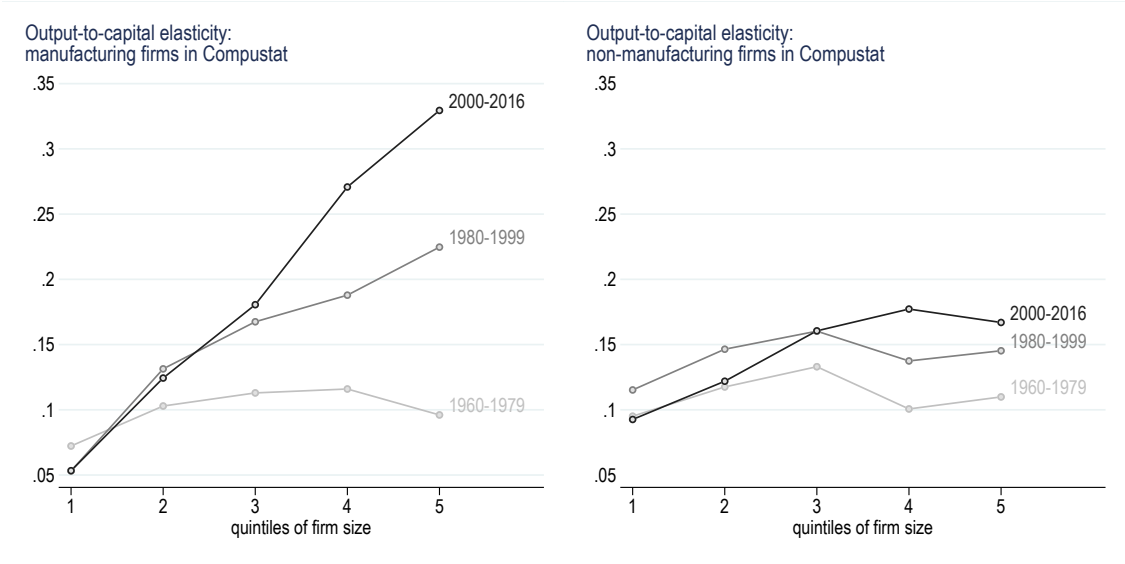

FIGURE A7: OUTPUT-TO-CAPITAL ELASTICITIES FOR COMPUSTAT FIRMS ESTIMATED UNDER THE ASSUMPTION THAT PRODUCTIVITY FOLLOWS A LINEAR MARKOV PROCESS. The left panel presents estimates for Compustat manufacturing firms. The right panel presents estimates for Compustat non-manufacturing firms. Firm-level elasticities are estimated under the assumption that productivity follows a linear Markov process, as explained in Appendix E.

FIGURE A8: DECOMPOSITION OF THE CONTRIBUTION OF WITHIN-FIRM CHANGES IN MARKUPS AND BETWEEN-FIRM REALLOCATION TO (PERCENT) CHANGES IN THE LABOR SHARE. See the main text for details on this decomposition. Firm-level markups are estimated under the assumption that productivity follows a linear Markov process, as explained in Appendix E. The left panel provides the decomposition for manufacturing firms in Compustat. The right panel provides the decomposition for Compustat firms in other economic sectors.

E.3 Implications for the average markup in the economy as a whole

Figure A9 plots the implied time series for the economy-wide aggregate markup, computed as a sales-weighted *harmonic* mean of firm-level markups. Our estimates for markups suggest that they have been stable over time at around 1.2. This is in line with our quantitative exercise, which points to a modest increase in markups.

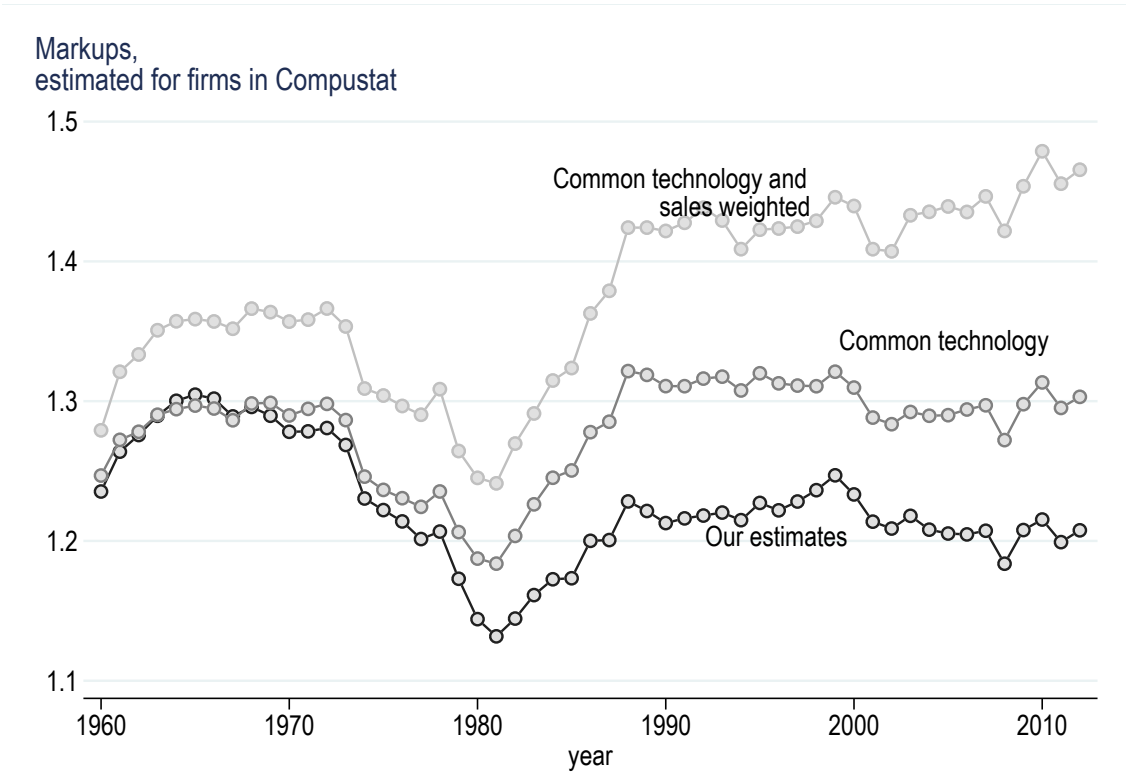

FIGURE A9: MARKUPS. The figure presents the aggregate markup for firms in Compustat. Our estimates are obtained as as a sales-weighted *harmonic* mean of firm-level markups. The figure also reports the aggregate markup that would result under the assumption of common output elasticities across firms in the same industry, and a version of these estimates that aggregates firms' markups using a sales-weighted arithmetic mean.

For comparison, we provide an alternative estimate of the aggregate markup obtained under the assumption that all firms in an industry operated technologies with the same capital intensity (as opposed to letting it vary by size class). This series reveals a mild secular increase in the aggregate markup from 1.25 in 1960 and 1.2 in 1980 to 1.3 in recent years, which aligns with the harmonic-mean estimates in Edmond, Midrigan and Xu (2022). We also provide estimates for an *arithmetic* mean of sales-weighted markups obtained under the assumption that all firms in a given industry operate technologies with the same capital intensity, which coincide with the series in De Loecker, Eeckhout and Unger (2020). Despite

its increasing trend over time, this series is inappropriate for understanding the contribution of markups to the decline in the labor share because it ignores differences in technology across firm-size classes and uses the wrong weights for aggregation.

E.4 Additional evidence from Compustat

This section provides additional descriptive statistics from Compustat that support the notion that large firms operate more capital-intensive technologies. In what follows, we estimate regression models of the form

$$\ln y_{tfi} = \alpha_{ti} + \beta_{tc(f)} + \varepsilon_{tfi}, \tag{A9}$$

where we explain different measures for the capital intensity y_{tfi} of firm f in industry i at time t as a function of industry and year fixed effects (the α_{ti}) and size class dummies ($\beta_{tc(f)}$) that are allowed to vary over time. In particular, we estimate different size-class dummies for the periods of 1960–1980, 1980–2000 and 2000–2016. We treat firms in the smallest size class of an industry as the excluded category and report estimates weighted by firm sales.

Figure A10 plots estimates of equation (A9) for firms' investment rates, defined as their investment (*CAPX* in Compustat) normalized by variable cost (top panel), employment (middle panel), and sales (bottom panel). The left panel provides estimates for manufacturing firms and the right panel for firms outside of manufacturing. For the 1980–2000 and 2000–2016 period, the largest manufacturing firms in each industry have had investment rates 60–140 log points higher than those of the smallest firms. Outside of manufacturing, the difference is less pronounced, with the largest firms having 10–90 log points higher investment rates than the smallest firms in their industries. In both cases, the gradient by size has become steeper over time.

Figure A11 plots estimates for firms' capital intensity, defined as their net capital stock (*PPENT* in Compustat) normalized by variable cost (top panel), employment (middle panel), and sales (bottom panel). For the 1980–2000 and 2000–2016 period, the largest manufacturing firms in each industry had a 55 log point higher capital to variable cost ratio, a 120 log point higher capital per worker, and a 45 log point higher capital to sales ratio than the smaller firms in their industries. Here too, we see some evidence of the gradient by size becoming steeper over time, though the gradient and its rotation are less pronounced outside of manufacturing.

Finally, Figure A12 plots estimates for firms' reliance on capital services. Along a bal-

anced growth path, the flow value of capital services used by a firm can be computed as[34]

$$\text{flow value of capital services} = (r - g) \cdot \text{net capital stock} + \text{capital expenditures}.$$

The figure provides estimates normalizing the flow value of capital services by variable costs (so that we get a measure of capital services relative to variable input services), employment, and sales (a measure of capital services in sales). In this exercise, we fix $r - g = 2.5\%$, which aligns with the calibration in Farhi and Gourio (2018). For the 1980–2000 and 2000–2016 period, the largest manufacturing firms in each industry had a 70 log point higher reliance on capital services vs. variable input services when compared to the smallest firms in their industries. Here too, we see some evidence of the gradient by size becoming steeper over time, with the gradient and its rotation being less pronounced outside of manufacturing.

[34] In particular, suppose the firm faces no adjustment costs. Then the PDV of capital services equal the PDV of capital costs. The PDV of capital costs are $(1+r)\cdot$net capital stock$+\frac{1+r}{r-g}\cdot$capital expenditure, which gives the cost of the initially installed capital and of financing it plus the PDV of capital expenditures. The flow value of capital services is $\frac{r-g}{1+r}\cdot$PDV of capital costs and we get the formula in the text.

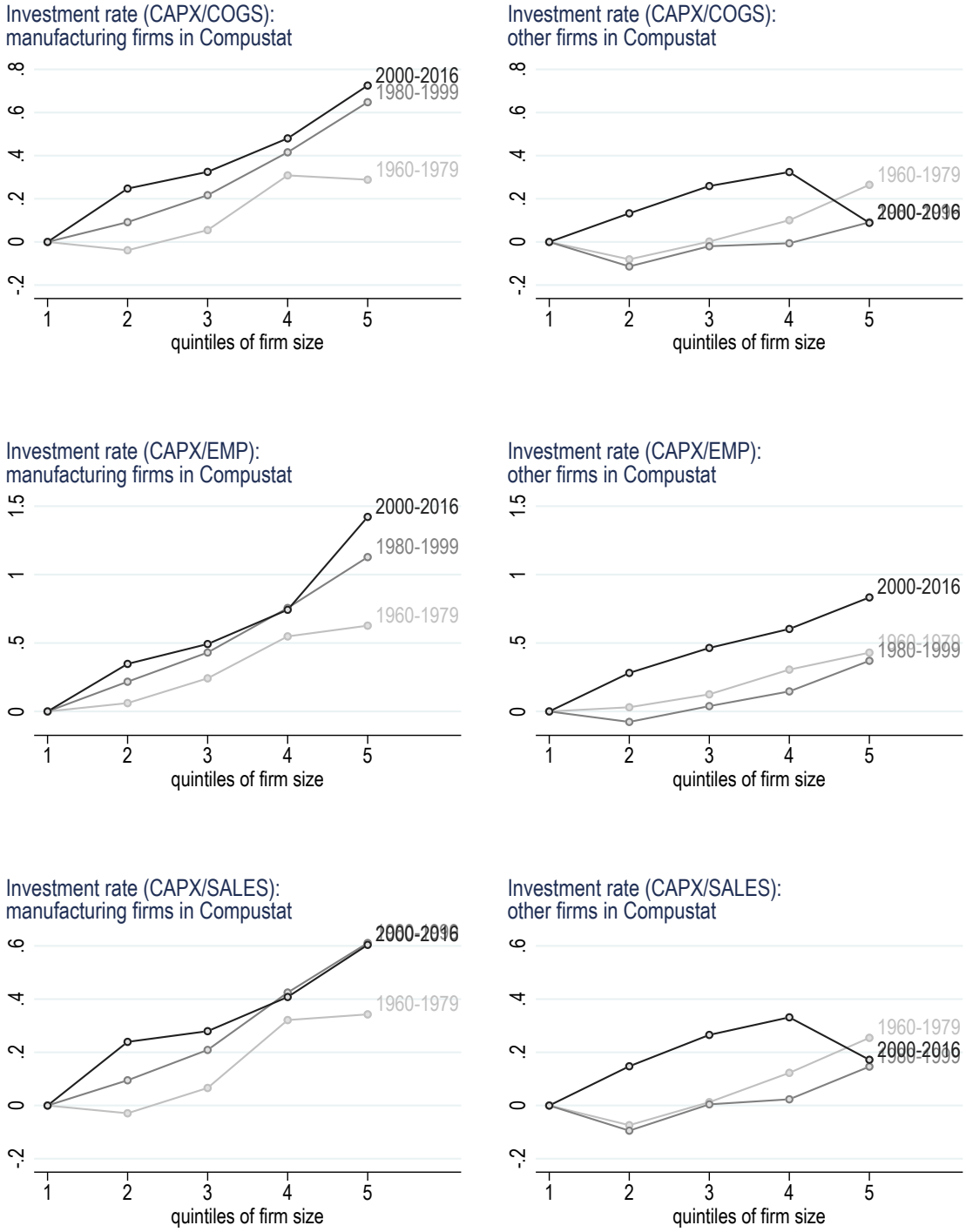

FIGURE A10: INVESTMENT RATES, COMPUSTAT. The figure presents estimates of the relative difference in investment rates by firm-size class using Compustat. The left panel presents estimates for Compustat manufacturing firms. The right panel presents estimates for Compustat non-manufacturing firms.

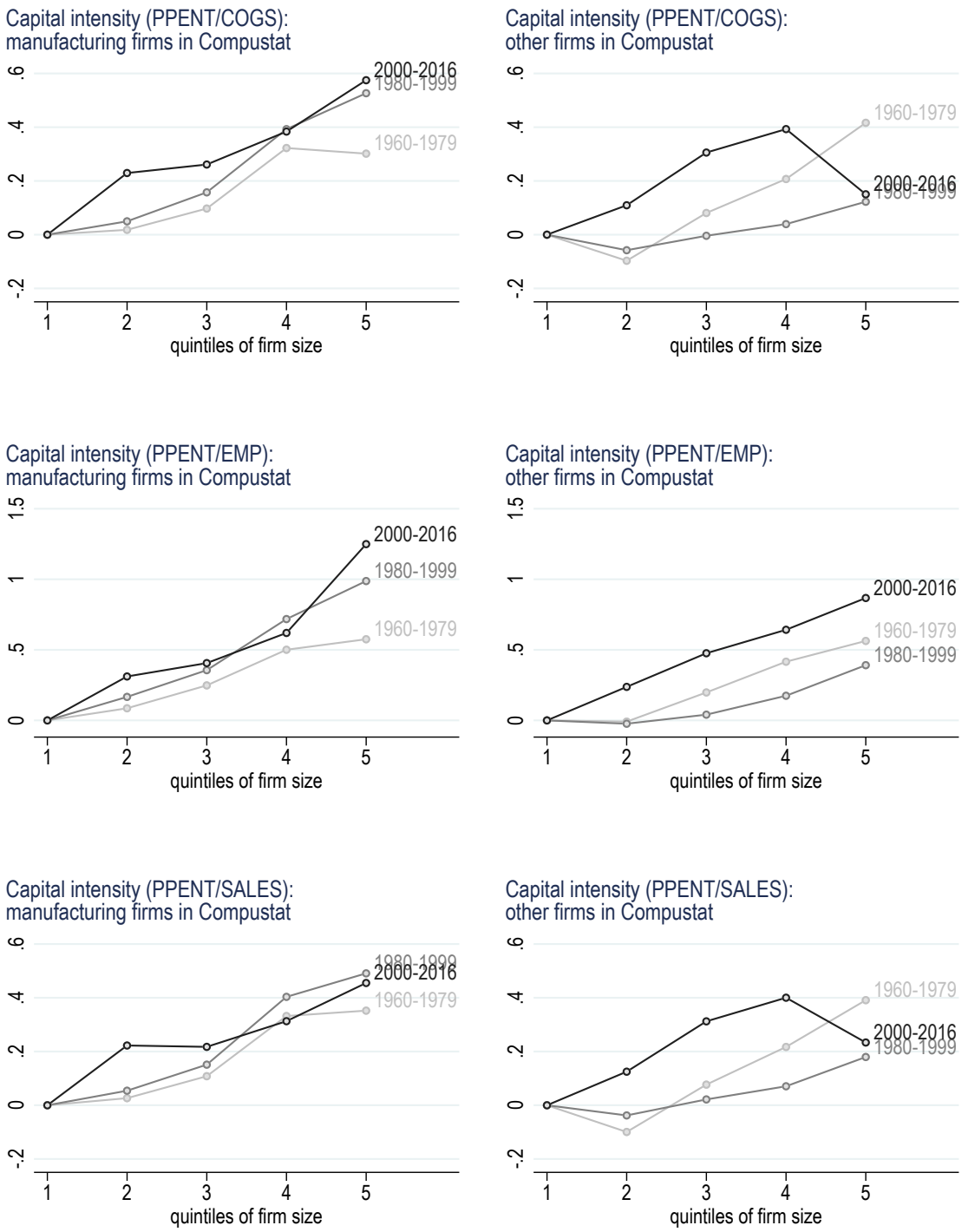

FIGURE A11: CAPITAL INTENSITY, COMPUSTAT. The figure presents estimates of the relative difference in capital intensity by firm-size class using Compustat. The left panel presents estimates for Compustat manufacturing firms. The right panel presents estimates for Compustat non-manufacturing firms.

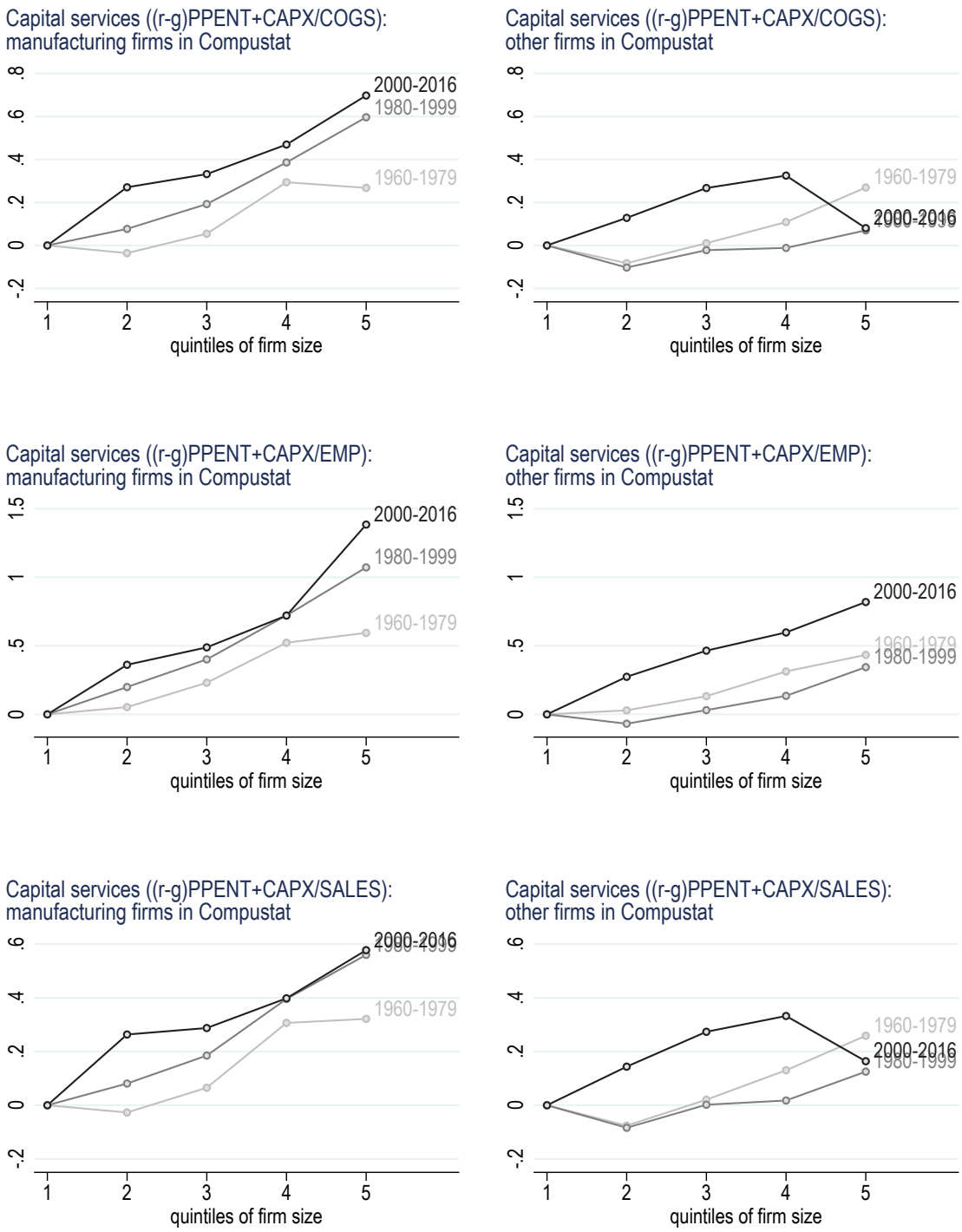

FIGURE A12: CAPITAL SERVICES, COMPUSTAT. The figure presents estimates of the relative difference in capital services by firm-size class using Compustat. The left panel presents estimates for Compustat manufacturing firms. The right panel presents estimates for Compustat non-manufacturing firms.

F Measurement of capital prices

We create measures of quality-adjusted capital prices at the sectoral level building on DiCecio (2009), Cummins and Violante (2002), and Gordon (1990).

In the first step, we obtain data on nonresidential asset prices and quantities from the BEA Fixed Asset Tables. These data cover 39 types of equipment, 32 types of structures, 3 types of software, and 22 types of other intellectual property products. We exclude other intellectual property products from our analysis since these would be considered part of the fixed cost of adopting new technologies in our model. The data cover the period from 1947 to 2020 and include information on investment at current nominal prices, investment at constant 2012 prices, stocks, and depreciation.

Using these series, we construct a price index for each detailed asset a as

$$p_{a,t} = \frac{\text{investment at current nominal prices}_{a,t}}{\text{investment at constant 2012 prices}_{a,t}}.$$

We let $\Delta \ln p_{a,t}$ denote the percent change in asset prices between time t and $t+1$.

Our second step involves adjusting the BEA prices for quality. We follow the work by DiCecio (2009) and Cummins and Violante (2002). These authors use the series for quality-adjusted investment prices from Gordon (1990), and which covered the postwar period up to 1983, and extend it from 1947 to 2011. Cummins and Violante (2002) estimate a statistical model explaining Gordon's quality-adjusted price indices as a function of those by the BEA/NIPA, their lags, and time trends. They then extrapolate this model to produce quality-adjusted price indices for 1947–2000. DiCecio (2009) follows the same procedure and creates an updated series up to 2011 for equipment and software. We use the estimates from DiCecio (2009) on the quality adjusted price of equipment and software, obtained via FRED (variable code PERICD), and denote the percent change in the quality-adjusted price of software and equipment as $\Delta \ln p^*_{t,\text{E\&S}}$. We then compute a user-cost weighted price index for software and equipment from the BEA data using a Törnqvist index

$$\Delta \ln p_{t,\text{E\&S}} = \sum_{a \in \text{E\&S}} \frac{1}{2} \cdot \left(s^{\text{E\&S}}_{t+1,a} + s^{\text{E\&S}}_{a,t} \right) \cdot \Delta \ln p_{a,t},$$

where $s^{\text{E\&S}}_{a,t}$ denotes the share of asset a in equipment and software capital services.[35] The

[35] We compute capital services derived from an asset a as

$$\text{capital services}_{a,t} = (r + \delta_a - \Delta \ln p_{a,t}) \cdot \text{stock asset}_{a,t},$$

implied quality adjustment for equipment and software is therefore equal to

$$\Delta \text{quality adjustment}_{\text{E\&S},t} = \Delta \ln p_{E\&S,t} - \Delta \ln p^{*}_{\text{E\&S},t}.$$

In the BEA data, the price of equipment and software declined by an average of 3.1% per year in 1980-2011. The quality-adjusted series from DiCecio (2009) shows a decline on 5.7% per year, which implies an improvement in the quality of equipment and software of 2.6% per year.

We then compute a quality adjusted series for the detailed equipment and software products in the BEA data as

$$\Delta \ln p^{*}_{a,t} = \Delta \ln p_{a,t} - \Delta \text{quality adjustment}_{\text{E\&S},t}.$$

This assumes a common quality adjustment for all types of software and equipment. For structures, we do not perform quality adjustment.

In the third step, we account for changes in taxation using the estimates in Acemoglu, Manera and Restrepo (2020) of effective taxes on equipment, software, and structures. These authors estimate that the effective tax on equipment decreased from 12.4% to 4.7% during 1981–2018, the effective tax on software decreased from 14.6% to 4.7% during 1981–2018, and the effective tax on structures increased from 8.3% to 9% during 1981–2018. These changes in taxes imply a further reduction in the cost of producing tasks with capital of close to 10% during the whole 1981–2018 period.

In the fourth step, we compute a measure for the relative price of capital by asset, $\Delta \ln p^{*,r}_{a,t}$ by taking our quality-adjusted price indices adjusted for taxes and subtracting changes in the BEA price of consumption expenditures index.

In the final step, we construct a sector-specific measure of capital prices using a user-cost weighted Törnqvist index

$$\Delta \ln p_{i,t} = \sum_{a} \frac{1}{2} \cdot \left(s^{i,k}_{a,t+1} + s^{i,k}_{a,t} \right) \cdot \Delta \ln p^{*,r}_{a,t},$$

where $s^{i,k}_{a,t}$ denotes the share of asset a in total capital services in sector i, computed also from the industry-level version of the BEA Fixed Asset Tables. This index provides the average decline in capital prices used in sector i over the 1980–2011 period.

where we take a required rate of return $r = 4\%$, and use the depreciation rate and change in capital prices from the BEA.

Our resulting sectoral price indices imply that the average price of capital used in manufacturing declined by 108 log points from 1980 to 2011. For retail, the average decline was of 86 log points, for utilities 68 log points, and for wholesale of 159 log points. These differences across sectors reflect the different bundles of capital goods used, with manufacturing and wholesale investing more heavily in equipment and software.

F.1 Separating intensive and extensive margins

As discussed in the text, there is no straightforward method for separating advances in capital at the intensive and extensive margin. The problem is that data on capital prices are available for coarse categories and do not distinguish between investments to replace old equipment (intensive margin) and investment in capital used in tasks previously assigned to labor (extensive margin).

Despite this limitation, the available data does point to a more limited decline in capital prices at the intensive margin, in line with our inferred q shocks for manufacturing and retail.

There are two ways of illustrating this point. The first is model dependent. Imagine that all advances in capital take place at the intensive margin. Our model implies that, in this case, one can compute an aggregate price index as an exact CES index of all capital price declines with an elasticity of substitution of η, which governs the substitution across the tasks benefiting from these advances. This is the correct way of aggregating all capital price declines taking place at the intensive margin. This index can be computed using "exact hat algebra" as

$$\frac{p_{i,t}^{CES}}{p_{i,t_0}^{CES}} = \left(\sum_a s_{a,t_0}^{i,k} \cdot \left(\frac{p_{a,t}^*}{p_{a,t_0}^*} \right)^{1-\eta} \right)^{\frac{1}{1-\eta}}.$$

Using a value of $\eta = 0.5$ from Humlum (2019), we estimate a 76 log point decrease in capital prices at the intensive margin for manufacturing and a 31 log point decrease in capital prices at the intensive margin for retail. In both cases, the data points to a minor share of capital advances at the intensive margin.

The second strategy involves classifying assets into "established" and "new" types of capital. We classify an asset as "established" in a sector if its net investment rate in the 70s was below the average sectoral rate of capital formation. We view a below-average net investment rate as an indication of an asset whose stock has already reached a high enough level. We classify an asset as "new" if its net investment rate in the 70s was above average. We view this as an indication of an asset whose stock was only being built.

In manufacturing, the price of established assets decreased by 48.6 log points and the price of new assets decreased by 166 log points. In retail, the price of established assets decreased by 36 log points and the price of new assets decreased by 117 log points.

The intuitive (but imperfect) idea is that price declines for established assets provide an indication of the size of advances at the intensive margin, and price declines for new assets provide an indication of the potential magnitude of advances at the extensive margin.

This mapping is necessarily imperfect, especially for the extensive margin. For example, industrial furnaces might have been in use in the 1980s while robots were not. Yet both would be aggregated into the BEA asset class "Special industrial machinery," causing us to mislabel advances in robotics as taking place at the intensive margin. This would cause us to over-estimate the extent of advances in capital at the intensive margin. Conversely, new types of capital and assets might be associated with new industries and products and are not necessarily used to automate tasks as in our model. For example, the stock of solar panels is a new asset that has nothing to do with automation. This would cause us to over-estimate the extent of advances in capital at the extensive margin.

For these reasons, our preferred interpretation of these estimates is as a reality check. The estimates support the idea that capital advances at the intensive margin were relatively modest. The estimates also point to larger capital advances in "new" types of capital, driven in part by software and computers. Both facts are necessary if we believe extensive margin advances in capital were a dominant force during this period. However, the missing pieces are that we do not know if in practice the "new" types of capital facilitated the substitution of capital for labor at the extensive margin, as in our model, or if these new types of capital had other uses. We also do not know if all forms of capital inside an established asset class operated at the intensive margin, even though this seems more plausible.

Both calculations reflect the fact that the decline in capital prices has been far from uniform, even across broad asset classes. Figure A13 illustrates this point. It plots the price decline per year for all non-residential fixed assets in the BEA Fixed Asset Tables. The figure shows a more pronounced decline for software and computer equipment.

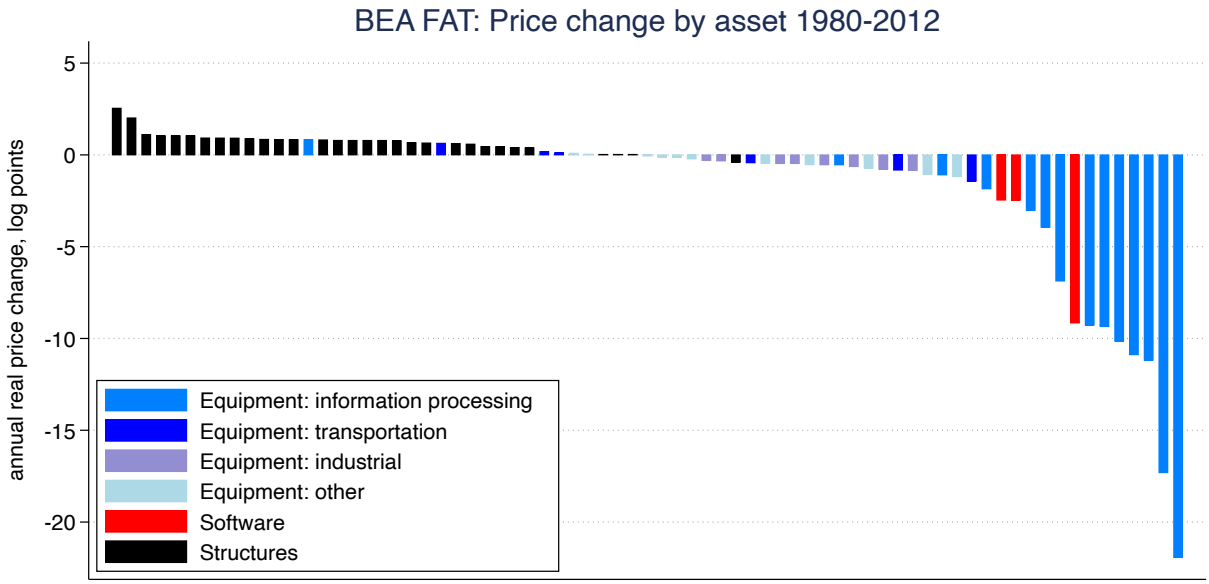

FIGURE A13: CAPITAL PRICE DECLINES BY BEA FIXED ASSET CLASS. The figure plots the average annual real quality-adjusted capital price change over 1980–2012, for private nonresidential fixed assets in the BEA Fixed Asset Tables.

G Historical behavior of labor shares

This section provides additional motivation for our focus on the 1982–2012 period. As a starting point, Figure A14 provide data on payroll shares by sector for 1947–1987 and 1987–2016 from the BEA industry accounts. We split the data into these two periods due to changes in industry definitions introduced by the BEA in 1987, as it switched from the *Standard Industry Classification* to the *North American Industry Classification System*. As discussed in the main text, Figure A14 shows that payroll shares were constant or increasing up to 1982, and then started a sharp decline both in manufacturing, retail and wholesale. Payroll shares differ from labor shares in that they exclude compensation and self employment. But looking at payroll shares has the advantage of allowing us to go back further in time.

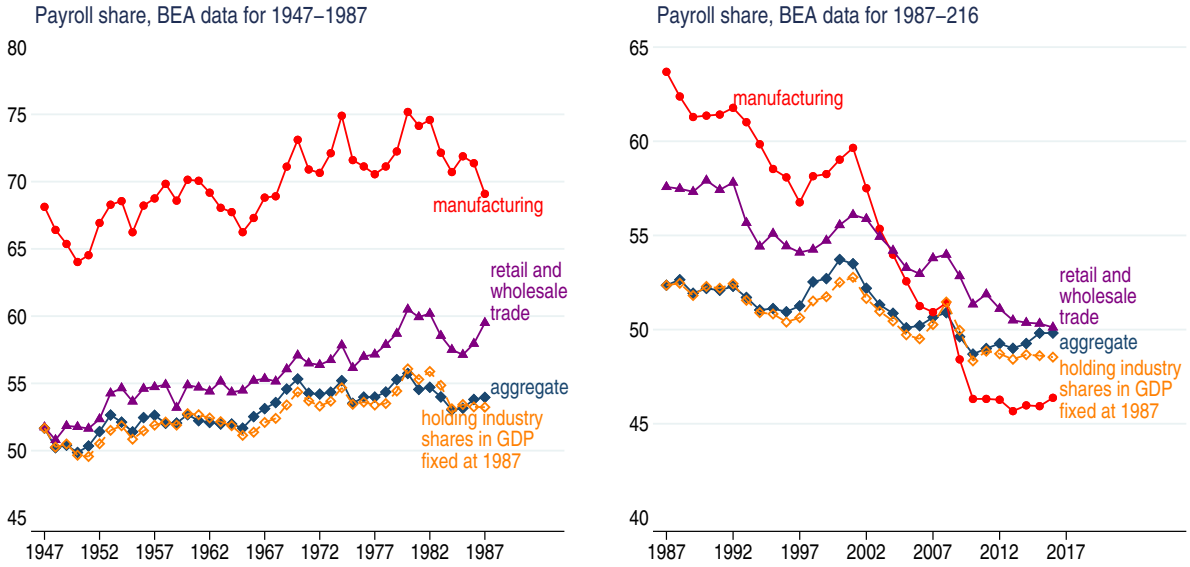

Figure A14: Payroll share in the US for 1947–2016. The figure plots the payroll share of value added, both for some specific sectors and the economy as a whole. Data from the BEA industry accounts. Industry definitions based on SIC in left panel, NAICS in right panel.

Labor shares (which also include non-wage compensation) are available starting in 1963 from the BEA-BLS integrated industry-level production account. Figure A15 confirms that labor shares exhibit the same trend behavior with a flat or slightly increasing trend until 1982 and a subsequent decline. This motivates our focus on the 1982–2012 period and supports our choice of 1982 as the steady state of the model.

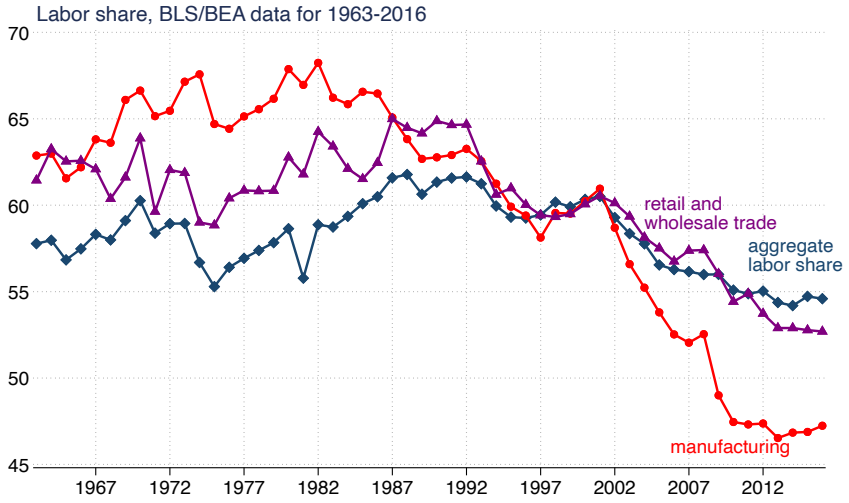

FIGURE A15: LABOR SHARE IN THE US FOR 1963–2016 The figure plots the labor share of value added, both for some specific sectors and the economy as a whole. Data from the BEA-BLS integrated industry-level production account.

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